MIL-A-62019B(AT)
24 January 1969
SUPERSEDING
MIL-A-62019A(AT)
14 February 1967

MILITARY SPECIFICATION

ARMORED RECONNAISANCE/AIRBORNE ASSAULT VEHICLE: FULL TRACKED, 152MM, M551; PROCESSING FOR STORAGE AND SHIPMENT OF

1. SCOPE

1.1 Scope. This specification covers processing of the M551 Assault Vehicle, Reconnaisance/Airborne, Armored, mounting a 152MM Gun, designated General Sheridan, for immediate use shipment and for domestic or oversea shipment, including car-loading; meeting the requirements of levels A, B and C processing (see 1.2).

1.2 Classification. Processing shall be of the following levels as specified (see 6.1):

Level A	- Processing for domestic or oversea shipment and any storage outside of buildings in excess of 90 days from date of processing (periodic care and preservation during storage required).
Level B and C	- Limited processing for immediate shipment and for domestic or oversea shipment (excluding open deck loading) and any storage not to exceed 90 days from date of processing.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

L-P-378	- Plastic Sheet and Strip (Polyolefin).
O-A-548	- Antifreeze, Ethylene Glycol, Inhibited.
O-E-760	- Ethyl Alcohol (Ethanol); Denatured Alcohol: and Proprietary Solvent.
O-I-490	- Inhibitor, Corrosion, Liquid Cooling System.
O-S-576	- Sodium Bicarbonate, Technical.

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P-D-220	-	Detergent, General Purpose.
V-T-295	-	Thread, Nylon.
FF-N-105	-	Nails, Wire, Brads, and Staples.
GG-T-250	-	Tester, Antifreeze Solutions.
QQ-A-200/8	-	Aluminum Alloy Bar, Rod, Shapes, Tube and Wire, Extruded, and Structural Shapes, 6061 and 6062.
QQ-A-225/8	-	Aluminum Alloy Bar, Rod, Wire, and Special Shapes; Rolled, Drawn, or Cold Finished, 6061.
QQ-P-416	-	Plating, Cadmium (Electrodeposited).
QQ-S-630	-	Steel, Bar, Carbon, Hot Rolled, (Merchant Quality).
QQ-S-631	-	Steel, Bar, Carbon, Hot Rolled, (Special Quality).
QQ-S-634	-	Steel, Bar, Carbon, Cold Finished, (Standard Quality).
QQ-S-698	-	Steel, Sheet and Strip, Low-Carbon.
QQ-S-741	-	Steel, Carbon: Structural Shapes, Plates, and Bars.
QQ-S-781	-	Steel Strapping, Flat.
QQ-T-830	-	Tube, Steel, (Carbon, Mechanical; Seamless and Welded).
QQ-W-461	-	Wire, Steel, Carbon, (Round, Bare and Coated).
QQ-Z-325	-	Zinc Coating, Electrodeposited, Requirements for.
RR-W-360	-	Wire Fabric, Industrial.
TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-E-485	-	Enamel, Semigloss, Rust-Inhibiting.
TT-E-529	-	Enamel, Alkyd, Semi-Gloss.
TT-P-636	-	Primer Coating, Alkyd, Wood and Ferrous Metal.
UU-P-268	-	Paper, Kraft, Untreated, Wrapping.
UU-T-81	-	Tags, Shipping and Stock.
UU-T-106	-	Tape, Pressure-Sensitive Adhesive, Masking, Paper.
VV-L-800	-	Lubricating Oil, General Purpose, Preservative (Water-Displacing, Low Temperature).
ZZ-T-416	-	Tire Rebuilding and Tire and Tube Repair Materials.
NNN-P-40	-	Paper, Lens Tissue.
PPP-B-636	-	Box, Fiberboard.
PPP-C-843	-	Cushioning Material, Cellulosic.
PPP-T-60	-	Tape: Pressure-Sensitive Adhesive, Waterproof for Packaging.
PPP-T-97	-	Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

Military

MIL-P-116	-	Preservation, Methods of.
MIL-B-117	-	Bags, Sleeves and Tubing, Interior Packaging.
MIL-A-148	-	Aluminum Foil.

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MIL-S-207 - Sulfuric Acid, Electrolyte: Packaging, Packing, and Marking for Shipment and Storage of.
MIL-C-450 - Coating Compound, Bituminous Solvent Type, Black (for Ammunition).
MIL-D-1000 - Drawings, Engineering and Associated Lists.
MIL-R-3065 - Rubber and Synthetic Rubber Compounds, General Purpose (Except Tires, Inner Tubes, Sponge Rubber and Hard Rubber).
MIL-P-3420 - Packaging, Materials, Volatile Corrosion Inhibitor Treated, Opaque.
MIL-M-3912 - Wood Parts, Fabricated: for Transport Vehicle Bodies and Similar Applications.
MIL-I-8574 - Inhibitors, Corrosion, Volatile, Utilization of.
MIL-G-10924 - Grease, Automotive and Artillery.
MIL-A-11755 - Antifreeze, Arctic Type.
MIL-S-13518 - Sealer, Surface, Wood Preservative.
MIL-P-13983 - Paint, Temporary, Lusterless, Gasoline Removable.
MIL-P-13988 - Paper, Lens, Tissue, Antitarnish, Wrapping.
MIL-C-16173 - Corrosion Preventive Compound, Solvent Cutback, Cold-Application.
MIL-D-16791 - Detergents, General Purpose (Liquid, Nonionic).
MIL-Z-17871 - Zinc, Coating (Hot-Dip Galvanizing).
MIL-C-20696 - Cloth, Coated, Nylon, Waterproof.
MIL-L-21260 - Lubricating Oil, Internal-Combustion Engine, Preservative.
MIL-I-22110 - Inhibitors, Corrosion, Volatile, Crystalline.
MIL-F-22191 - Film, Transparent, Flexible Heat Sealable for Packaging Applications.
MIL-L-46002 - Lubricating Oil, Contact and Volatile Corrosion Inhibited.

STANDARDS

Federal

FED. STD. NO. 595 - Colors.
FED. STD. NO. 751 - Stitches, Seams, and Stitchings.

Military

MIL-STD-8 - Dimensioning and Tolerancing.
MIL-STD-9 - Screw Thread Conventions and Methods of Specifying.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1261 - Welding Procedures for Constructional Steels.
MS 20230 - Grommets, Metallic, Plain and Spur, with Washers, Type I and Type III.

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MS 20995 - Wire, Lock.
MS 24665 - Pin, Cotter.
MS 27183 - Washer, Flat-Round, Steel, Cadmium Plated, General Purpose.
MS 35338 - Washer, Lock-Spring, Helical, Regular (Medium) Series.
MS 35751 - Bolt, Square Neck, Round Head, (Carriage).
MS 35752 - Bolt, Square Neck, Truss Head, (Step).
MS 51108 - Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Shank Drilled for Cotter Pin, Alloy Steel, Grade 8 Phosphate Coated, UNF-2A.
MS 51967 - Nut, Plain, Hexagon-Carbon Steel, Cadmium Plated UNC-2B.
MS 90727 - Screw, Cap Hexagon Head-(Finished Hexagon Bolt), Alloy Steel, Cadmium Plated, UNF-2A.
MS 90728 - Screw, Cap, Hexagon Head-(Finished Hexagon Bolt), Alloy Steel, Cadmium Plated, UNC-2A.

Air Force - Navy Aeronautical

AN 970 - Washer, Flat, for Wood.

DRAWINGS

Army

538456 - Fastener, Slide, Interlocking, Non-Separating, Medium Heavy.
8710565 - Rope.
10941915 - Washer, Flat, Metal, Round, Hardened.
10954699 - Positioning and Tie Down, Vehicle.
11604683 - Closure Kit, Vehicle Protective.

PUBLICATIONS

Packaging Data Sheet

8736408 - OVE, Packaging of.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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ASSOCIATION OF AMERICAN RAILROADS PUBLICATION

Section No. 1 - General Rules Governing Loading of Commodities On Open Top Cars.
Section No. 6 - Rules Governing the Loading of Department of Defense Materiel on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van Buren, Chicago, Illinois 60605.)

3. REQUIREMENTS

3.1 Level A.

3.1.1 First production processed vehicle. Unless otherwise specified (see 6.1), the first production processed vehicle shall be subjected to the inspection specified in 4.2. Approval of the first production processed vehicle shall not relieve the contractor of his obligation to process all vehicles in accordance with this specification. Except as otherwise specified by the procuring activity, any changes to materials or design after approval of the first production processed vehicle, shall require submission of an additional production processed vehicle to the inspection specified in 4.2.

3.1.2 One of the next 10 production processed vehicles. After approval of the first production processed vehicle (see 3.1.1), one of the next 10 production processed vehicles shall be selected by the Government representative for inspection specified in 4.2.

3.1.3 Materials.

3.1.3.1 Stock numbers. Materials to be obtained through Government supply channels shall be ordered by stock numbers or specifications.

3.1.4 Preparation prior to processing. Except as otherwise specified herein, and to the maximum extent consistent with production efficiency, economy, and safe storage and shipment, vehicle shall be prepared for storage and shipment in a completely assembled condition after test runs, and completion and approval of all necessary repairs. Specified equipment shall be installed and all adjustments made so that the vehicle may be operated, shipped, and placed into service with a minimum of delay.

3.1.4.1 Processing records. Records of vehicle processing shall be maintained and readily available for review by the Government representative.

3.1.4.2 Disassembly. Parts vulnerable to damage and pilferage, and projecting parts whose removal will accomplish the desired reduction in cube shall be removed from the vehicle. Removed parts shall be preserved, packaged, and packed.

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The packed parts shall be identified and placed in a protected location in the vehicle and secured in a manner to prevent movement and damage during shipment and storage. Removed bolts, nuts, screws, pins, and washers shall be placed in one of the mating parts and secured.

3.1.4.2 Matchmarking. Parts removed from the vehicle shall be matchmarked when necessary to facilitate reassembly. Matchmarking information shall be on cloth shipping tags conforming to type of UU-T-81, or metal tags marked with soluble paint and attached to mating parts. The marked cloth shipping tags shall be waterproofed in accordance with MIL-T-[illegible]-12[illegible].

3.1.5 Cleaning and drying.

3.1.5.1 Interior of vehicle. Vehicle interior surfaces shall be cleaned in accordance with process C-1 of MIL-P-116, except that cleaning compounds shall be in accordance with detergents conforming to P-D-220 or MIL-D-16791, in warm water solution. Liquids under pressure shall be used only in the engine compartment. Drying shall be in accordance with any applicable procedure of MIL-P-116.

3.1.5.1.1 Battery supports and retainers. Battery supports and retainers shall be cleaned with a solution composed of 1/2 pound of sodium bicarbonate, conforming to O-S-576, per gallon of water, and flushed with clear water and dried. Battery supports and retainers shall be coated with compound conforming to MIL-C-450.

3.1.5.1.2 Backrests and seats. Installed backrests and cushioned components of seats shall be washed with a solution of soap and water, rinsed with clear water, dried, and immediately packaged (see 3.1.15).

3.1.5.1.3 Fire control items. Exposed optical glass components of fire control items shall be cleaned by blowing on exposed optical glass surfaces with air from a hand syringe, or by use of a clean camel-hair brush, followed by the use of ethyl alcohol conforming to O-E-760. In cases of contamination not removable by alcohol, cleaning shall be accomplished by use of a solution consisting of two ounces of detergent conforming to MIL-D-16791, 1/2 gallon of alcohol conforming to O-E-760, and one gallon of distilled water. Using a swab made of lens tissue conforming to type I or II of NNN-P-40, optical surfaces shall be washed with the cleaning agent; and washing shall be repeated, using a clean swab each time, until no dirt or other foreign matter remains on the surface. Cleaning shall be accomplished with a minimum of pressure and rubbing, and without the use of cloth or rubber materials, in order to prevent damage to lens coatings. Immediately after cleaning, the optics shall be covered as specified in 3.1.13.1.

3.1.5.1.4 Fuel filters. Except when fuel filter elements are new and clean, and prior to engine preservation (see 3.1.9), fuel filter shells shall be drained, removed and cleaned in accordance with process C-1 of MIL-P-116 and dried by any applicable procedure specified therein. New filter elements shall be installed if necessary. Fuel filter shells shall then be reinstalled using new gaskets.

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3.1.5.2 Exterior of vehicle. Using a mild detergent, vehicle exterior surfaces including the engine compartment shall be cleaned in a manner which will assure removal of all foreign matter, including excess cleaning material. Cleaning shall be followed by a complete rinsing with clear water, then thorough drying. Vehicle shall be cleaned without directing steam or water jet at or above the ring opening between turret and upper hull or against armament. Caution shall be exercised to prevent the entry of water into vehicle.

3.1.5.2.1 Gun-launcher. When inspection indicates the need for reprocessing of gun-launcher (see 4.5.2.3), gun-launcher shall be cleaned in accordance with process C-3 and thoroughly dried in accordance with procedure D-4 of MIL-P-116.

3.1.6 Relubrication. When vehicle has been operated in excess of 50 miles since previous lubrication, or after a detergent or steam cleaning, vehicle shall be lubricated using materials in accordance with drawings, specifications or lubrication order pertinent to the vehicle. All exposed oil can points such as hinges, latches, clevises and pins, and control cable pulley pins shall be lubricated with oil conforming to VV-L-800. Excess lubricating material shall be removed after lubrication.

3.1.7 Transmission. Transmission shall contain preservative oil only. Preservative oil in transmission shall conform to type I, grade 10 of MIL-I-21260, filled to operating level. DD Form 1397 (see 3.1.19), shall be completed to indicate grade of preservative oil used.

3.1.8 Engine crankcase and generator drive gear box. Engine crankcase and generator drive gear box shall be filled with lubricating oil conforming to MIL-L-21260, type II, grade 10 or 30, as specified by the applicable drawing, specification or lubrication order. DD Form 1397 shall be completed to indicate type and grade of preservative oil used.

3.1.9 Engine preservation. Engine preservation as specified in 3.1.9.1 through 3.1.9.4 shall be accomplished without interruption, except for cooling as specified in 3.1.9.2.

3.1.9.1 Preservation through fuel system. Engine fuel supply line shall be disconnected at the most convenient point nearest the fuel tank. A line from a portable container containing preservative oil conforming to grade 1 of MIL-L-46002, shall be connected to the disconnected fuel supply line leading to the engine. Engine fuel return line shall be disconnected at the most convenient point and a transparent plastic line connected to the engine fuel return line. The other end of the transparent line shall be inserted into a recovery container to collect the returned fuel. The engine shall be started and operated at approximately 1200 revolutions per minute (rpm) until undiluted preservative oil is flowing into the recovery container. Engine shall then be stopped. Manifold flame heater switch shall be operated concurrently with fuel system preservation. Auxiliary fuel return line shall be disconnected and original fuel return line reconnected. Recovered fuel and preservative oil mixture shall be discarded.

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3.1.9.2 Preservation through combustion chamber. Prior to continuation of engine processing, the engine shall be allowed to cool for 24 hours or until engine temperature is 100°F or less, measured at injector nozzle flange surface of all cylinders. Induced air currents or flushing cold water through cooling system may be used to accelerate cooling of engine. Rocker arm covers or side access plates shall be removed and the exhaust valves of each cylinder shall be mechanically depressed and held in open position by using 0.025 inch spacers or any method which will hold the valves open without damage to the engine.

CAUTION: Special precautions shall be taken to assure that clearance between piston and valve is not exceeded. Prior to actual processing of first engine, with valves depressed, engine shall be manually rotated, if possible, or rotated by starter if manual turning is not possible. Examination shall be made to assure there is no contact of piston and depressed valve.

The throttle shall be placed at maximum open position and engine cranked with starting motor for 30 seconds.

CAUTION: Special precautions shall be taken to assure that the amount of oil injected into the combustion chambers will not result in a hydrostatic lock. Prior to processing additional engines, the first engine shall be processed as specified above and allowed to stand for 12 hours. Engine shall then be manually rotated or rotated by the starter if manual turning is not possible, to assure that amount of oil injected permits free rotation of the engine.

Portable container shall be removed and vehicle line reconnected. Valve depressing devices shall be removed. Interior surfaces of rocker arm covers, side access plates, valves, valve springs, and rocker arms shall be atomized sprayed with preservative oil conforming to grade 1 of MIL-L-46002. Damaged gaskets shall be replaced and rocker arm covers or side access plates reinstalled.

3.1.9.3 Preservation through air intake and exhaust system. After preservation through fuel system (3.1.9.1) and combustion chamber (3.1.9.2), 15 grams of volatile corrosion inhibitor compound (VCI), conforming to type I of MIL-I-22110, shall be fogged into the exhaust opening. Opening shall then be sealed with tape conforming to class 1 of PPP-T-60. Hose shall be disconnected at the air intake and 15 grams of VCI shall be fogged into the air intake opening. The air intake opening and engine crankcase breathers shall then be sealed with a plastic plug or with tape conforming to class 1 of PPP-T-60. A red warning tag bearing the information: "ENGINE PRESERVED WITH VCI - DO NOT CRANK" and "BEFORE CRANKING, REMOVE CAPS, TAPE OR PLUGS AND REINSTALL HOSE" shall be placed in a conspicuous location within the driver's compartment. DD Form 1397 shall be marked to indicate engine is preserved with VCI compound and oil.

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3.1.9.4 Preservation through dipstick shroud opening. After preservation specified in 3.1.9.1, 3.1.9.2 and 3.1.9.3, dipstick shall be removed and six ounces of lubricating oil conforming to grade 1 of MIL-L-46002 shall be atomized sprayed into the crankcase through the dipstick shroud opening. An extension nozzle of sufficient length to allow the spraying nozzle to be within the crankcase shall be used. Nozzle shall not be submerged in the oil within the crankcase. Dipstick shall be reinstalled. All openings leading to interior of the engine, including dipstick shroud opening and oil filler cap shall be sealed with tape conforming to class 1 of PPP-T-60.

3.1.10 Fuel tank. Fuel shall be removed from the fuel tank to the maximum extent possible. When fuel tank cap and filler screen cannot be adequately preserved in place, fuel tank cap and filler screen shall be removed and coated with preservative oil conforming to VV-L-800. One quart of preservative oil conforming to VV-L-800, shall be poured into the fuel tank and an additional quart for each five gallons of residual fuel remaining in the fuel tank. If removed, fuel tank cap and filler screen shall be reinstalled.

3.1.11 Cooling system. Prior to, or concurrently with, the other preservation requiring engine operation, cooling system shall be protected by one of the following procedures, depending on the condition of shipment and storage, as specified (see 6.1). If no procedure is specified, cooling system shall be protected in accordance with 3.1.11.1. DD Form 1397 shall be completed to indicate coolant used.

3.1.11.1 Water and antifreeze procedure. Cooling system shall contain a clean solution consisting of equal parts by volume of antifreeze (ethylene glycol) conforming to type I of O-A-548, and water, filled to capacity. Engine shall be operated until temperature has been reached that causes thermostat to open to assure complete mixing and even distribution of the antifreeze solution. A warning tag, bearing the information "COOLING SYSTEM FILLED WITH WATER AND ANTIFREEZE SOLUTION (ETHYLENE GLYCOL) IN EQUAL PARTS BY VOLUME - DO NOT DRAIN", shall be securely attached to radiator filler neck. When tested as specified in 4.5.2.4, coolant shall show protection to minus 40°F.

3.1.11.2 Antifreeze compound procedure. For shipment and storage in areas where the temperature drops below minus 40°F, cooling system shall be filled with antifreeze compound conforming to MIL-C-11755. The compound shall be used without dilution. A warning tag, bearing the information "COOLING SYSTEM FILLED WITH ANTIFREEZE (ARCTIC-TYPE) - DO NOT DRAIN", shall be securely attached to radiator filler neck.

3.1.11.3 Water and corrosion inhibitor procedure. For shipment and storage within the bounds of 30 degrees north latitude and 30 degrees south latitude, except Continental United States, cooling system shall be preserved as follows:

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Cooling system shall be filled with clear water up to, but not including, the radiator upper tank. A corrosion inhibitor conforming to O-I-490 shall be added in the proportion of five ounces of the inhibitor for each 10 quarts of water. The inhibitor shall be dissolved in two quarts of warm water and poured into the radiator while the engine is idling. More water shall be added, if necessary, to fill the radiator to operating level. A warning tag, bearing the information "COOLING SYSTEM DOES NOT CONTAIN ANTIFREEZE - FILLED WITH WATER AND INHIBITOR", shall be securely attached to radiator filler neck.

3.1.12 Batteries, cables, and electrolyte.

3.1.12.1 Dry charge batteries and cables. Dry charged batteries shall be installed and secured in the vehicle battery carrier. Battery cables shall be secured to battery carrier with 3/4 inch tape conforming to type IV of PPP-T-97. Filler cap openings shall be sealed by placing a 2 inch wide by 3 mil thick sheet of film, conforming to type I of MIL-F-22191, over all filler cap openings with cap removed. The sheet shall be of sufficient length to allow sheet to be depressed into the filler cap opening to the same depth as the filler plug. Filler caps shall be screwed or inserted into filler openings to form a complete seal without damaging the sheet.

3.1.12.2 Electrolyte. Electrolyte shall be packaged, packed and marked in accordance with the requirements prescribed for type IV, class 1 or 2 unit, as applicable, of MIL-S-207.

3.1.12.2.1 Stowage. Electrolyte shall be stowed with OVE, secured independently in a manner to prevent movement and damage in transit and permitting easy removal at ports when special stowing is required by maritime regulations.

3.1.13 Fire control items.

3.1.13.1 Quadrant, periscopes, telescopes and vision blocks. Quadrant, periscopes, telescopes and vision blocks shall remain installed. Driver's periscope quick release latches shall be safety-wired to prevent accidental release. Immediately after cleaning (see 3.1.5.1.3), optics shall be covered with a four layer thickness of lens tissue conforming to MIL-P-13988. Lens tissue shall be secured with tape conforming to UU-T-106. All exposed, unpainted metal surfaces shall be coated with grease conforming to MIL-G-10924.

3.1.14 Gun-launcher and related items.

3.1.14.1 Gun-launcher. Immediately after cleaning (see 3.1.5.2.1), bore and chamber of the gun-launcher shall be coated with preservative oil conforming to VV-L-800. Excess preservative shall be allowed to drain from coated surfaces. A strip of VCI treated carrier material conforming to type I, class 3, style A of MIL-P-3420 shall be cut and rolled into a tube with the VCI treated surface on the outside. The carrier material shall be of a size that will provide a continuous cover for the bore and chamber surfaces. The rolled carrier material

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shall be inserted into gun-launcher extended entire length of bore and chamber. The carrier tube shall not be forced or kinked in a manner which would obstruct the chamber. The gun-launcher shall remain in battery with turret travel lock secured in the lock position.

3.1.14.1.1 Muzzle plug. A plug shall be provided for the muzzle end of the gun-launcher. Muzzle plug shall be completely overwrapped with aluminum foil conforming to MIL-A-148, positioned in muzzle end of gun-launcher and secured in place with tape conforming to class 1 of PPP-T-60. The joint around muzzle plug and gun-launcher shall be completely sealed with tape conforming to class 1 of PPP-T-60. A bag fabricated of material conforming to grade A, finish 1 of L-P-378 shall be provided. Bag shall be 16 inches long, of applicable width and 0.006 inch thick. Bag shall be installed over muzzle end of gun-launcher and secured in place with four strips of tape conforming to class 1 of PPP-T-60. Tape shall run lengthwise on the gun-launcher placed at top, bottom, and each side of gun-launcher. Tape shall be 12 inches long and a minimum one inch in width, applied equally six inches onto bag and six inches onto painted surfaces of the gun-launcher. Bag shall be sealed to cannon with tape conforming to class 1 of PPP-T-60. Tape shall be of applicable length, to provide a continuous seal around circumference of gun-launcher, and six inches in width. Tape shall be applied equally three inches onto bag and three inches onto painted surfaces of the gun-launcher. Two additional one inch strips of tape conforming to class 1 of PPP-T-60 shall be applied at equal intervals between muzzle end of gun-launcher and area where bag is sealed to gun-launcher. Tape shall be applied completely around circumference of bag to provide additional securement of bag to gun-launcher.

3.1.14.1.2 Breech mechanism and breech coupling. All unpainted surfaces, including phosphated surfaces of the breech coupling, breech chamber, breech operating and firing mechanisms, shall be coated with grease conforming to MIL-G-10924.

3.1.14.2 Exercising of recoil mechanism. Prior to processing as specified in 3.1.14.2.2 and when the recoil mechanism has not been exercised, proof fired, overhauled, or manufactured within four months prior to preparation for storage or shipment, the recoil mechanism shall be exercised a minimum of three extensions of the recoil piston. Extension shall be a minimum of six inches. Record of exercising shall be entered on DA Form 2408-4 "Weapon Record Data" and proof testing of the weapon shall be entered on DA Form 2408-9 "Proof Acceptance Record" (see 3.1.19).

3.1.14.2.1 Processing of recoil mechanism after exercising. Dust shield shall be removed and accessible surface of chrome plated recoil sleeve coated with grease conforming to MIL-G-10924. The surface of the recoil mechanism immediately forward of breech coupling shall be coated with the same type grease. Application of grease shall be made while gun-launcher is out of battery during exercising and upon the last extension prior to return to battery.

3.1.14.2.2 Counter recoil buffer. Without disassembly, exposed unpainted machined surfaces of the counter recoil buffer shall be cleaned, then coated with grease conforming to MIL-G-10924.

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3.1.14.3 Exercising of replenisher. Replenisher shall be exercised coincidently with the recoil mechanism (see 3.1.14.2).

3.1.14.3.1 Processing of replenisher after exercising. Gun-launcher shall be elevated to 15 degrees (265 mils). Reservoir oil level shall be checked and filled, to one quart mark, if required, with the specified oil.

3.1.14.4 Elevating and traversing mechanism. Exposed unpainted surfaces of elevating mechanism trunnions, and turret traversing ring gear and elevating hand crank shaft shall be coated with grease conforming to MIL-G-10924. Handle lock pin and turret traverse lock shall be secured in locked position.

3.1.14.5 Turret ring bearing. Plugs shall be removed from grease fitting openings in turret ring bearing and grease fittings installed. Grease conforming to MIL-G-10924 shall be pumped through grease fittings until grease is visible between the two sections around the circumference of the ring. The turret shall be rotated five times in both directions and again lubricated until all surfaces on the circumference of the ring are lubricated. Grease fittings shall be removed and plugs reinstalled. Excess grease shall be removed from the outside surface of the rail. DD Form 1397 shall be marked with the information "TURRET BEARING COMPLETELY FILLED WITH GREASE - MIL-G-10924."

3.1.15 Backrests and seats. Immediately after drying (see 3.1.5.1.2), backrests and cushioned components of seats shall be covered with paper, having a minimum basis weight of 60 pounds, conforming to grade B of UU-P-268. Paper shall be secured in place with tape conforming to type I of UU-T-106.

3.1.16 Hatches and doors. Rubber seals around hatches and doors shall be coated with powdered talc conforming to ZZ-T-416. Hatches and doors shall be closed and locked from the inside except that for shipment, loader's hatch shall be secured using an approved railroad type seal placed through locking lugs. Driver's hatch lever shall be taped in position with tape conforming to class 1 of PPP-T-60.

3.1.17 Miscellaneous preservation. Except as otherwise specified herein, all exposed, unpainted machined metal surfaces on the interior of the vehicle shall be coated with preservative conforming to grade 4 of MIL-C-16173. All exposed, unpainted machined metal surfaces on the exterior of the vehicle, except tracks, shall be coated with preservative conforming to grade 4 of MIL-C-16173.

3.1.18 On vehicle equipment. Unless otherwise specified (see 6.1), all spare parts, tools and equipment shall be preserved, packaged and packed in accordance with Packaging Data Sheet 8736408. Pack shall be identified to pertinent vehicle by serial number and, except during shipment, shall be stored inside buildings.

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3.1.19 Record forms. A copy of "The Equipment Log Book" consisting of Binder (FSN 7510-889-3494) with required forms, and two copies of "Processing and Deprocessing Record for Shipment, Storage and Issue of Vehicles and Spare Engines", DD Form 1397, and single copies of DA Form 2408-4 Weapon Record Data and DA Form 2408-9 Proof Acceptance Record (see 6.3) shall be furnished and completed with each vehicle. Instructions for preparation and attachment or location of "The Equipment Log Book" shall be supplied by the Government representative. One copy of DD Form 1397 shall be placed in a bag conforming to type II, class B of MIL-B-117. Bag shall be closed by stapling, identified and secured to headlamp. The duplicate copy shall be placed in the bag with the log book.

3.1.20 Vehicle closure. Vehicle shall be provided with a vehicle protective closure kit in accordance with Drawing 11604883. The closure cover shall be fabricated, assembled and installed in accordance with figures 1 through 64 of Closure Frame and Cover Assemblies. All sharp corners of framework and bows where cloth cover will make contact, such as corners on base of the frame, shall be cushioned with a 3/4 inch minimum thickness of cushioning material conforming to PPP-C-843, secured in place with tape conforming to class 1 of PPP-T-60. To lift vehicle for loading, cover shall be rolled away from front and rear bows to expose vehicle lifting eyes.

3.1.20.1 Ventilation. All access plates and gaskets on underside of vehicle shall be removed for ventilation. Unpainted metal surfaces shall be coated with preservative conforming to grade 1 of MIL-C-16173. Items shall be placed in a box conforming to PPP-B-636, identified and stowed within vehicle. Threaded portions exposed by removal of these items shall be coated with preservative conforming to grade 1 of MIL-C-16173. The following information shall be stenciled in a conspicuous location on the exterior of vehicle: "REMOVE SCREENS, INSTALL ACCESS PLATES, COVERS AND GASKETS, BEFORE VEHICLE OPERATION". Stenciling color shall be white or yellow paint conforming to MIL-P-13903. Characters shall be a minimum of 3/4 inch in height.

3.1.20.2 Screens. Screens constructed of wire cloth, 0.047 inch diameter, 4 by 4 mesh, conforming to type I, class 1 of RR-W-360 shall be installed in access cover openings. Screens shall be installed in accordance with figures 1 through 10 of Ventilation Kit.

3.1.21 Fire extinguishers. Fire extinguishers shall have a minimum of 90 percent of the rated full charge. All seals shall be intact. DA Form 253 shall be completed and securely attached to each extinguisher (see 6.2).

3.1.22 OVE saddle. Vehicle shall be provided with an OVE saddle constructed, assembled and installed in accordance with figures 1 through 23 of OVE Saddle Assembly.

3.1.23 Tow shackles. Tow shackles and related hardware shall be removed for shipment, placed in a box conforming to V3c of PPP-B-636, sealed, identified, then stowed inside vehicle.

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3.2 Level B and C. Vehicles shall be processed the same as specified for level A with the following exceptions:

3.2.1 Transmission. Transmission shall contain lubricant in accordance with applicable drawings, specifications, or lubrication order, filled to operating level; except when the unit contains preservative oil conforming to type I, grade 10, 30 or 50 as applicable, of MIL-L-21260, operating level shall be attained by the addition of the applicable grade of preservative oil. Operating lubricant and preservative oil shall not be mixed. DD Form 1397 shall be completed to indicate grade of lubricant or preservative oil used.

3.2.2 Engine crankcase and generator drive gear box. Engine crankcase and generator drive gear box shall contain lubricant in accordance with applicable drawings, specification, lubrication order, or preservative oil conforming to type I, grade 10, 30 or 50 as applicable of MIL-L-21260, filled to operating level. Operating lubricant and preservative oil shall not be mixed. DD Form 1397 shall be completed to indicate grade of lubricant or preservative oil used.

3.2.2.1 Engine preservation. Engine preservation shall be in accordance with 3.1.9.3 and 3.1.9.4 only.

3.2.3 Fuel tank. Unless otherwise specified (see 6.1), vehicles shall be shipped without draining residual fuel from the fuel tank.

3.2.4 Hatches and doors. Hatches and doors shall be closed and locked from the inside except that for shipment, loader's hatch shall be secured using an approved railroad type seal placed through locking lugs.

3.3 All levels.

3.3.1 Loading. Vehicles shall be blocked, braced and anchored on open top railroad cars in accordance with the Association of American Railroads publications (see 2.2) and Drawing 10954099. Parking brake shall be set and secured in that position.

3.3.2 Reprocessing engine after loading. If engine is operated in connection with loading, or moving vehicle to loading area, the engine shall be reprocessed as specified in 3.1.9 or 3.2.2.1 as applicable. An auxiliary fuel tank shall supply the fuel. If installed, vehicle cover shall be rolled away from the first two front and first two rear bow sections. After reprocessing of engine, vehicle cover shall be restored to its original position.

3.3.3 Marking. In addition to any special marking required in the contract or order (see 6.1), vehicles shall be marked in accordance with MIL-STD-129.

3.3.3.1 Lifting points. Information, "LIFT HERE" shall be stenciled adjacent to each lifting eye, using white enamel conforming to TT-E-529. Stenciling shall be in characters a minimum of 3/4 inch high.

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3.3.3.2 Closure marking. The information "TO LIFT VEHICLE FOR LOADING, UNFASTEN AND ROLL COVER FROM FRONT AND REAR BOWS EXPOSING LIFTING EYES", shall be stenciled in a conspicuous location on the front and rear of vehicle closure, using white enamel conforming to TT-E-529. Stenciling shall be in characters a minimum of 3/4 inch high. The information "CLOSURE KIT, VEHICLE PROTECTIVE - M551 - DO NOT DESTROY - REINSTALL FOR SHIPMENT OR STORAGE - OR SHIP WITH OVE", shall be stenciled in a conspicuous location on one end and one side of vehicle closure, using white enamel conforming to TT-E-529. Stenciling shall be in characters a minimum of two inches high.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First production processed and one of the next 10 production processed vehicles. The first production processed vehicle (see 3.1.1) and one of the next 10 production processed vehicles (see 3.1.2) shall be subjected to inspection specified in 4.5 through 4.5.2.5.

4.3 Production processed vehicles. Except as otherwise specified, all production processed vehicles shall be subjected to inspection specified in 4.5 through 4.5.2.3.

4.4 Rejection. Failure of the first production, one of the next 10, or any production processed vehicle to conform to the requirements of this specification shall be cause for rejection of the vehicle, and refusal to continue acceptance of vehicles, by the Government, until objective evidence has been provided, by the contractor, that corrective action has been taken to eliminate the conditions which caused rejection.

4.5 Quality conformance inspection.

4.5.1 Materials. Except for materials which have been Government inspected at the source, all materials to be used in processing of vehicles shall be inspected in accordance with the material specification; or certified inspection and laboratory test reports shall be furnished which show that materials, as furnished, conform to the detailed specification.

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4.5.2 Processing. All vehicle processing shall be inspected to determine conformance to this specification. Inspection of processing shall include all items specified in table I, and 4.5.2.1 through 4.5.2.3.

Table I. Processing inspection
(See indicated paragraphs for level A, level B, and level C requirements)

	Cleaning	Preservation		Packaging
	Levels A,B and C	Level A	Levels B and C	Levels A,B and C
Preparation prior to processing				3.1.4
Processing records				3.1.4.1
Disassembly				3.1.4.2
Matchmarking				3.1.4.2.1
Interior of vehicle	3.1.5.1			
Battery supports and retainers	3.1.5.1.1	3.1.5.1.1	3.1.5.1.1	
Backrests and seats	3.1.5.1.2			3.1.15
Fire control items	3.1.5.1.3	3.1.13.1	3.1.13.1	3.1.13.1
Fuel filters	3.1.5.1.4			
Exterior of vehicle	3.1.5.2			
Gun-launcher	3.1.5.2.1	3.1.14.1	3.1.14.1	
Relubrication		3.1.6	3.1.6	
*Transmission		3.1.7	3.2.1	
*Engine crankcase and generator drive gear box		3.1.8	3.2.2	
*Engine preservation		3.1.9	3.2.2.1	
Preservation through fuel system		3.1.9.1	3.1.9.1	
Preservation through combustion chamber		3.1.9.2	3.1.9.2	
*Preservation through air intake and exhaust system		3.1.9.3	3.2.2.1	
Preservation through dipstick shroud opening		3.1.9.4	3.2.2.1	
Fuel tank		3.1.10	3.2.3	
*Cooling system		3.1.11	3.1.11	
Water and antifreeze procedure		3.1.11.1	3.1.11.1	
Antifreeze compound procedure		3.1.11.2	3.1.11.2	
Water and corrosion inhibitor procedure		3.1.11.3	3.1.11.3	
Dry charge batteries and cables				3.1.12.1
Electrolyte				3.1.12.2
Gun-launcher		3.1.14.1	3.1.14.1	
Muzzle plug		3.1.14.1.1	3.1.14.1.1	
Breech mechanism and breech coupling		3.1.14.1.2	3.1.14.1.2	

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Table I. Processing inspection (cont'd)
(See indicated paragraphs for level A, level B, and level C requirements)

	Cleaning	Preservation		Packaging
	Levels	Level	Levels	Levels
	A,B and C	A	B and C	A,B and C
**Exercising of recoil mechanism		3.1.14.2	3.1.14.2	
Processing of recoil mechanism after exercising		3.1.14.2.1	3.1.14.2.1	
Counter recoil buffer		3.1.14.2.2	3.1.14.2.2	
Processing of replenisher after exercising		3.1.14.3.1	3.1.14.3.1	
Elevating and traversing mechanism		3.1.14.4	3.1.14.4	
Turret ring bearing		3.1.14.5	3.1.14.5	
Hatches and doors		3.1.16	3.2.4	
Miscellaneous preservation		3.1.17	3.1.17	
On vehicle equipment		3.1.18	3.1.18	3.1.18
Record forms				3.1.19
Vehicle closure				3.1.20
Ventilation		3.1.20.1	3.1.20.1	3.1.20.1
Screens				3.1.20.2
Fire extinguishers				3.1.21
OVE saddle				3.1.22
Tow shackles				3.1.23
Loading				3.3.1
Reprocessing engine after loading		3.3.2	3.3.2	
Marking				3.3.3
Lifting points				3.3.3.1
Closure marking				3.3.3.2

*NOTE: Inspect DD Form 1397
**NOTE: Inspect DA Form 2408-4 and DA Form 2408-9

4.5.2.1 Cleaning. To determine conformance to 3.1.5.1, interior of vehicles shall be examined for cleanliness, and one vehicle each day shall be tested for cleanliness in accordance with the applicable provisions of MIL-P-116. To determine conformance to 3.1.5.2, exterior of vehicle shall be examined for cleanliness. Surfaces to which tape is to be applied shall be examined for cleanliness before applying tape.

4.5.2.2 Fuel tank. To determine conformance to 3.1.10, interior of fuel tank shall be examined and visual inspection shall assure that complete processing, as specified, has been accomplished.

4.5.2.3 Gun-launcher. Gun-launcher shall be examined to determine condition and effectiveness of processing. When reprocessing has been accomplished, it shall be examined for conformance to 3.1.5.2.1 and 3.1.14.1.

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6.3 Forms. A copy of "The Equipment Log Book" and all required forms will be furnished the contractor by the Government at least 30 days before shipment of the equipment as required by the contract delivery schedule (see 3.1.19).

Custodian:
Army - AT

Preparing activity:
Army - AT

Project No. 2350-A106

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FIG. NO.	NOMENCLATURE	QTY REQD
2	MATERIAL LIST	—
3	CLOSURE FRAME ASSEMBLY	—
4	FRAME ASSEMBLY-PLAN VIEW	—
5	FRAME ASSEMBLY-LEFT ELEVATION	—
6	FRAME ASSEMBLY-FRONT ELEVATION	—
7	FRAME ASSEMBLY-REAR ELEVATION	—
8	CORNER PLATE (WELDMENT)	4
8A	PLATE	4
8B	TUBE	4
9	BOW SECTION	2
10	BOW SECTION	2
11	BOW SECTION	2
12	BOW SECTION	5
13	BOW SECTION	5
14	BOW SECTION	2
15	BOW SECTION	1
16	BEAM, FRAME	2
17	BEAM, FRAME	2
18	BEAM, FRAME	2
19	BEAM, FRAME	1
20	BEAM, FRAME	1
21	KNEE BRACE	4
22	"U" CLAMP	7
23	PLATE	7
24	SUPPORT, FRAME	2
25	CONNECTOR, BEAM	4
26	FASTENER (SEE DRAWING 10922144)	36
27	BRACE, FRAME SUPPORT	4
28	BRACKET	4
28A	PAD	4
29	HARDWARE, CLOSURE FRAME ASSEMBLY SEE HARDWARE LIST	—

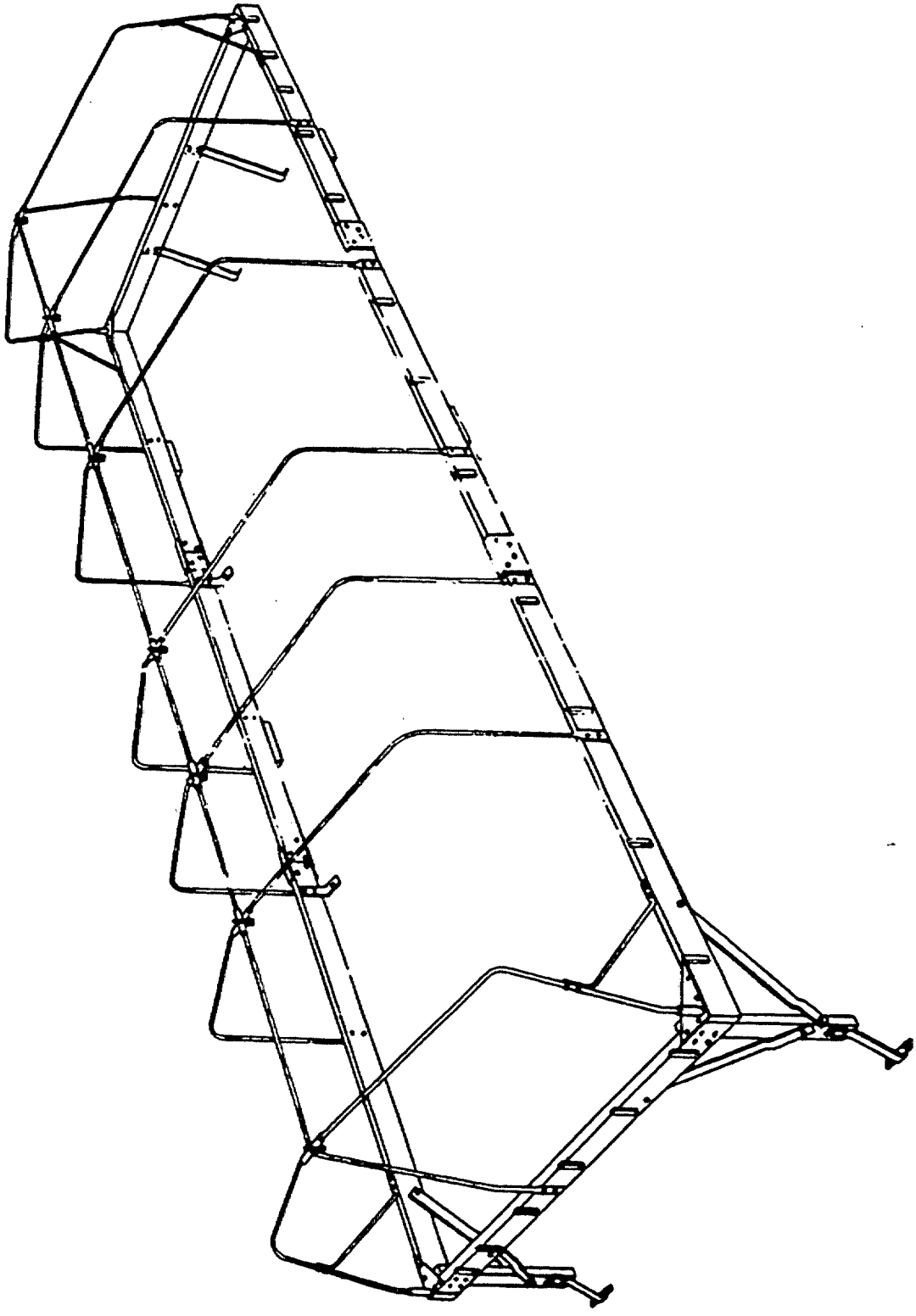
PARTS LIST-CLOSURE FRAME ASSEMBLY
FIGURE 1 PAGE 20

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DESCRIPTION	QTY REQD
1 OD X .065 WALL THICKNESS, TUBE, STEEL	130 FEET LONG
2 X 4 WOOD (COMMERCIAL SIZE)	61 FEET LONG
1 1/4 OD X .109 WALL THICKNESS, TUBE, STEEL	15 INCHES LONG
.187 STEEL PLATE, 8 INCHES WIDE	42 INCHES LONG
4 X 3 X 1/4 ALUMINUM ANGLE	9 INCHES LONG
.250 DIA, ROD, STEEL	54 INCHES LONG
.187 STEEL STRIP, 1 1/2 WIDE	75 INCHES LONG
1/8 STEEL STRIP, 1 WIDE	24 INCHES LONG
.76 X 1.62 WOOD	25 INCHES LONG

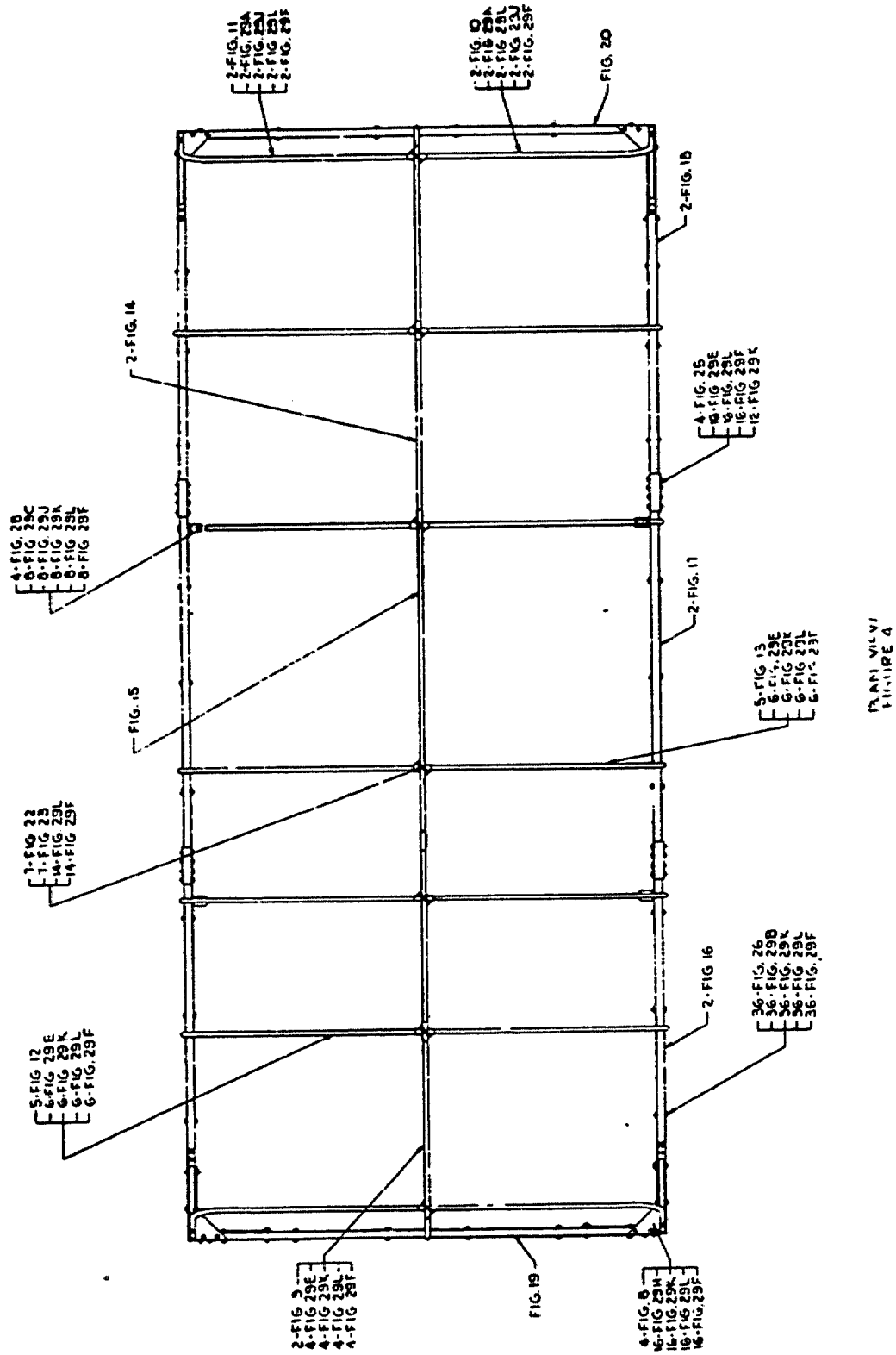
MATERIAL LIST-CLOSURE FRAME ASSEMBLY
FIGURE 2

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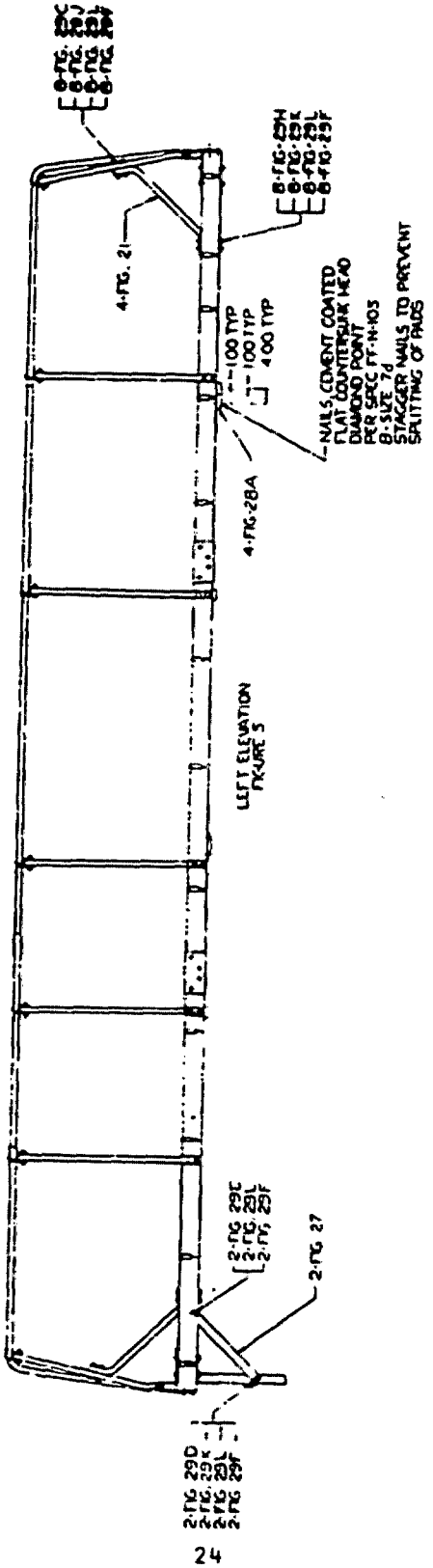


CLOSURE FRAME ASSEMBLY WITH INSTALLATION BRACKETS (PERSPECTIVE VIEW)
FIGURE 3

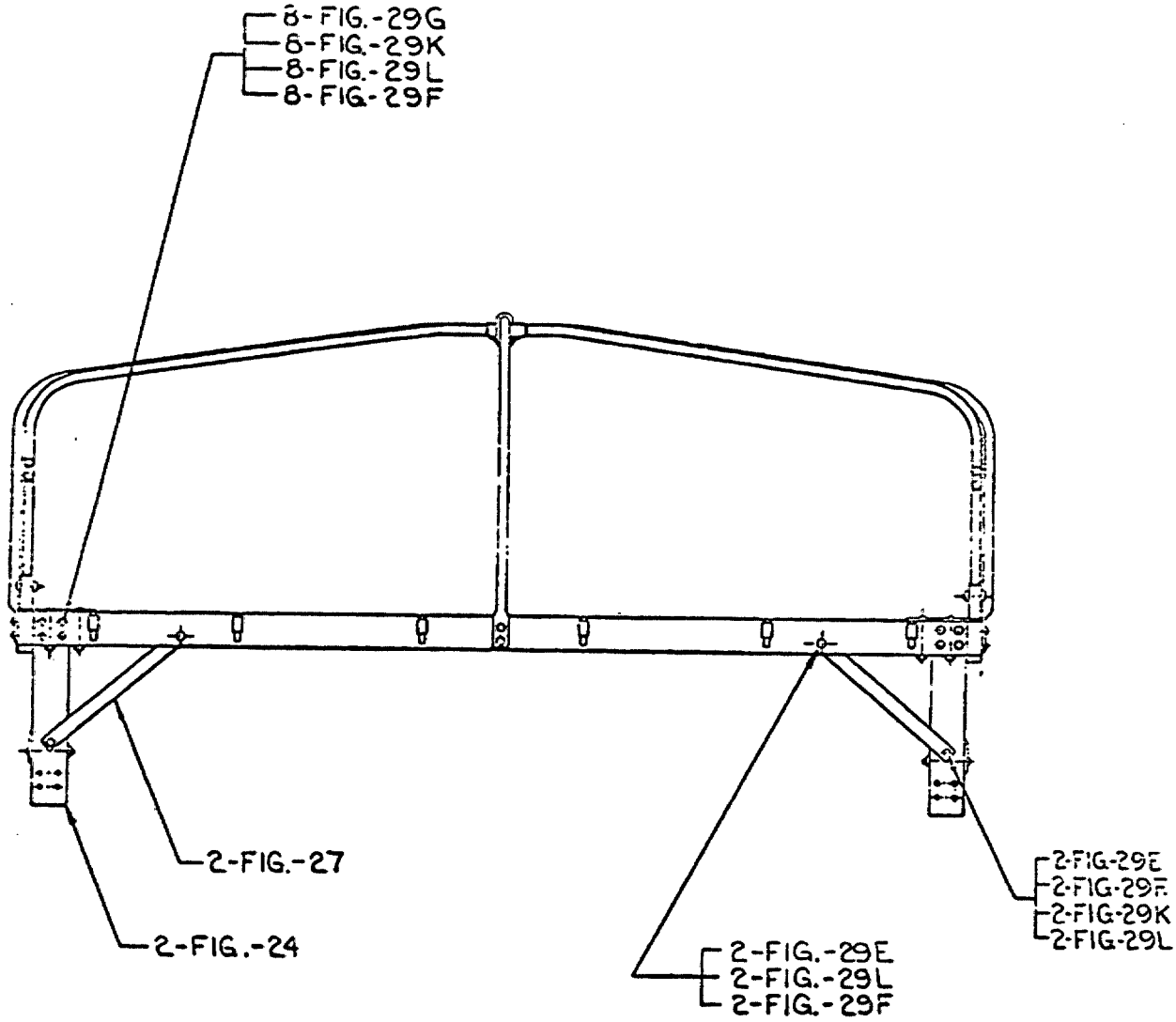
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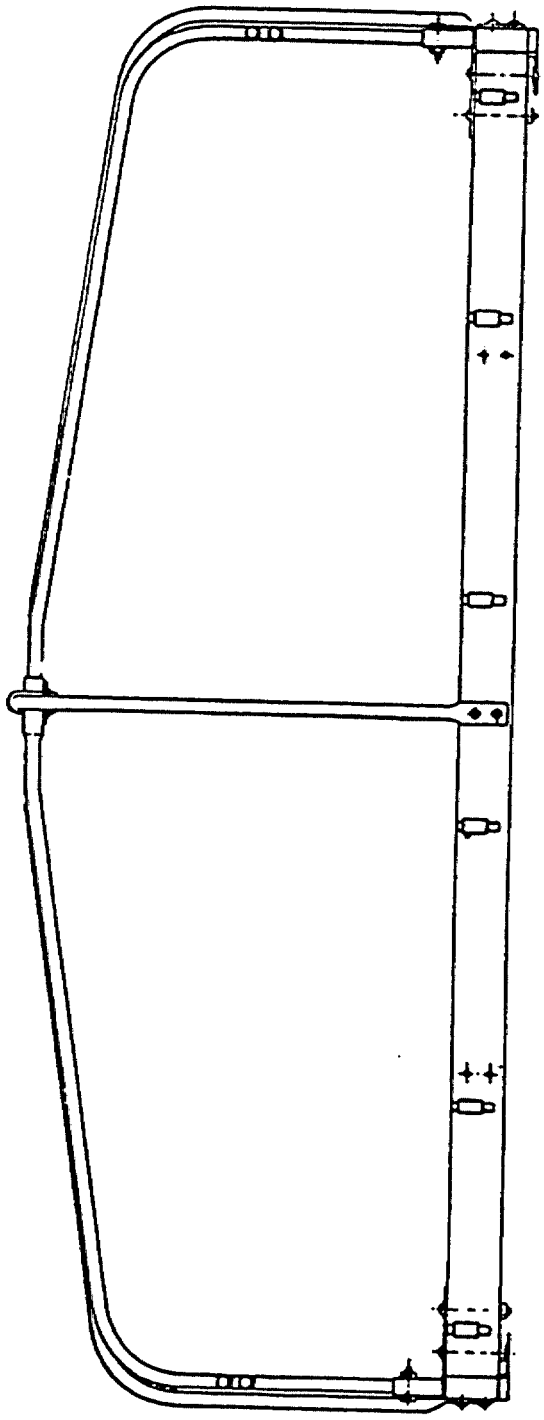


FRONT ELEVATION
FIGURE 6

DRAWING NO.
F 11604079

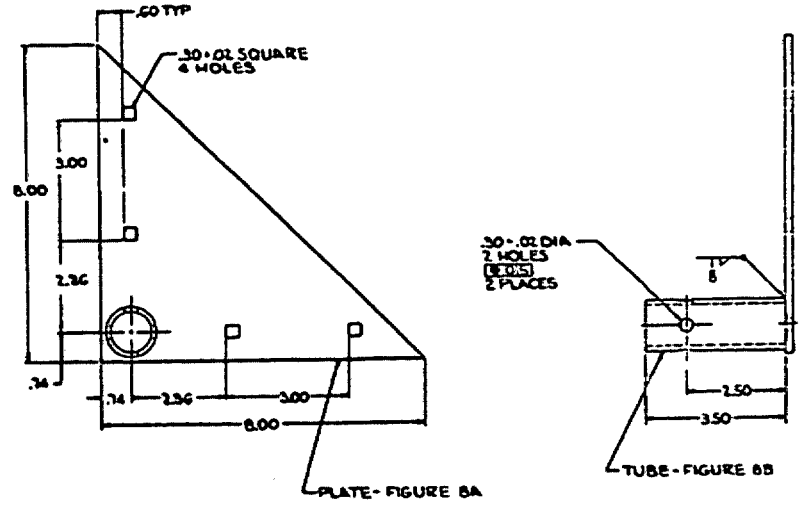
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DRAWING NO.
F 11604879



REAR ELEVATION
FIGURE 7

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FINISH NOTE
TREAT PER TYPE I OR II
SPEC TT-C-490
PRIME PER
SPEC TT-P-636
ENAMEL OD
SPEC TT-E-529 OR TT-E-485

MUST BE FREE OF BURRS
AND SHARP EDGES

STEEL, WELD PER CLASS I
SPEC MIL-STD-1261

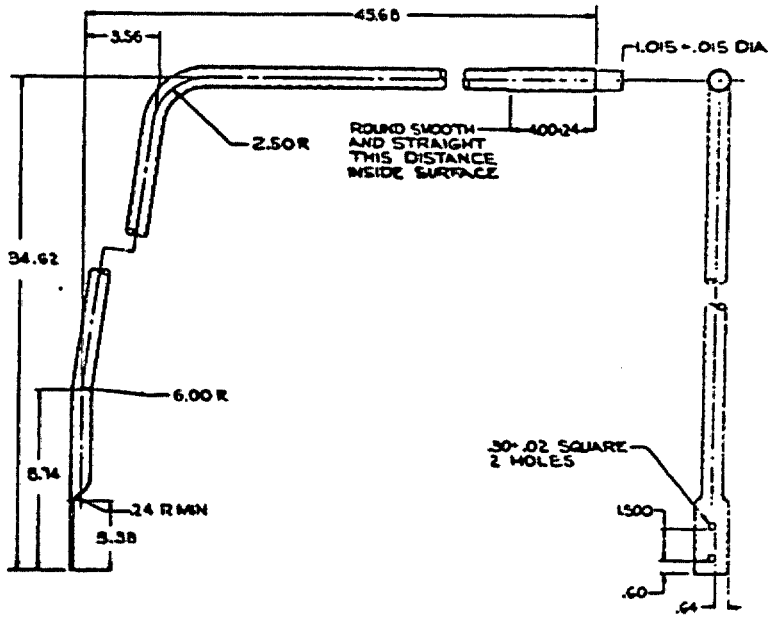
MATERIAL NOTE - TUBE
TUBE, CARBON STEEL
COMPOSITION M1010 THRU M1020
TYPE OPTIONAL
CONDITION CD
SPEC QQ-T-830
1¼ OD BY .109 WALL THICKNESS

MATERIAL NOTE - PLATE
STEEL, CARBON
HRCO, PCO OR CR
TEMPER OPTIONAL
SPEC QQ-S-698
.119 THICK

CORNER PLATE
FIGURE 8

DRAWING NO.
F 11604879

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BOW SECTION
FIGURE 8

DEVELOPED LENGTH TO APPROX.

FINISH NOTE
TREAT PER TYPE I OR III
SPEC TT-C-490
PRIME PER
SPEC TT-P-636
ENAMEL OD
SPEC TT-E-529 OR TT-E-485

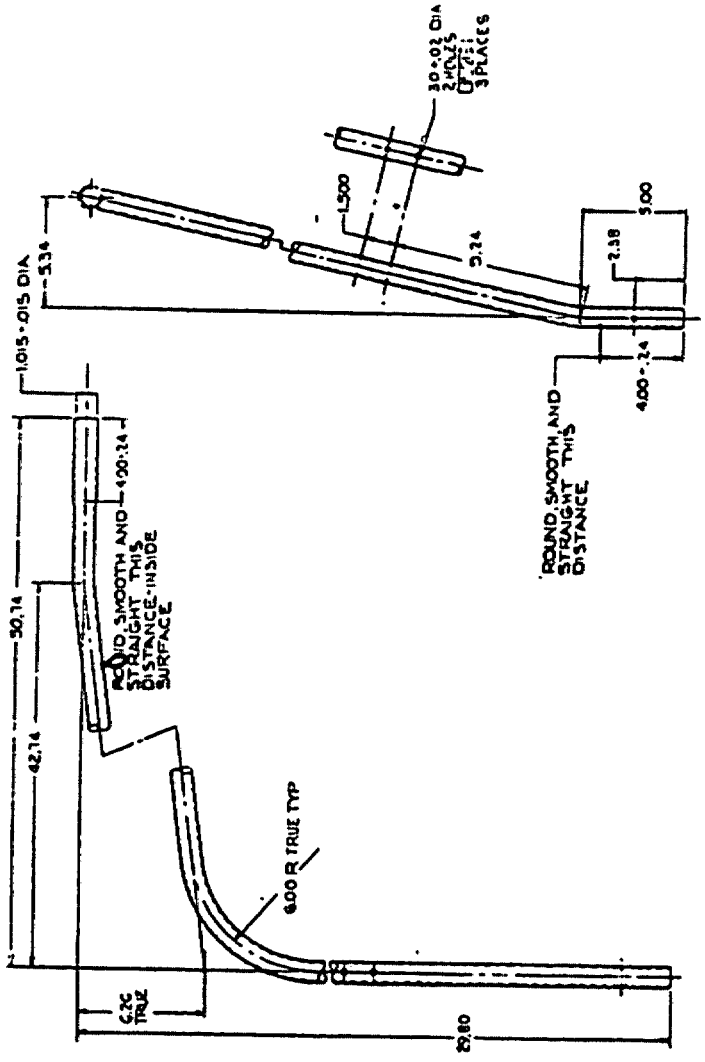
MUST BE FREE OF BURRS
AND SHARP EDGES

MATERIAL NOTE
TUBE, CARBON STEEL
COMPOSITION MT1010 THRU MT1020
TYPE OPTIONAL
CONDITION OPTIONAL
SPEC QQ-T-830
1.00 x .065 WALL THICKNESS

DRAWING NO.
F 11604879

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DRAWING NO.
F 11604879



BOW SECTION
FIGURE 10

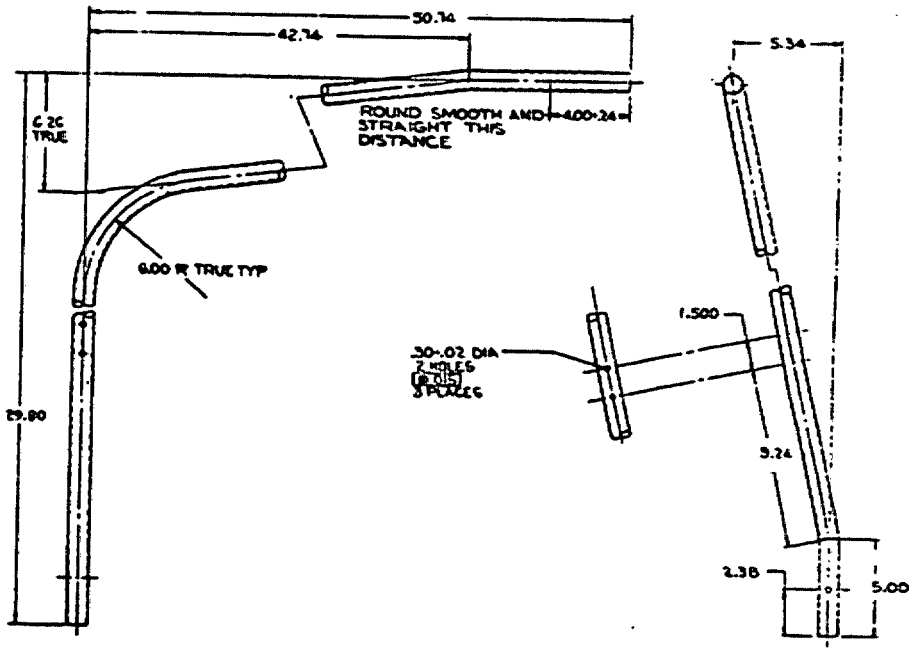
DEVELOPED LENGTH 73.74 APPROX

FINISH NOTE
TREAT PER TYPE I OR III
SPEC TTC-490
PRIME PER
SPEC TT-P-636
ENAMEL OD
SPEC TT-E-515 OR TT-E-485

MUST BE FREE OF BURRS
AND SHARP EDGES

MATERIAL NOTE
TUBE CARBON STEEL
COMPOSITION M1010 THRU M11020
TYPE OPTIONAL
CONDITION OPTIONAL
SPEC QQ-T-830
.100-.065 WALL THICKNESS

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BOW SECTION
FIGURE II

DEVELOPED LENGTH 73.74 APPROX

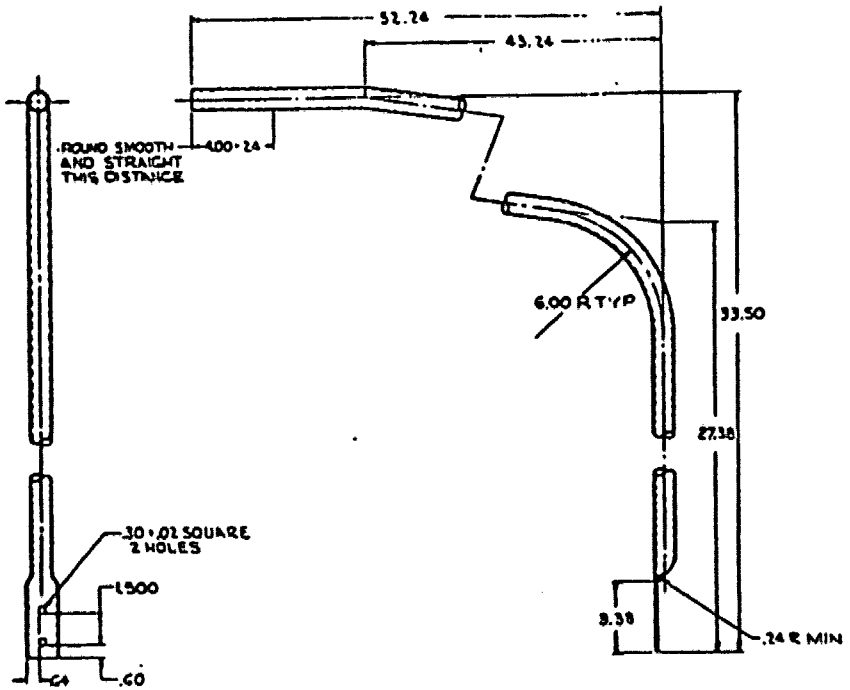
FINISH NOTE
TREAT PER TYPE I OR III
SPEC TT-C-490
PRIME PER
SPEC TT-P-636
ENAMEL O.D.
SPEC TT-E-529 OR TT-E-489

MUST BE FREE OF BURRS
AND SHARP EDGES

MATERIAL NOTE
TUBE CARBON STEEL
COMPOSITION MT1010 THRU MT1020
TYPE OPTIONAL
CONDITION OPTIONAL
SPEC QQ-T-850
1.00 = .065 WALL THICKNESS

DRAWING NO.
F 11604679

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BOW SECTION
FIGURE 12

DEVELOPED LENGTH 78.32 APPROX.

FINISH NOTE
TREAT PER TYPE I OR III
SPEC. TT-C-490
PRIME PER
SPEC. TT-P-636
ENAMEL, OD
SPEC. TT-E-529 OR TT-E-485

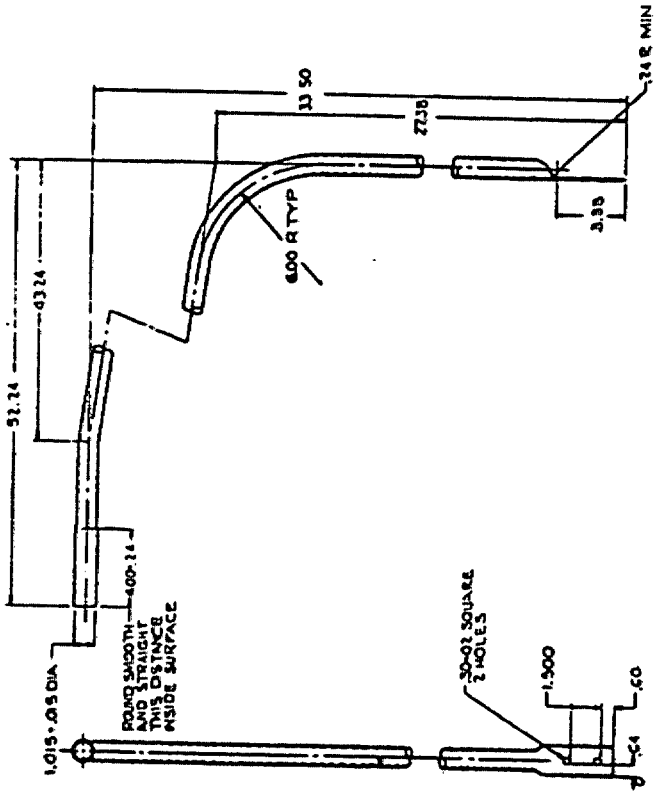
MUST BE FREE OF BURRS
AND SHARP EDGES

MATERIAL NOTE
TUBE CARBON STEEL
COMPOSITION MT-010 THRU MT1020
TYPE OPTIONAL
CONDITION OPTIONAL
SPEC. QQ-T-830
100-.065 WALL THICKNESS

DRAWING NO.
F 11604879

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DRAWING NO.
F 11604879



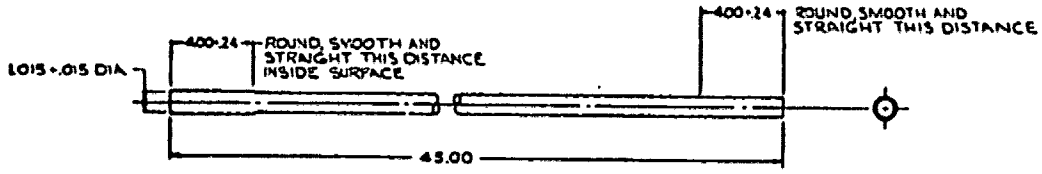
BOW SECTION
FIGURE 13

DEVELOPED LENGTH 78.31 APPROX

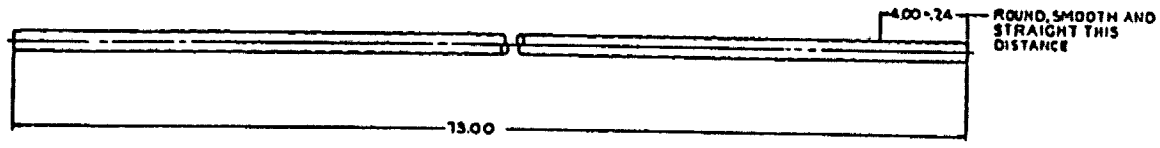
BWSN NOTE
TREAT PER TYPE I OR III
SPEC TT-C-490
PRIME PER TT-P-636
SPEC TT-P-636
ENAMEL PER TT-E-489 OR TT-E-485
MUST BE FREE OF BURRS AND SHARP EDGES

MATERIAL NOTE
TUBE, CARBON STEEL, COMPOSITION MT1020 THRU MT1026
TYPE OPTIONAL
CONDITION OPTIONAL
SPEC QQ-T-830
.120-.065 WALL THICKNESS

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BOW SECTION
FIGURE 14



BOW SECTION
FIGURE 15

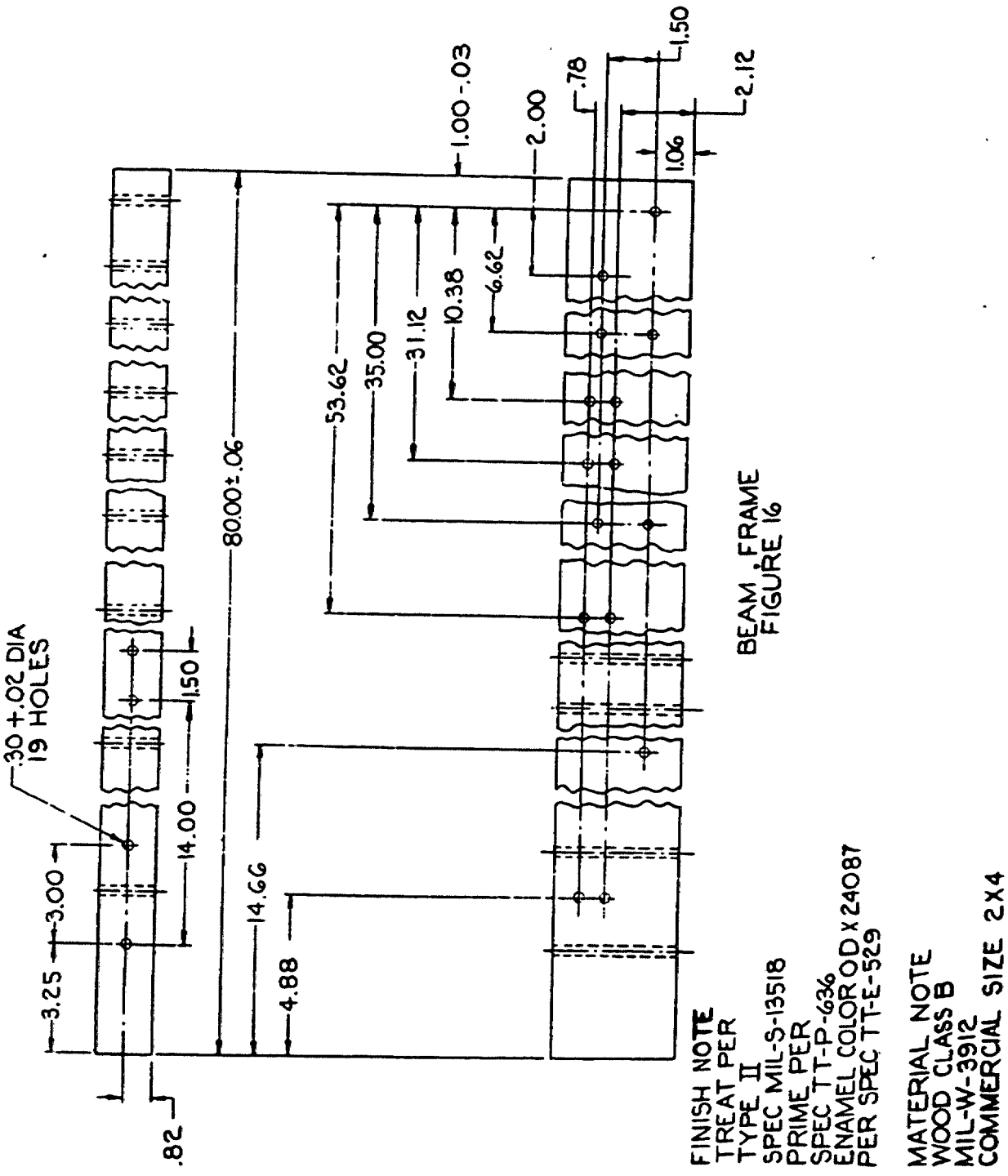
FINISH NOTE
TREAT PER TYPE I OR III
SPEC TT-C-490
PRIME PER
SPEC TT-P-636
ENAMEL, OD
SPEC TT-E-529 OR TT-E-489

MUST BE FREE OF BURRS
AND SHARP EDGES

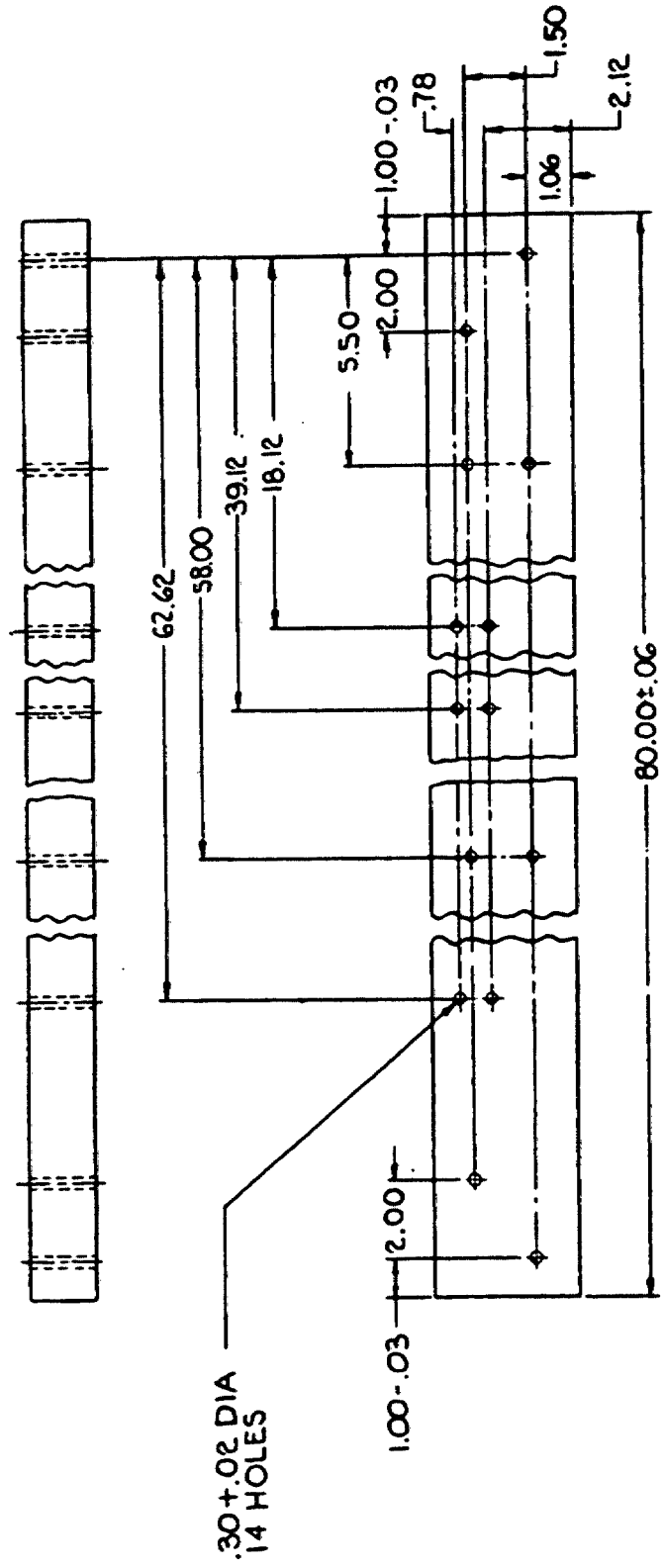
MATERIAL NOTE
TUBE, CARBON STEEL
COMPOSITION MT1010 THRU MT1020
TYPE OPTIONAL
CONDITION OPTIONAL
SPEC QQ-T-830
1.00 ±.065 WALL THICKNESS

DRAWING NO.
F 11604879

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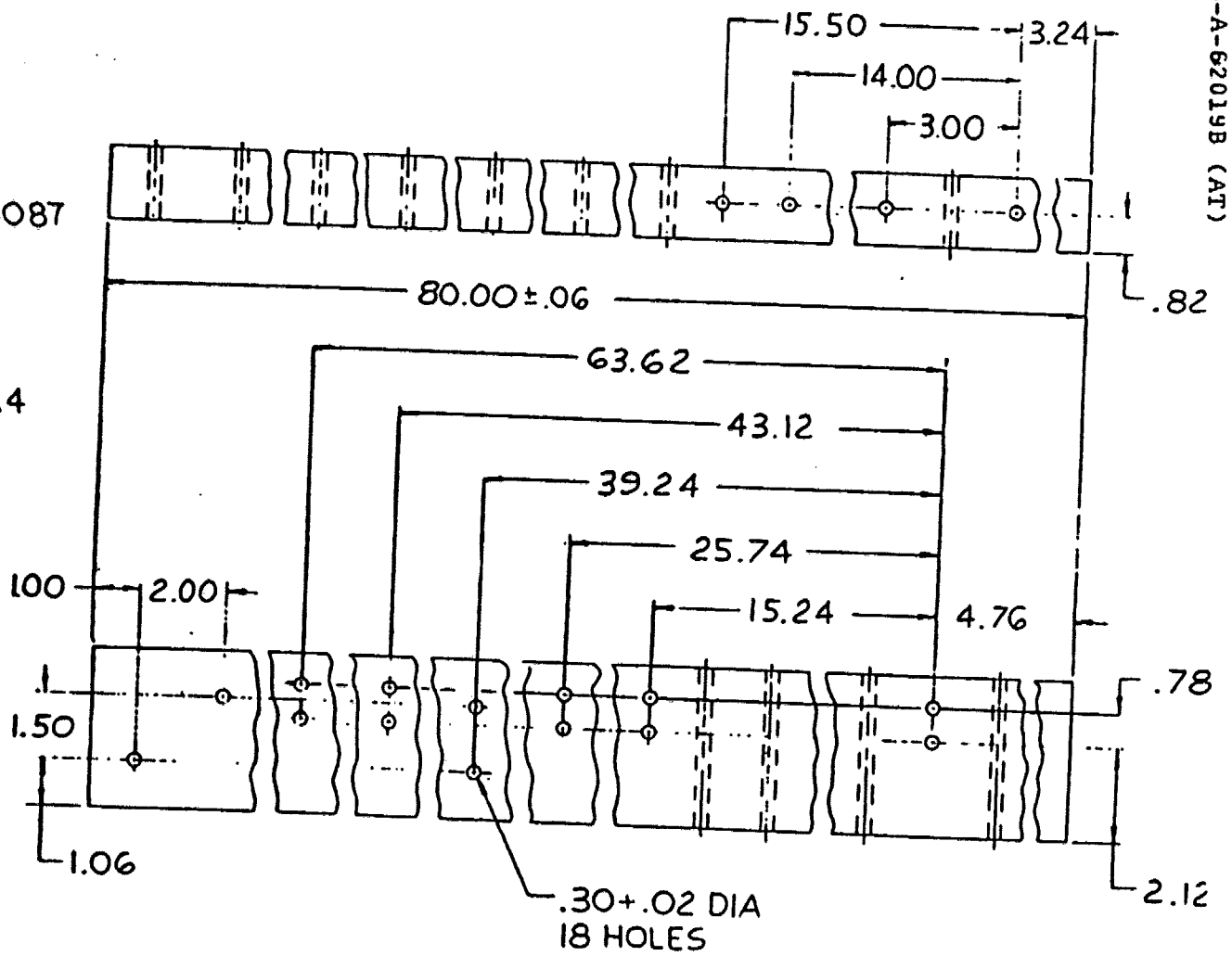
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FINISH NOTE
TREAT PER
TYPE II
SPEC MIL-S-13518
PRIME PER
SPEC TT-P-636
ENAMEL COLOR OD X 24087
PER SPEC TT-E-529

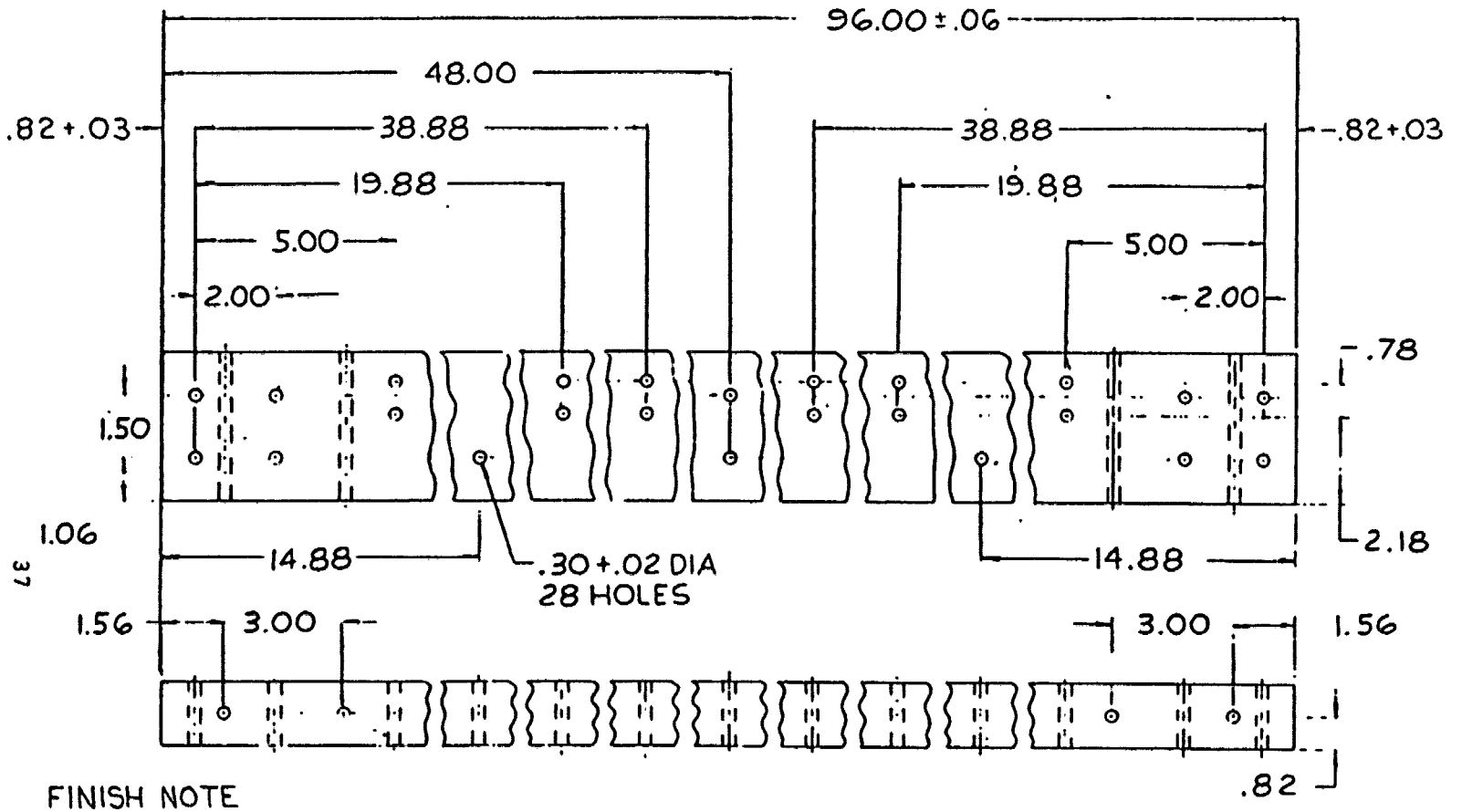
MATERIAL NOTE
WOOD CLASS B
MIL-W-3912
COMMERCIAL SIZE 2X4



BEAM, FRAME
FIGURE 18

DRAWING NO.
F 11604879

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
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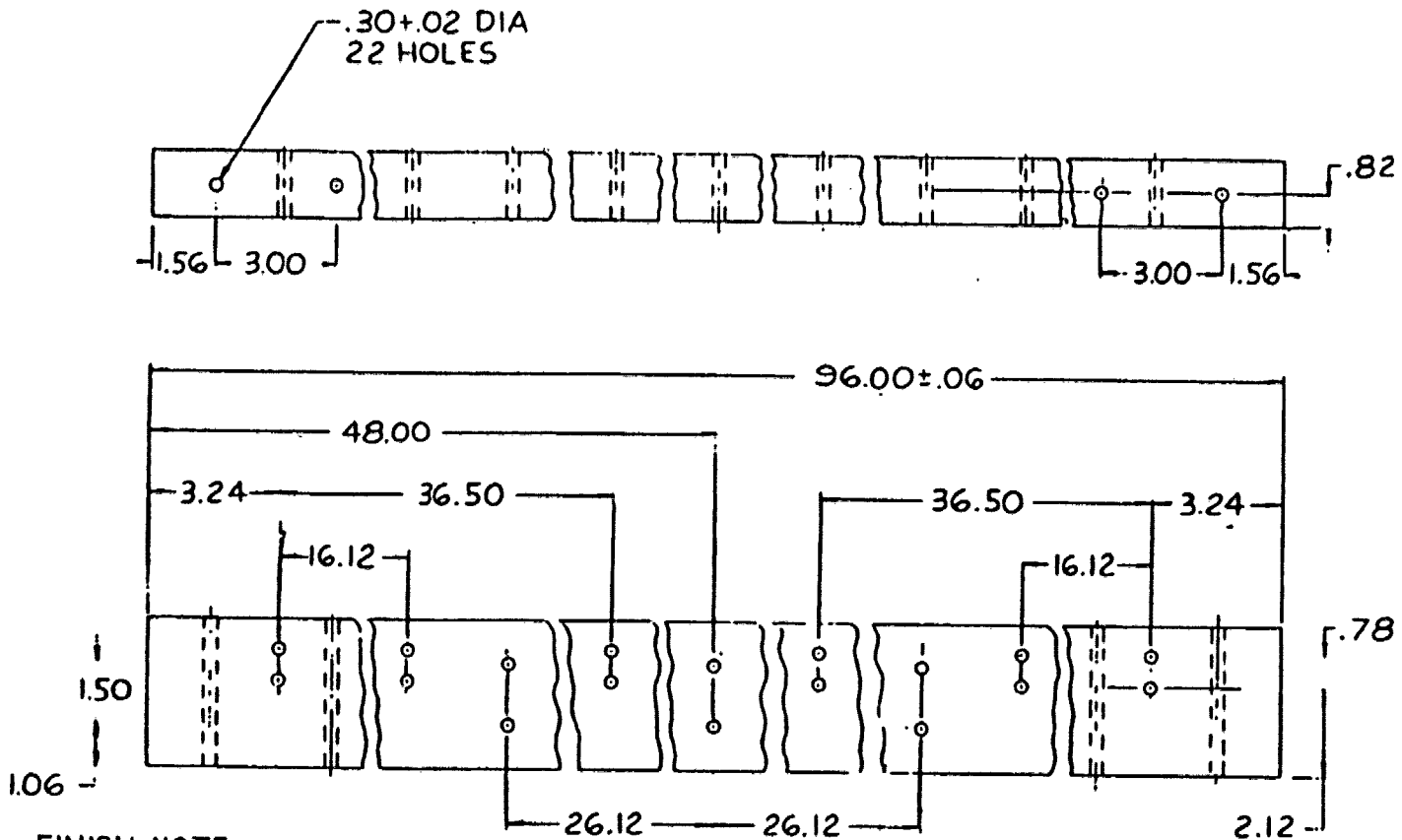
FINISH NOTE
TREAT PER
TYPE II
SPEC MIL-S-13518
PRIME PER
SPEC TT-P-636
ENAMEL COLOR OD X 24087
PER SPEC TT-E-529

MATERIAL NOTE
WOOD CLASS B
MIL-W-3912
COMMERCIAL SIZE 2X4

BEAM, FRAME
FIGURE 19

DRAWING NO.
F 11604879

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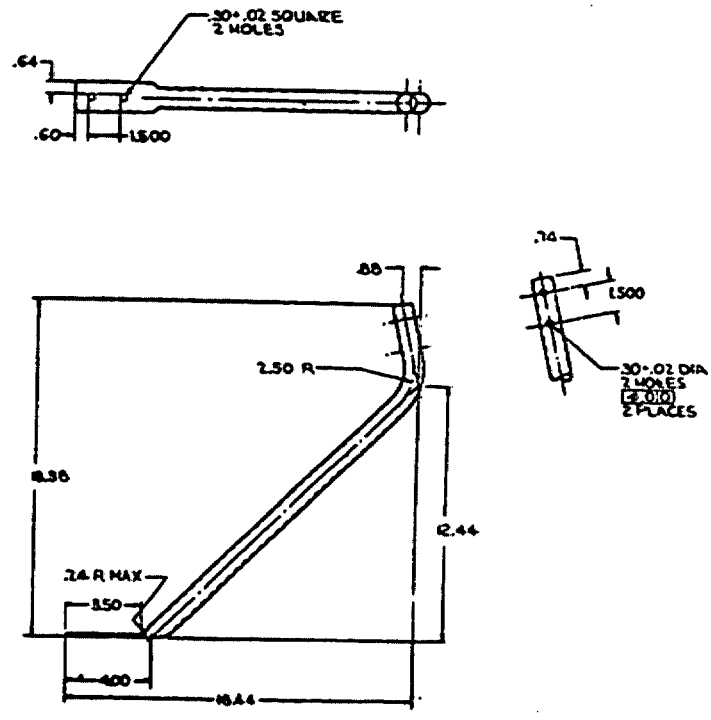
FINISH NOTE
TREAT PER
TYPE II
SPEC MIL-S-13518
PRIME PER
SPEC TT-P-636
ENAMEL COLOR OD X 24087
PER SPEC TT-E-529

MATERIAL NOTE
WOOD CLASS B
MIL-W-3912
COMMERCIAL SIZE 2X4

BEAM, FRAME
FIGURE 20

DRAWING NO.
F 11604879

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Check the source to verify that this is the current version before use.



KNEE BRACE
FIGURE 21

MUST BE FREE OF BURRS AND SHARP EDGES

FINISH NOTE
TREAT PER TYPE I OR II
SPEC TT-C-490
PRIME PER
SPEC TT-P-636
ENAMEL OD
SPEC TT-E-529 OR TT-E-485

DEVELOPED LENGTH 25.74 APPROX

MATERIAL NOTE
TUBE, CARBON STEEL
COMPOSITION MT1040 THRU MT1020
TYPE OPTIONAL
CONDITION OPTIONAL
SPEC QQ-T-830
1.00 • .065 WALL THICKNESS

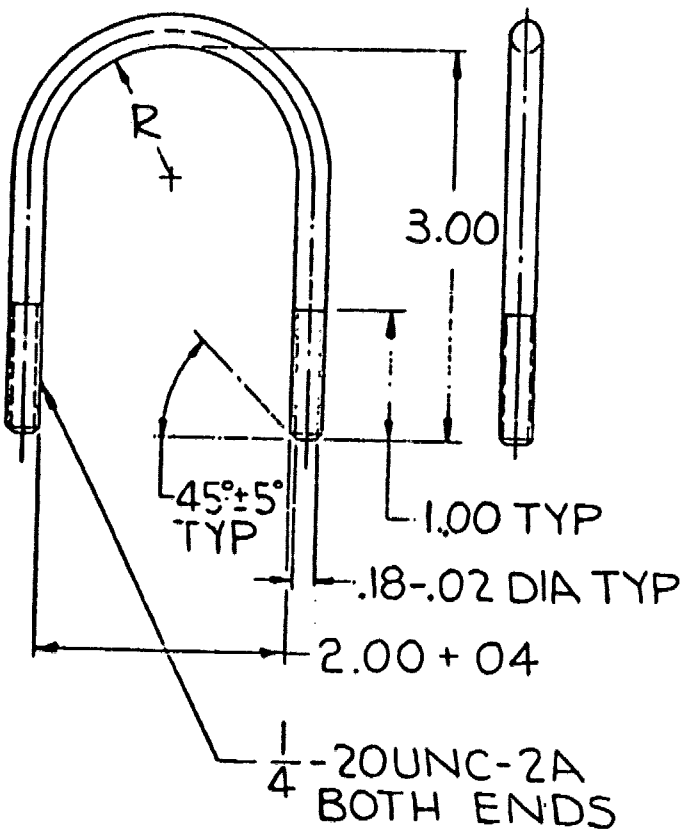
DRAWING NO.
F 11604879

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

FINISH NOTE
ZINC
CLASS 3 TYPE I
SPEC QQ-Z-325

OPTIONAL MATERIAL
STEEL, CARBON
C1010 THRU C1020
SPEC QQ-S-630

MATERIAL NOTE
WIRE, STEEL 1010-1020
FINISH 1
TEMPER OPTIONAL
SPEC QQ-W-461
.250 DIA



U CLAMP
FIGURE 22

DRAWING NO.
F 11604879

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
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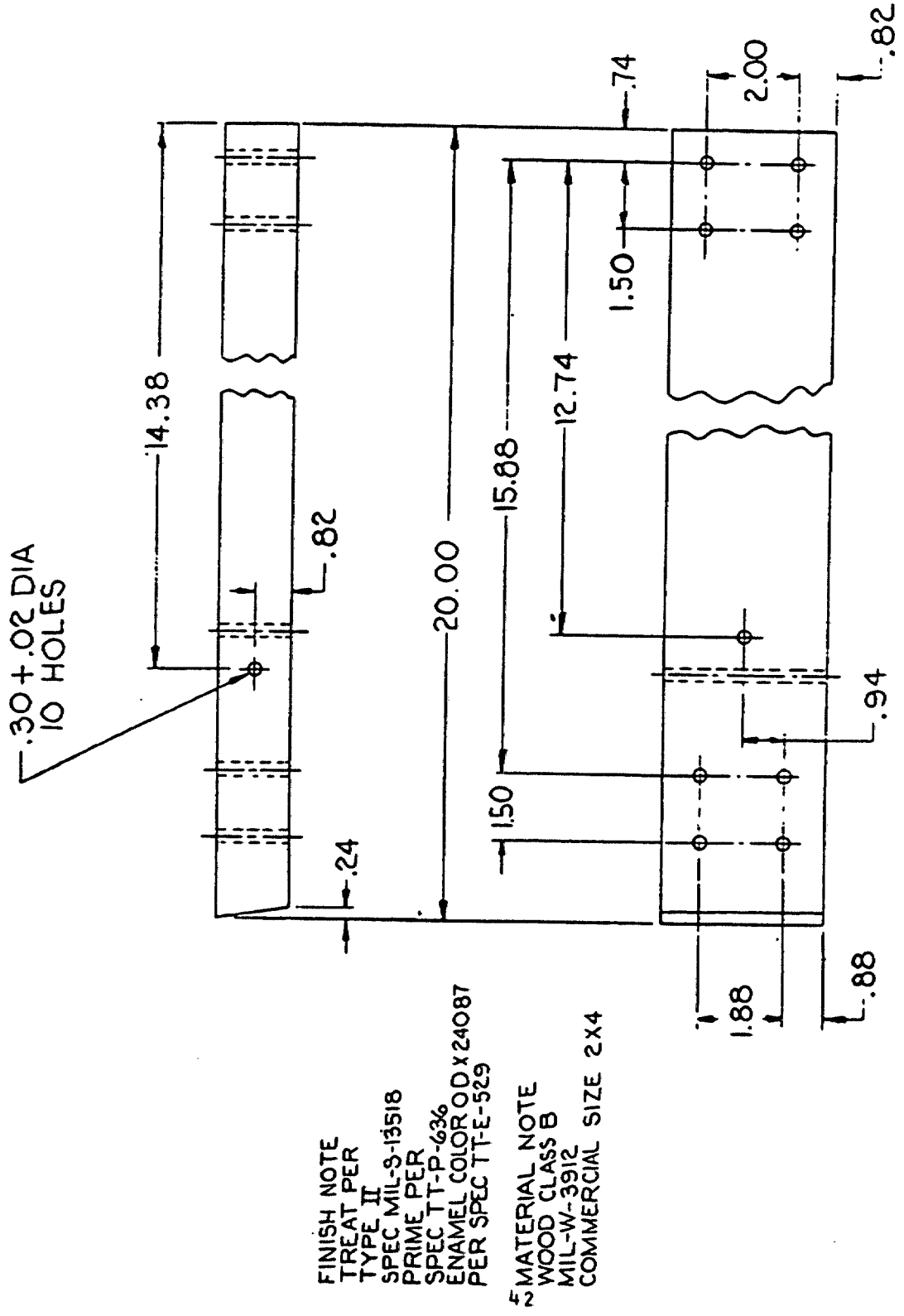
DRAWING NO
F 11604879

PLATE
FIGURE 23

FINISH NOTE
ZINC (HOT DIP)
SPEC MIL-Z-17871

MATERIAL NOTE
STEEL, CARBON
HRCQ, P&O OR CRCQ
SPEC QQ-S-698
.120 THICK

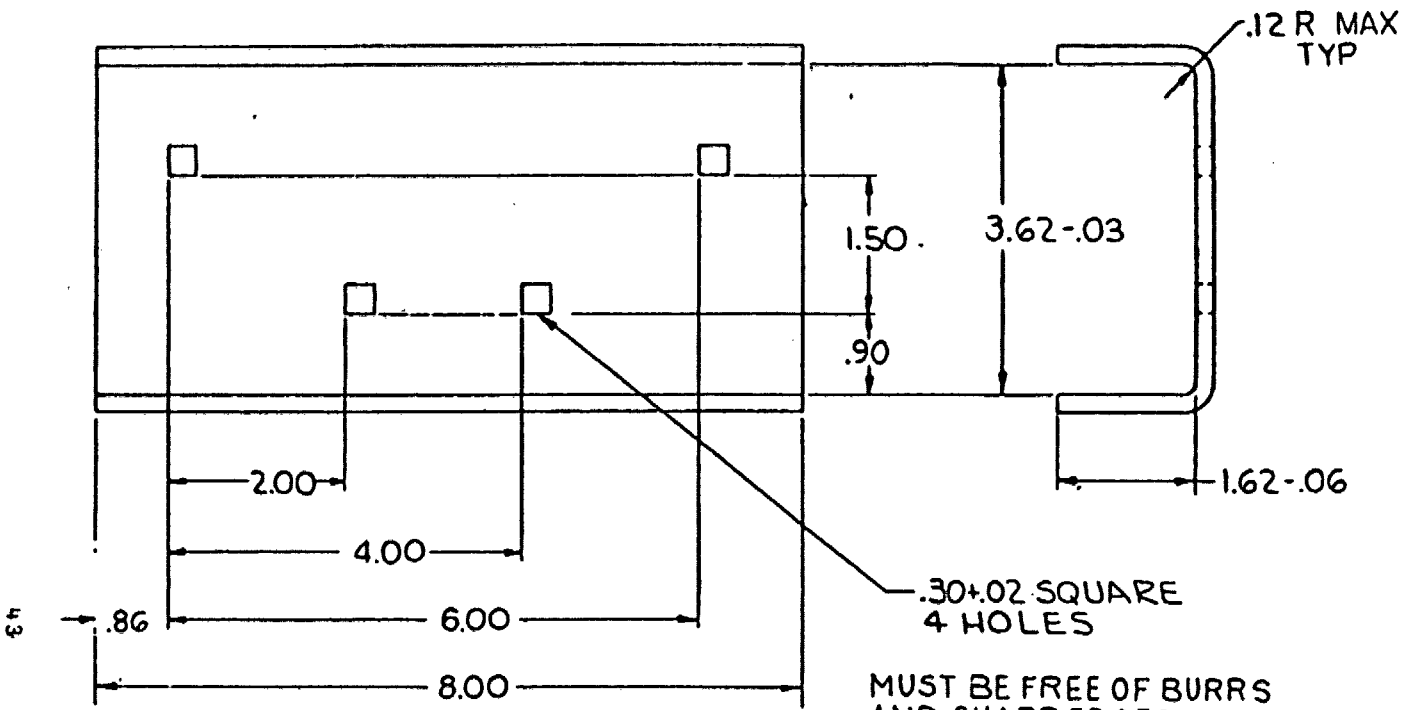
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



SUPPORT, FRAME
FIGURE 24

DRAWING NO.
F11604879

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



MUST BE FREE OF BURRS
AND SHARP EDGES

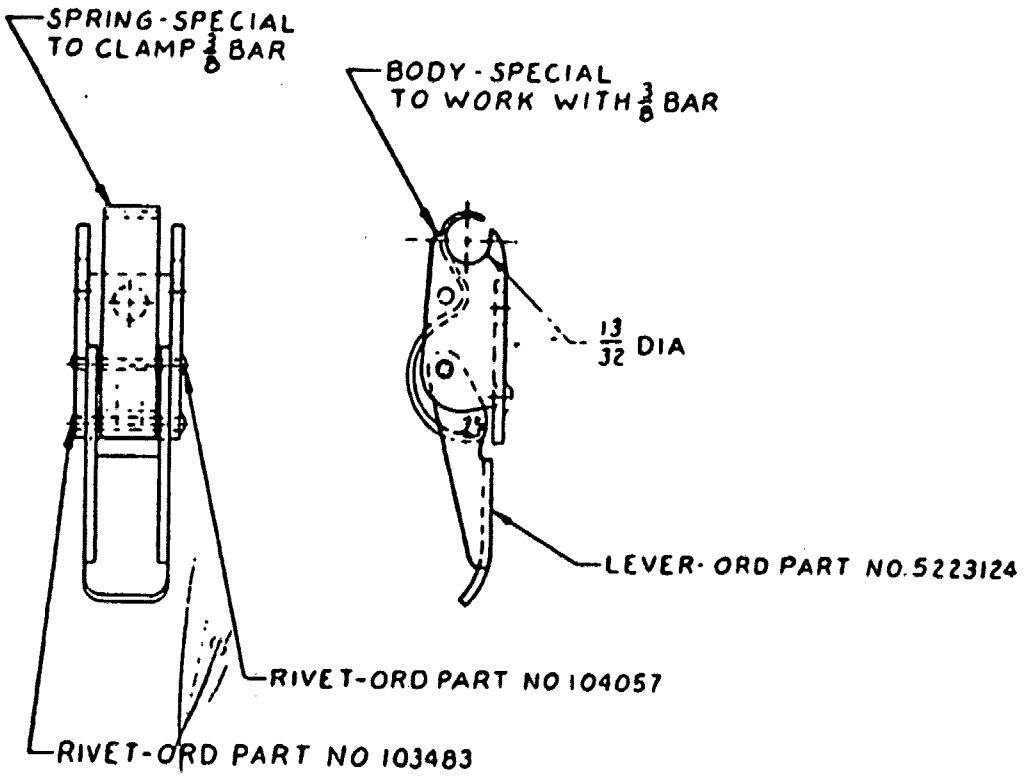
FINISH NOTE
TREAT PER TYPE I OR III
SPEC TT-C-490
PRIME PER
SPEC TT-P-636
ENAMEL, OD
SPEC TT-E-529 OR TT-E-485

MATERIAL NOTE
STEEL, CARBON
HRCQ P&O OR CR
TEMPER 4 OR 5
SPEC QQ-S-698
179 THICK

FIGURE 25 CONNECTOR, BEAM

DRAWING NO.
F 11604879

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Check the source to verify that this is the current version before use.



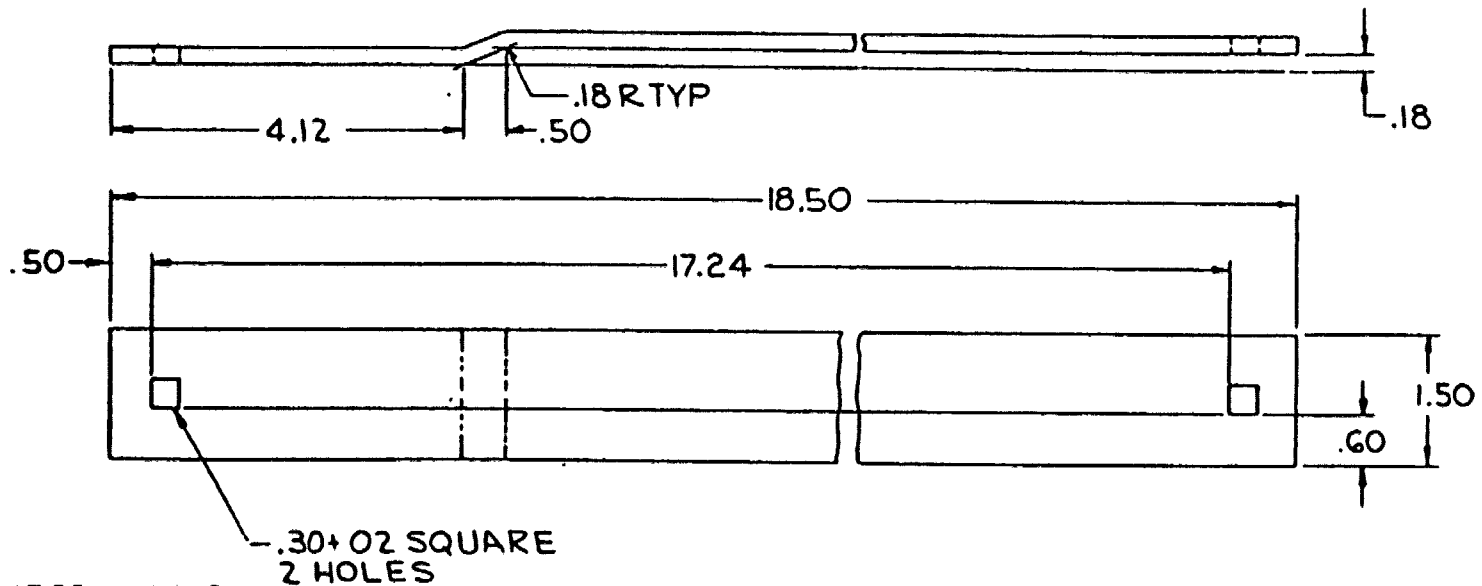
APPROVED SOURCE
DETROIT HINGE MFG. CO.
DETROIT, MICHIGAN
PART NO. 12661.

ALL SOURCES MUST COMPLY WITH DRAWING REQUIREMENTS AND IN ADDITION THE ITEM MUST BE FUNCTIONALLY EQUIVALENT TO THE MANUFACTURER'S ITEM INDICATED ORDNANCE CORPS APPROVAL REQUIRED.

FASTENER
FIGURE 26

PART NUMBER
10922144

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



MUST BE FREE OF BURRS
AND SHARP EDGES

FINISH NOTE
TREAT PER TYPE I OR III
SPEC TT-C-490
PRIME PER
SPEC TT-P-636
ENAMEL, OD
SPEC TT-E-529 OR TT-E-485

MATERIAL NOTE
STEEL, CARBON
HRCQ P+O OR CR
TEMPER OPTIONAL
SPEC QQ-S-698
.179 THICK

FIGURE 27 BRACE, FRAME SUPPORT

DRAWING NO
F 11604879

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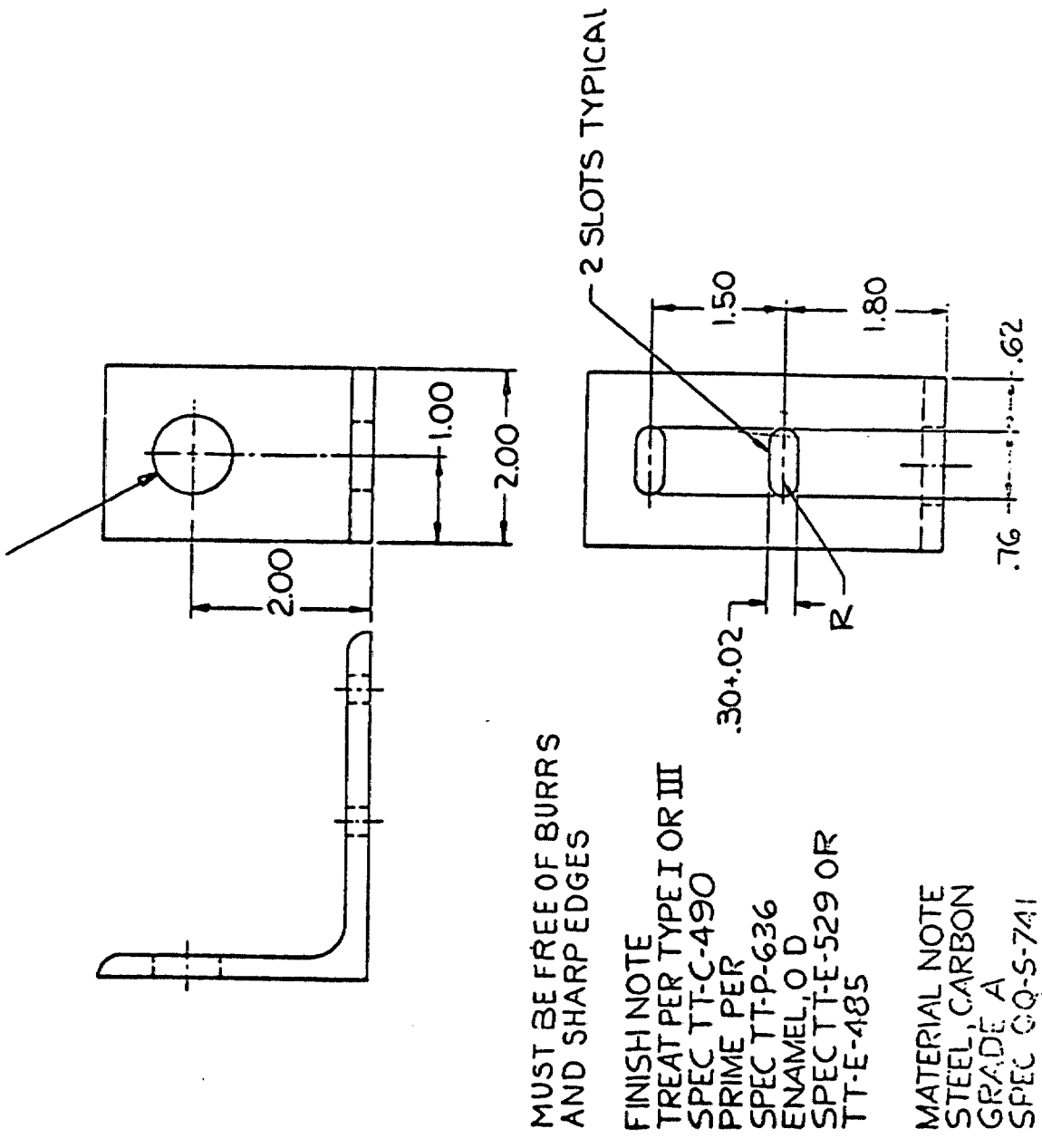


FIGURE 28 BRACKET

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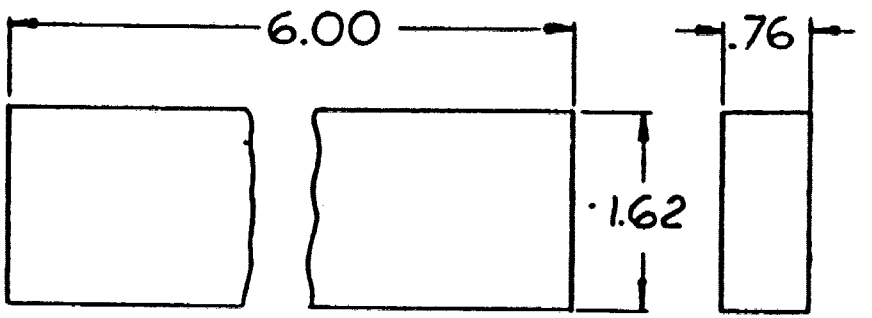


FIGURE 28 A PAD

FINISH NOTE
TREAT PER
TYPE II
SPEC MIL-S-13518
PRIME PER
SPEC TT-P-636
ENAMEL COLOR OD X 24087
PER SPEC TT-E-529

MATERIAL NOTE
WOOD CLASS B
MIL-W-3912

DRAWING NO
F 11604879

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

FIG. NO.	ARMY PART NO.	DESCRIPTION	MATERIAL	QTY REQD
29A	MS90728-14	SCREW, CAP 1/4-20UNC-2A X 2	ALLOY STEEL	4
29B	MS90728-15	SCREW, CAP 1/4-20UNC-2A X 2 1/4	ALLOY STEEL	36
29C	MS90728-16	SCREW, CAP 1/4-20UNC-2A X 2 1/2	ALLOY STEEL	16
29D	MS35751-202	BOLT, SQ. NECK 1/4-20UNC-2A X 4 1/4	CARBON STEEL	2
29E	MS35752-14	BOLT, SQ. NECK 1/4-20UNC-2A X 2 1/2	CARBON STEEL	46
29F	MS51967-3	NUT, HEX 1/4-20UNC-2B	CARBON STEEL	150
29G	MS35752-18	BOLT, SQ. NECK 1/4-20UNC-2A X 4	CARBON STEEL	8
29H	MS35752-19	BOLT, SQ. NECK 1/4-20UNC-2A X 4 1/2	CARBON STEEL	24
29J	MS27183-10	WASHER, FLAT 1/4	CARBON STEEL	20
29K	MS63040-4	WASHER, FLAT 1/4	CARBON STEEL	120
29L	MS35338-63	WASHER, LOCK 1/4	CARBON STEEL	150

HARDWARE -CLOSURE FRAME ASSEMBLY
FIGURE 29

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
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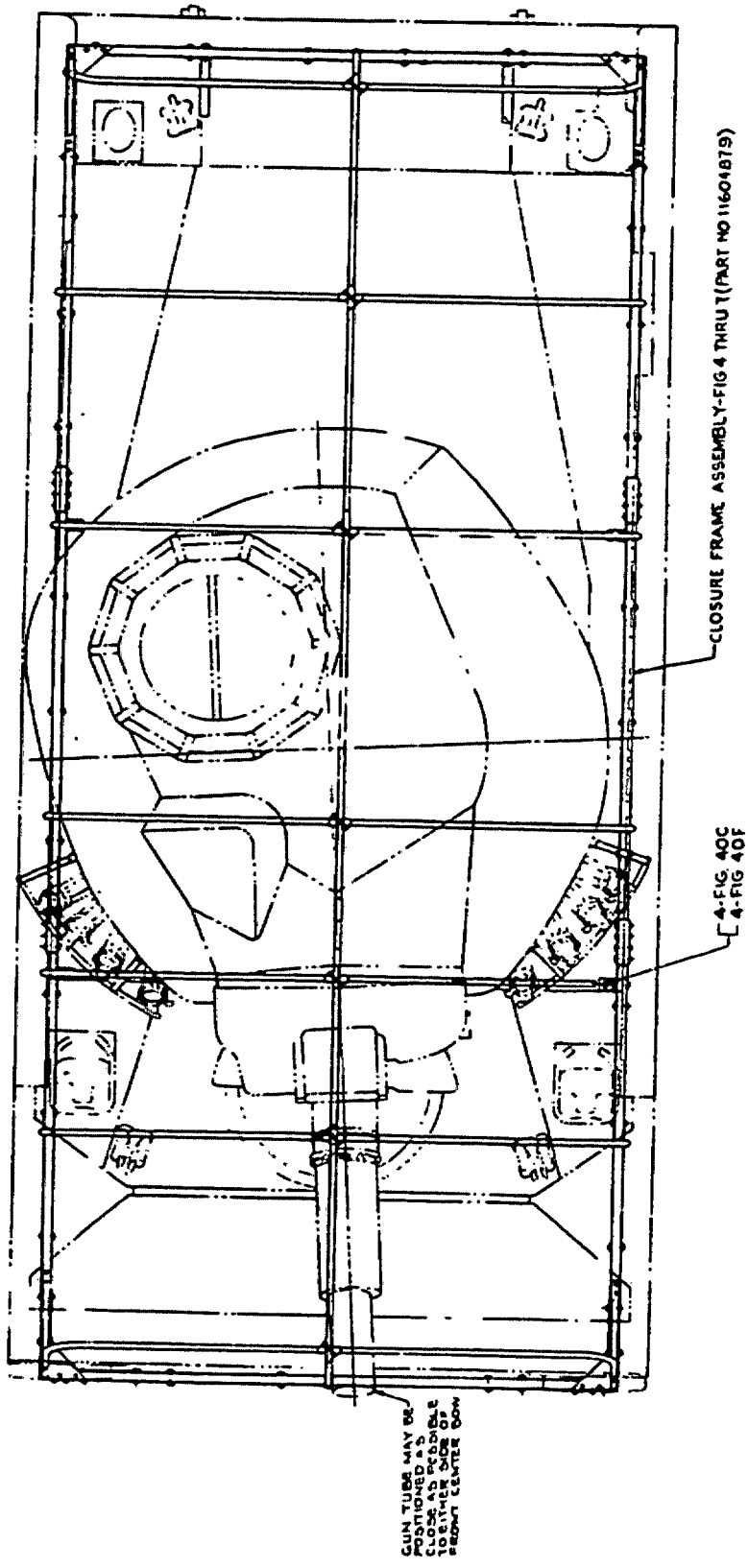
FIG NO.	NOMENCLATURE	QTY REQD	SHEET
31	MATERIAL LIST	—	4
32	FRAME INSTALLATION-PLAN VIEW	—	1
33	FRAME INSTALLATION-LEFT ELEVATION	—	2
34	FRAME INSTALLATION-FRONT ELEVATION	—	3
35	FRAME INSTALLATION-REAR ELEVATION	—	3
36	BRACKET	2	5
37	BRACKET, FRONT (WELDMENT)	2	6
38	BRACKET	2	6
39	PLATE	2	6
40	HARDWARE, CLOSURE FRAME INSTALLATION SEE HARDWARE LIST	—	4

PARTS LIST-CLOSURE FRAME INSTALLATION
FIGURE 30

DESCRIPTION	QTY REQD
1/4 THICK X 1 1/4 WIDE STEEL STRIP	13 INCHES LONG
1/4 THICK X 1 1/2 WIDE STEEL STRIP	63 INCHES LONG

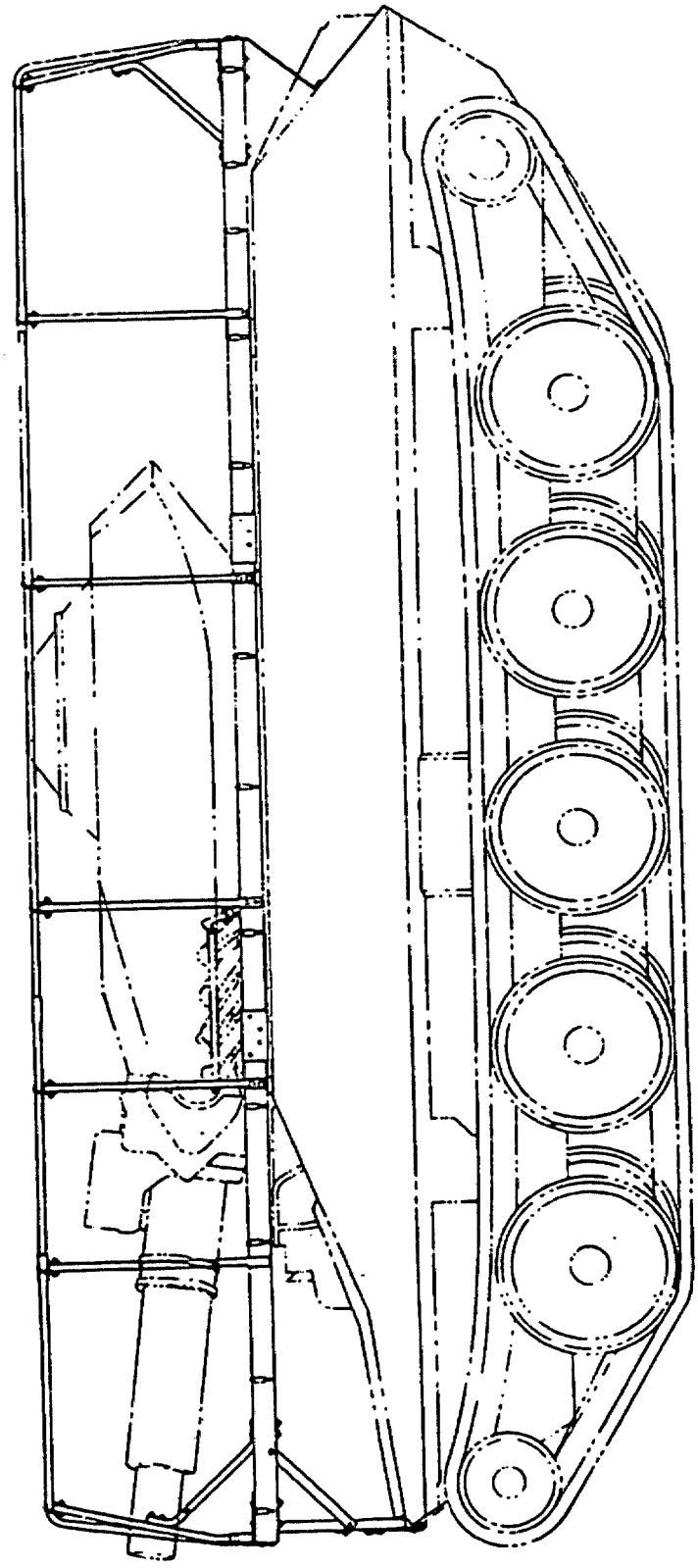
MATERIAL LIST-CLOSURE FRAME INSTALLATION
FIGURE 31

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



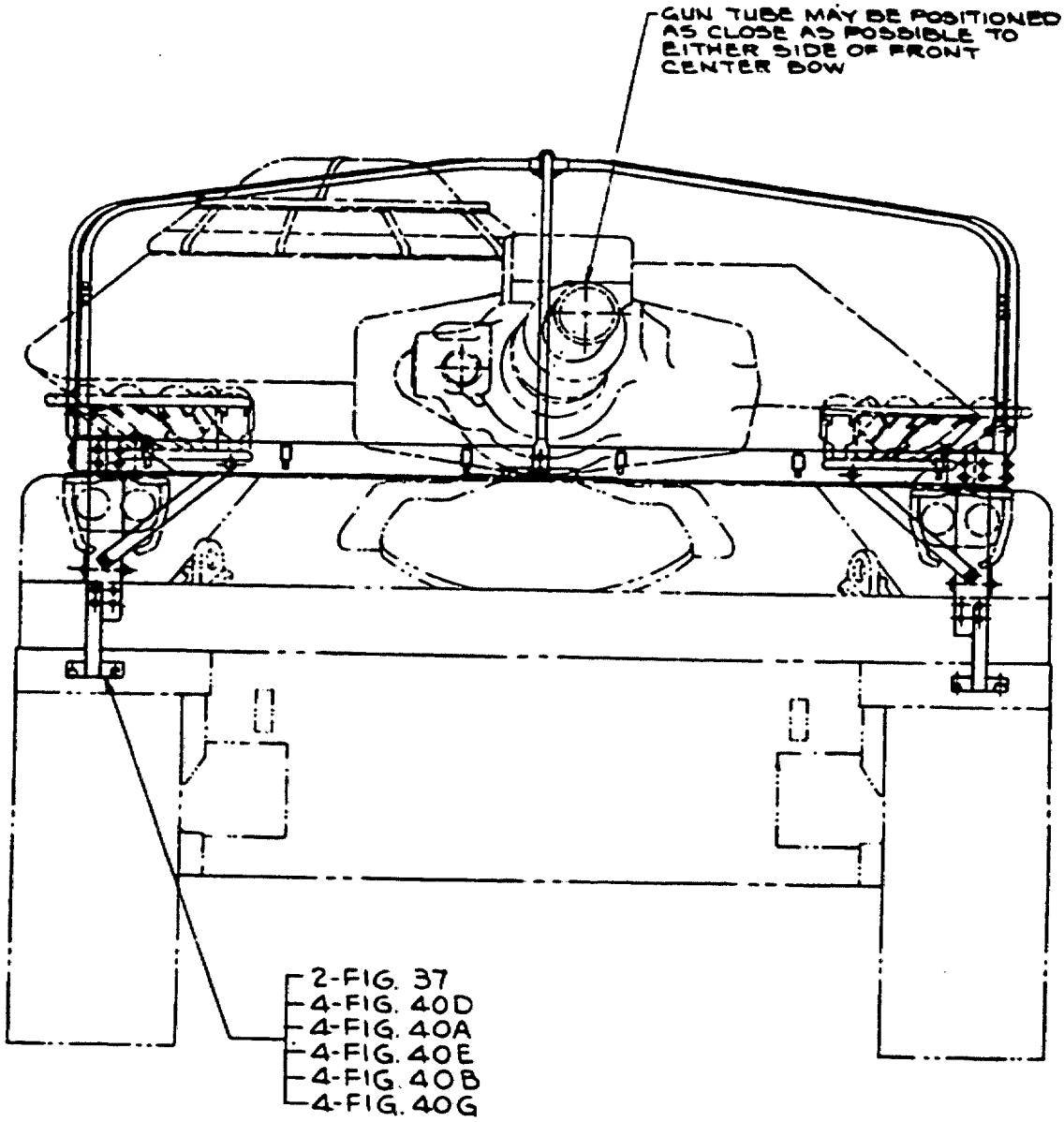
PLAN VIEW
FIGURE 32

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Check the source to verify that this is the current version before use.



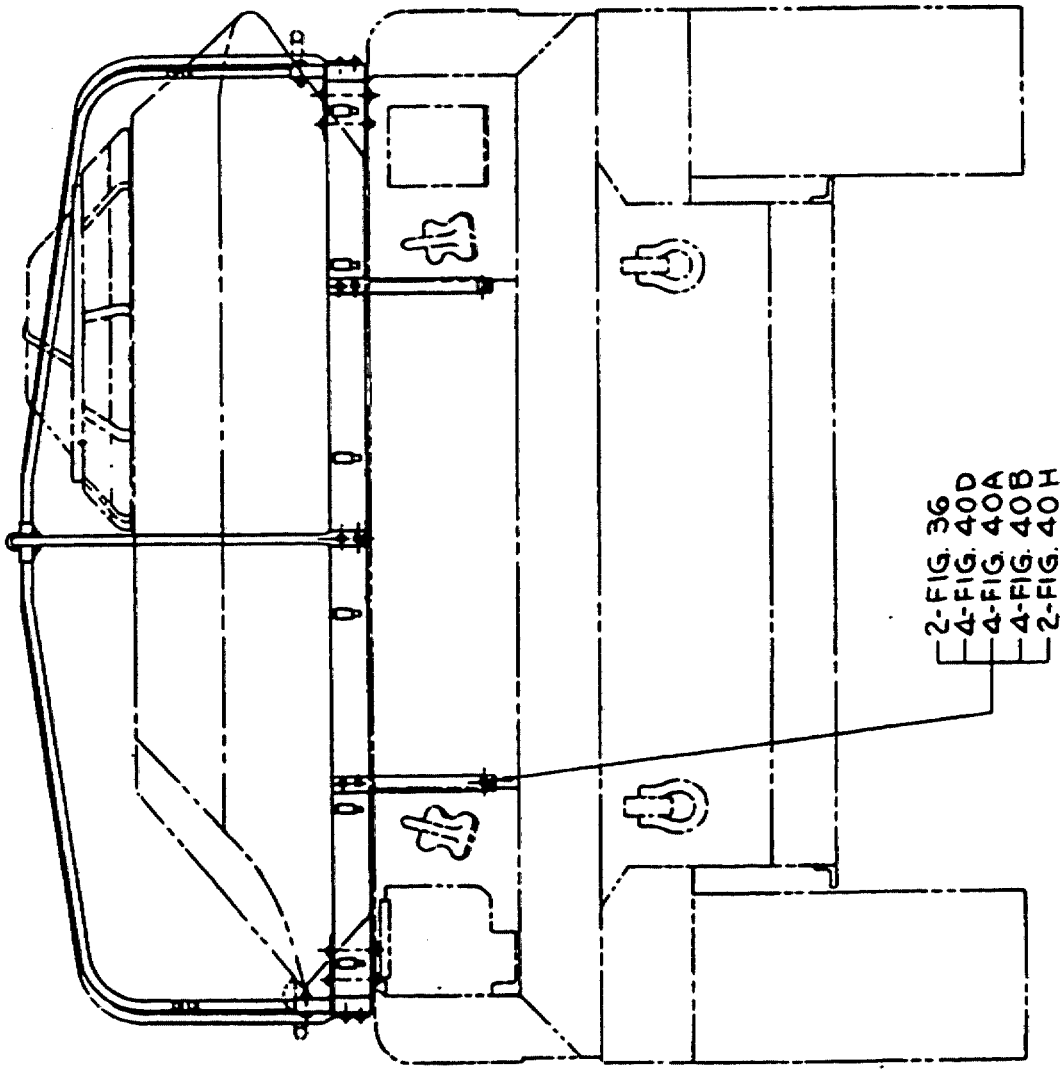
LEFT ELEVATION
FIGURE 53

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Check the source to verify that this is the current version before use.



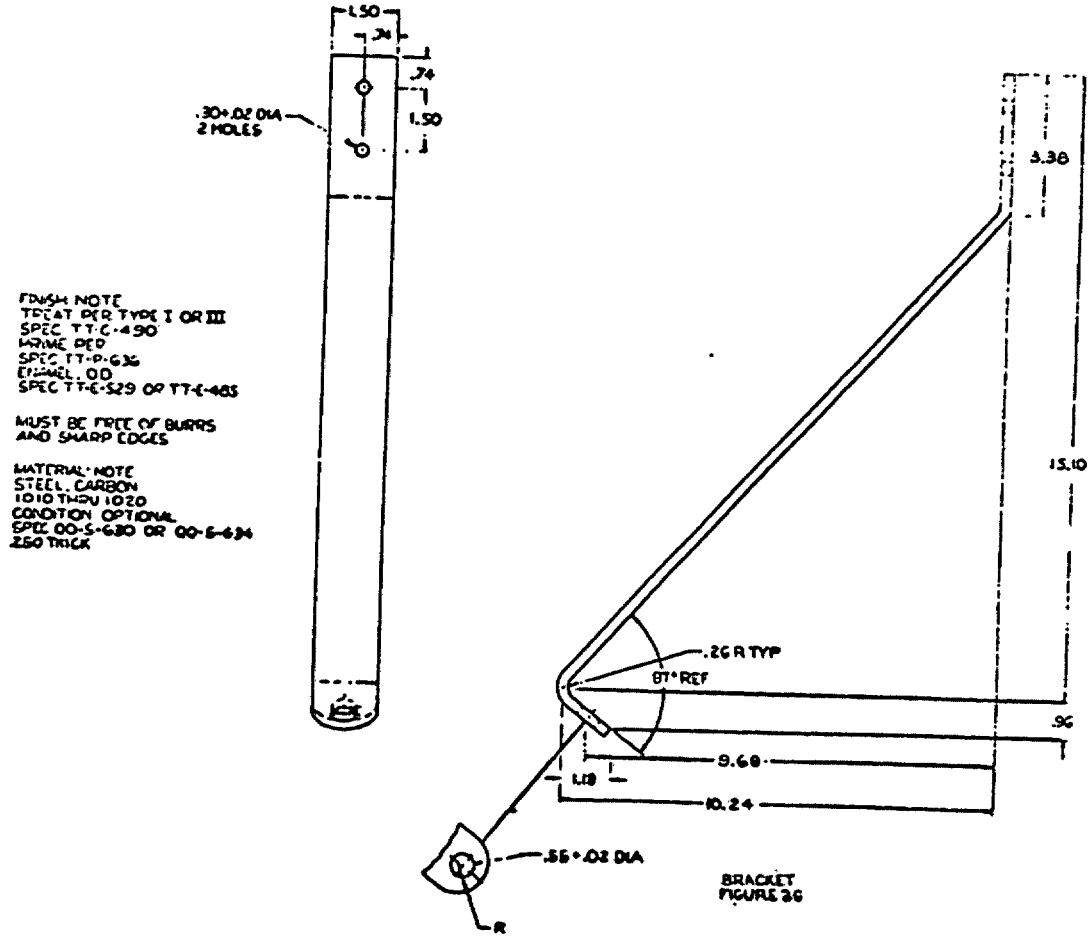
FRONT ELEVATION
FIGURE 34

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
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REAR ELEVATION
FIGURE 35

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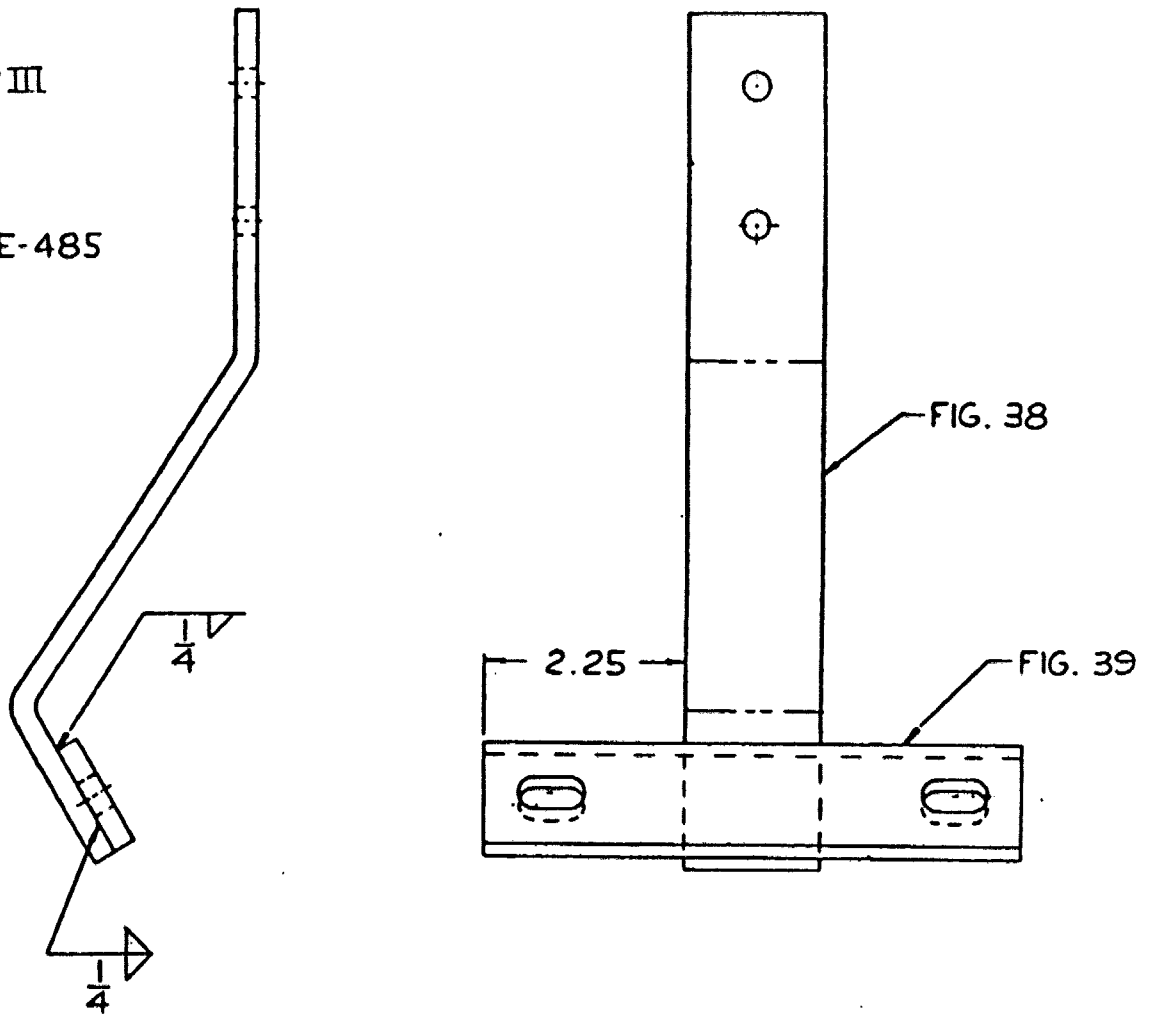


DRAWING NO
F 11604300

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FINISH NOTE
TREAT PER TYPE I OR III
SPEC TT-C-490
PRIME PER
SPEC TT-P-636
ENAMEL, OD
SPEC TT-E-529 OR TT-E-485

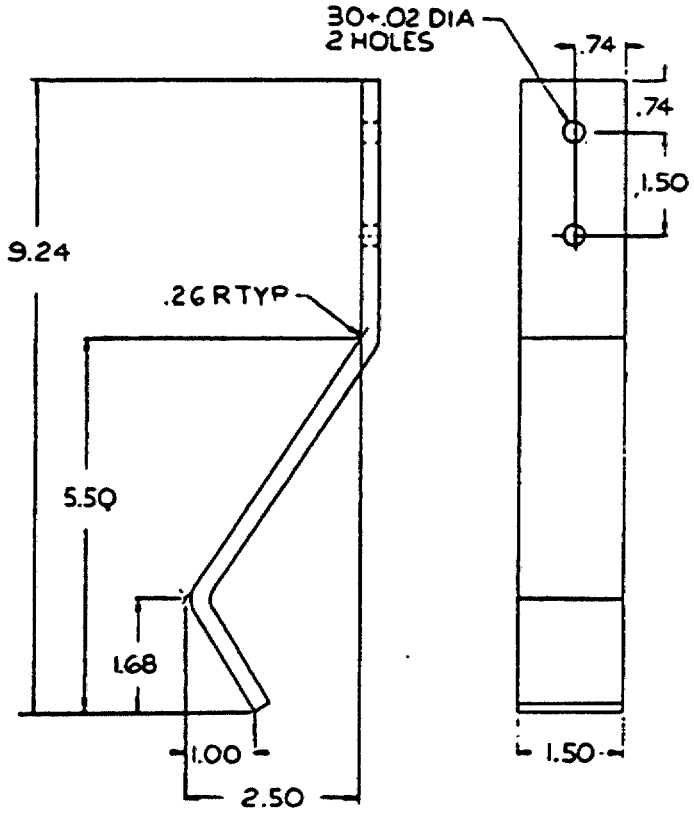
STEEL, WELD PER
CLASS 1
SPEC MIL-STD-1261



BRACKET FRONT (WELDMENT)
FIGURE 37

DRAWING NO.
F 11604880

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



MUST BE FREE OF BURRS
AND SHARP EDGES

MATERIAL NOTE
STEEL, CARBON
1010 THRU 1020
CONDITION OPTIONAL
SPEC QQ-S-630 OR 634
.250 THICK

BRACKET
FIGURE 38

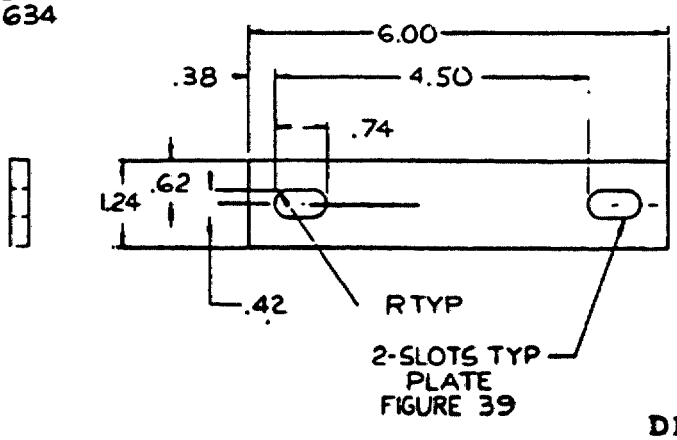


PLATE
FIGURE 39

DRAWING NO.
F 11604880

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

FIG. NO.	ARMY PART NO.	DESCRIPTION	MATERIAL	QTY REQD
40A	MS35338-63	WASHER, LOCK, 1/4	CARBON STEEL	8
40B	MS51967-3	NUT, HEX, 1/4-20UNC-2B	CARBON STEEL	8
40C	MS90728-109	SCREW, CAP, 1/2-13UNC-2A X 1	ALLOY STEEL	4
40D	MS35752-13	BOLT, SQ.NECK, 1/4-20UNC-2A X 2 1/4	CARBON STEEL	8
40E	MS63040-4	WASHER, FLAT, 1/4	CARBON STEEL	4
40F	10941915-5	WASHER, FLAT, HARDENED, 1/2	ALLOY STEEL	4
40G	MS90727-65	SCREW, CAP, 3/8-24UNF-2A X 1 3/4	ALLOY STEEL	4
40H	MS90727-112	SCREW, CAP, 1/2-20UNF-2A X 1 3/8	ALLOY STEEL	2

HARDWARE-CLOSURE FRAME INSTALLATION
FIGURE 40

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

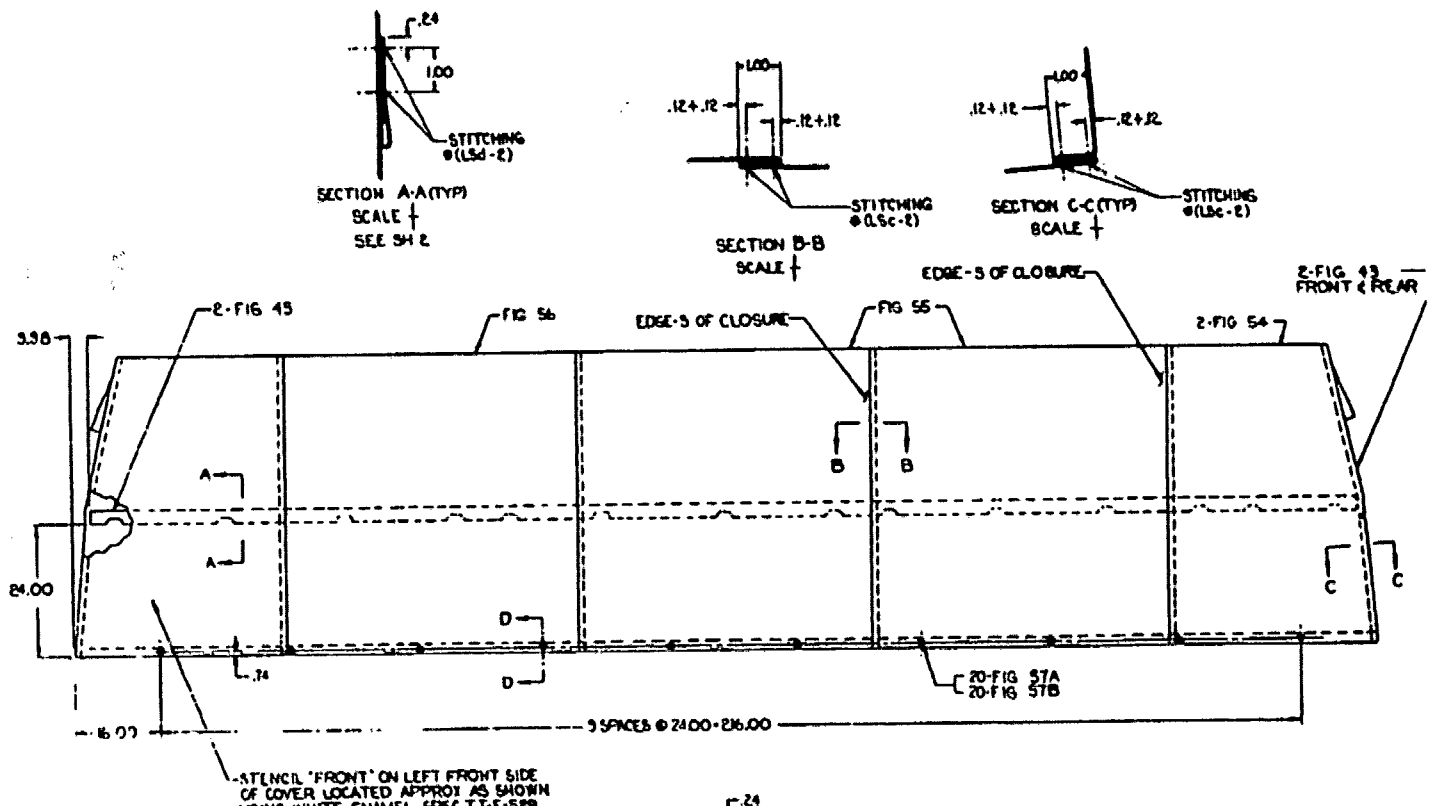
	FIG NO.	NOMENCLATURE	QTY REQD
	42	COVER,CLOSURE-LEFT SIDE ELEVATION	—
	43	COVER,CLOSURE-FRONT & REAR ELEVATION	—
	44	FLAP	4
	45	FLAP	2
	46	REINFORCEMENT	2
	47	COVER,END	2
	48	COVER,END	2
⊕	49	HOOD,VENT ASSEMBLY	2
⊕	50	HOOD,VENT	2
⊕	51	ROD ASSEMBLY	2
⊕	52	ROD,HOOD	2
⊕	53	TIP,ROD END	4
	54	COVER,SIDE	2
	55	COVER,SIDE	2
	56	COVER,SIDE	1
	57	HARDWARE-CLOSURE COVER-SEE HARDWARE LIST	

PARTS LIST-CLOSURE COVER
FIGURE 41

⊕

CONVERSION LIST	
FIG NO	PART NO.
49	10913563
50	10913537
51	10913561
52	10913542
53	10913551

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

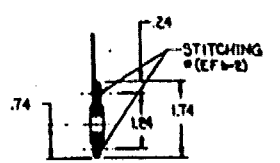


ALL PARTS TO BE IDENTIFIED BY FIGURE NUMBER

MANUFACTURER'S IDENTIFICATION MUST APPEAR ON INSIDE OF COVER

*STITCHING NOTE
STITCH CLASS 300, TYPE 301
SEAM CLASS LS, TYPE AS NOTED
STITCHING CLASS EF, TYPE AS NOTED
PER FED-STD-751
8-10 STITCHES PER INCH
USE SMALLEST NEEDLE SIZE
FEASIBLE TO ASSURE
WEATHERPROOF SEAMS

MATERIAL NOTE
THREAD, NYLON
TYPE II, CLASS 1
SIZE B COLOR OD
SPEC V-T-295



SECTION D-D
SCALE
SEE SH 2

LEFT SIDE ELEVATION

CLOSURE COVER
FIGURE 42

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

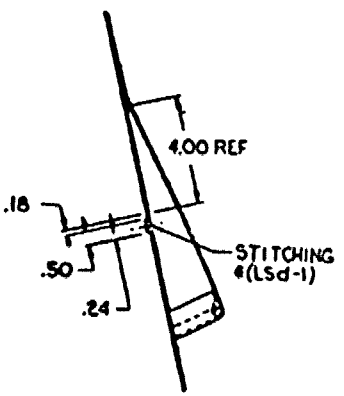
ALL PARTS TO BE IDENTIFIED BY FIGURE NUMBER

MANUFACTURER SHALL BE RESPONSIBLE FOR FINAL FIT OF COVER

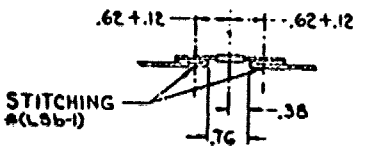
SEAM ALLOWANCE MAY VARY TO SUIT MANUFACTURER

STITCHING NOTE
STITCH CLASS 300, TYPE 301
SEAM CLASS SS, TYPE AS NOTED
SEAM CLASS LS, TYPE AS NOTED
PER FED-STD-751
8-10 STITCHES PER INCHES
USE SMALLEST NEEDLE SIZE FEASIBLE TO ASSURE WEATHERPROOF SEAMS

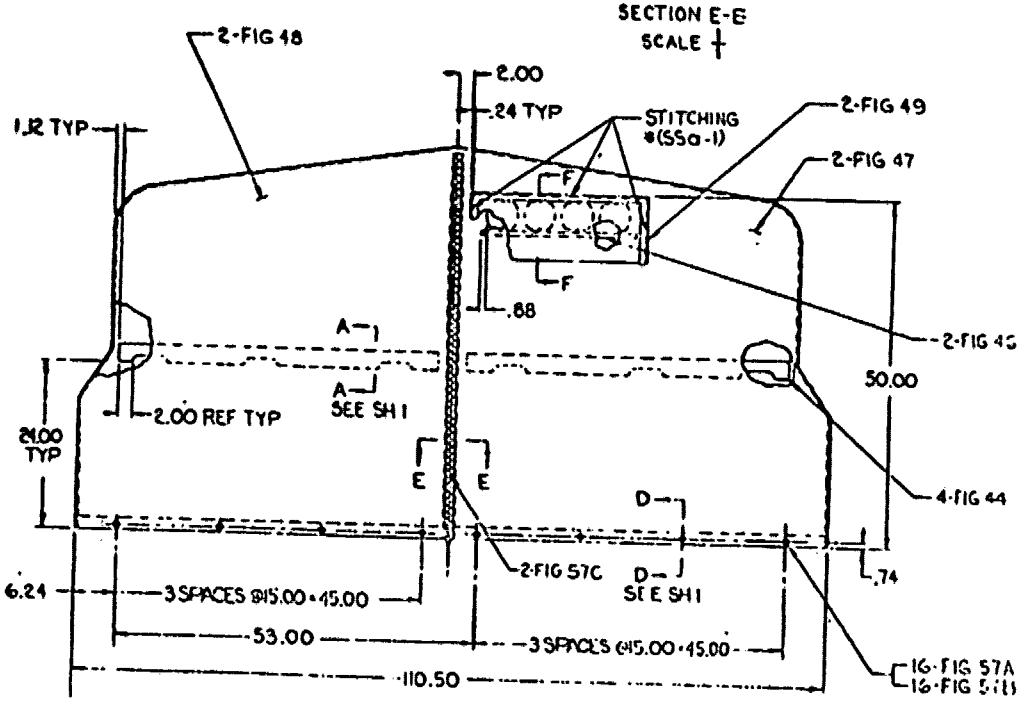
MATERIAL NOTE
THREAD, NYLON
TYPE II CLASS 1
SIZE B COLOR OD
SPEC V-T-295



SECTION F-F
SCALE 1/2



SECTION E-E
SCALE 1

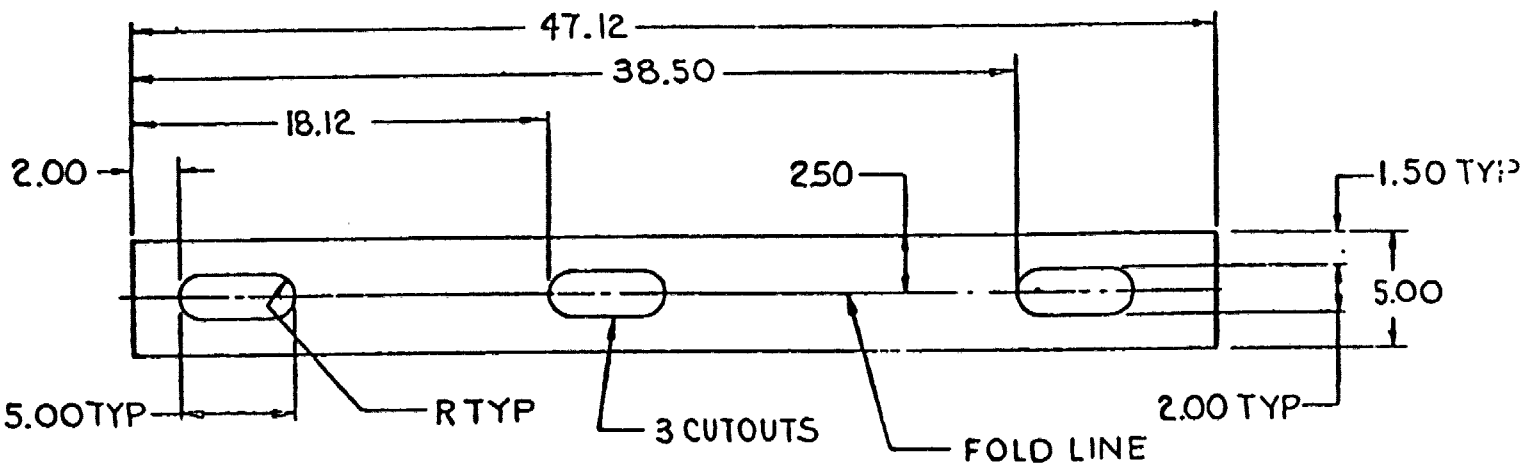


FRONT AND REAR ELEVATION

COVER, CLOSURE
FIGURE 43

DRAWING NO.
F 11604681

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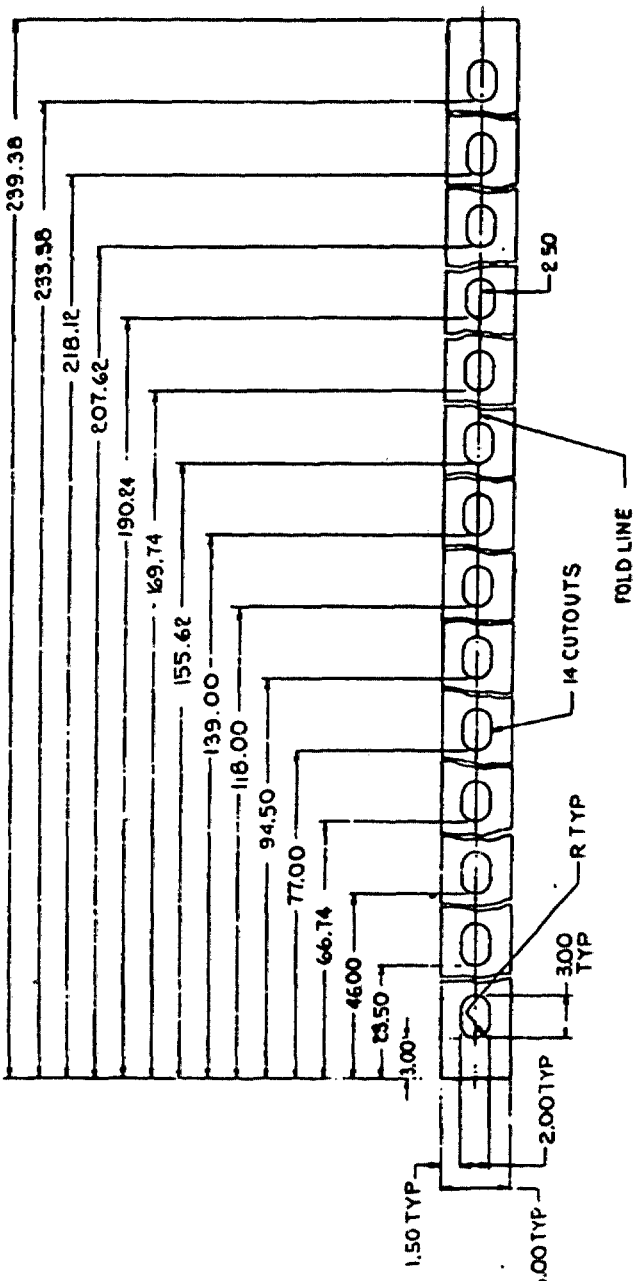


FLAP, FRONT AND REAR, CLOSURE COVER
FIGURE 44

MATERIAL NOTE
CLOTH, COATED NYLON
TYPE II CLASS 3
BALANCED COATING
SPEC MIL-C-20696
MIN. OVERALL WT TO BE
14 OZ PER SQ YD
COLOR OD NO. X34087 PER
FEDERAL-STD-595

DRAWING NO.
F 11604881

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Check the source to verify that this is the current version before use.

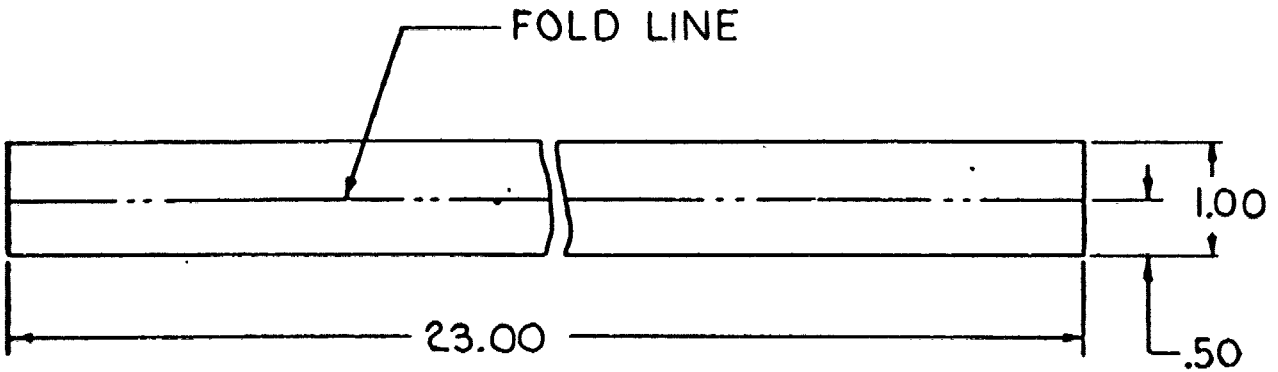


FLAP, SIDE, CLOSURE COVER
FIGURE 45

MATERIAL NOTE
CLOTH, COATED NYLON
TYPE II CLASS 3
BALANCED COATING
SPEC MIL-C-20696
MIN. OVERALL WT TO BE
14 OZ PER SQ YD
COLOR: CC NO X34087 PER
FEDERAL-STD-595

INTERPRET DRAWING IN ACCORDANCE WITH
STANDARDS PRESCRIBED BY MIL-D-1000

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

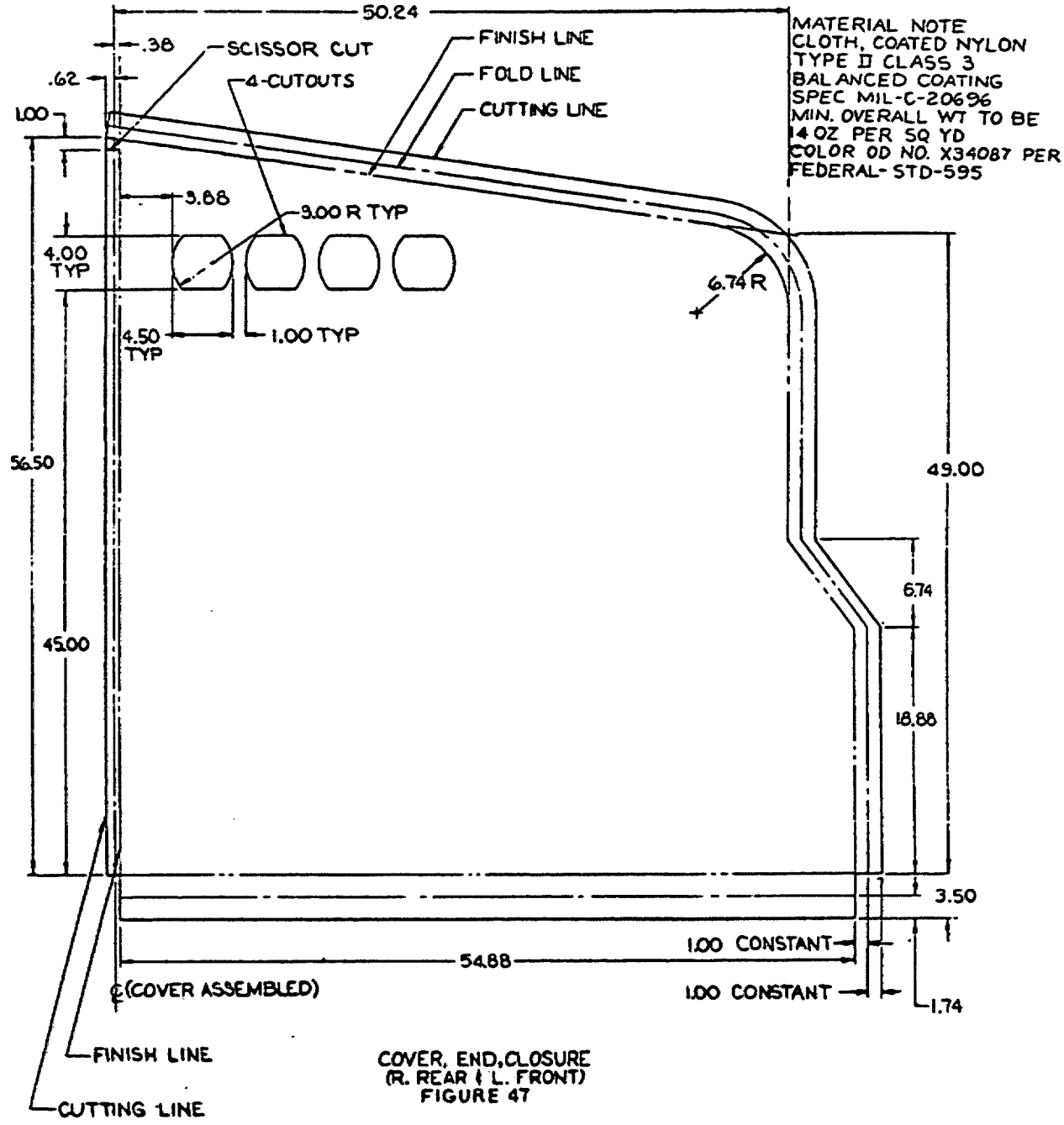


MATERIAL NOTE
CLOTH, COATED NYLON
TYPE II CLASS 3
BALANCED COATING
SPEC MIL-C-20696
MIN. OVERALL WT TO BE
14 OZ PER SQ YD
COLOR OD NO. X34087 PER
FEDERAL-STD-595

REINFORCEMENT, CLOSURE COVER
FIGURE 46

DRAWING NO.
F 11604881

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

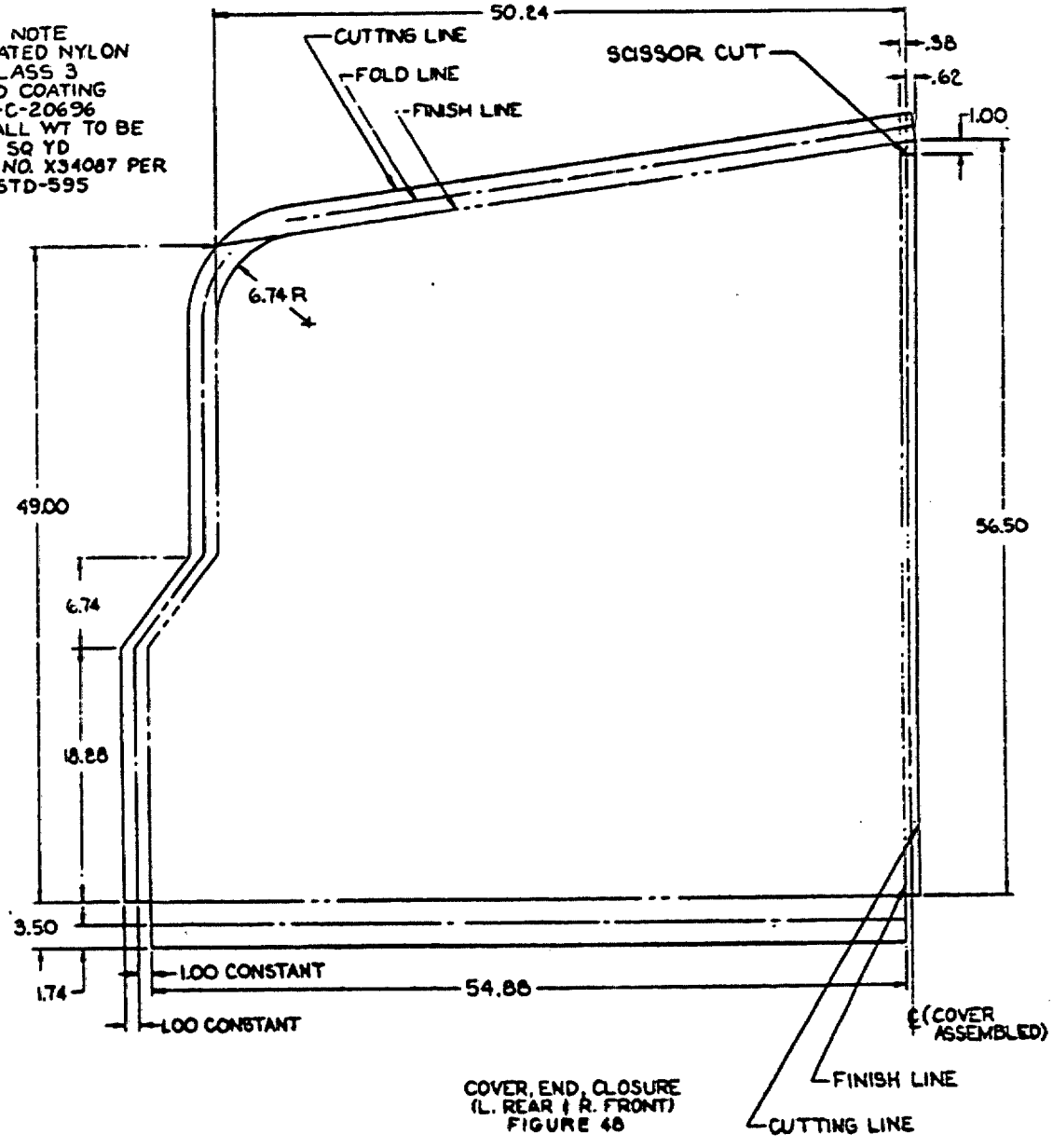


COVER, END, CLOSURE
(R. REAR & L. FRONT)
FIGURE 47

DRAWING NO.
F 11604881

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

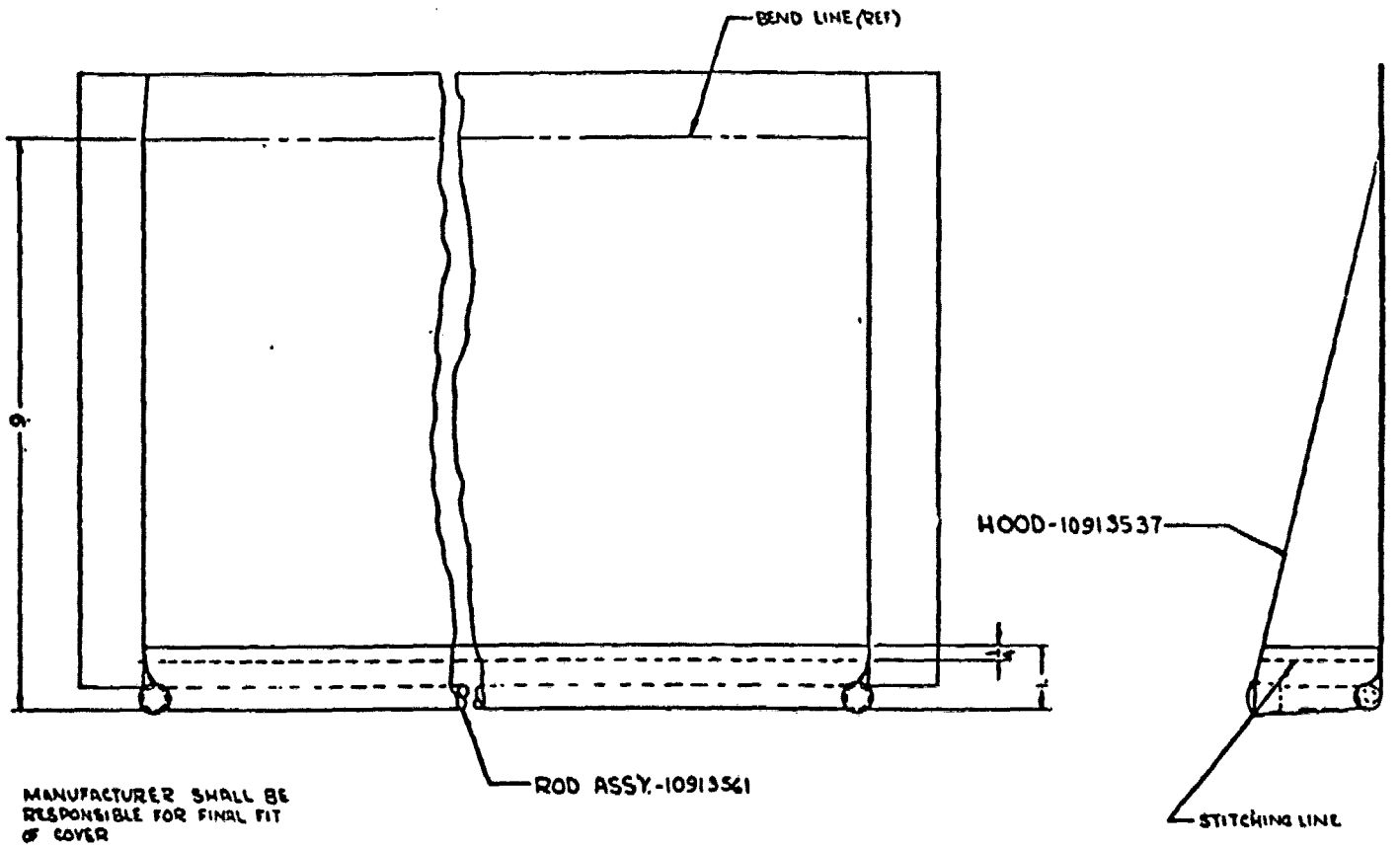
MATERIAL NOTE
CLOTH, COATED NYLON
TYPE II CLASS 3
BALANCED COATING
SPEC MIL-C-20696
MIN. OVERALL WT TO BE
14 OZ PER SQ YD
COLOR OD NO. X34087 PER
FEDERAL-STD-595



COVER, END, CLOSURE
(L. REAR & R. FRONT)
FIGURE 48

DRAWING NO.
F 11604881

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



MANUFACTURER SHALL BE RESPONSIBLE FOR FINAL FIT OF COVER

SEAM ALLOWANCE MAY VARY TO SUIT MANUFACTURER

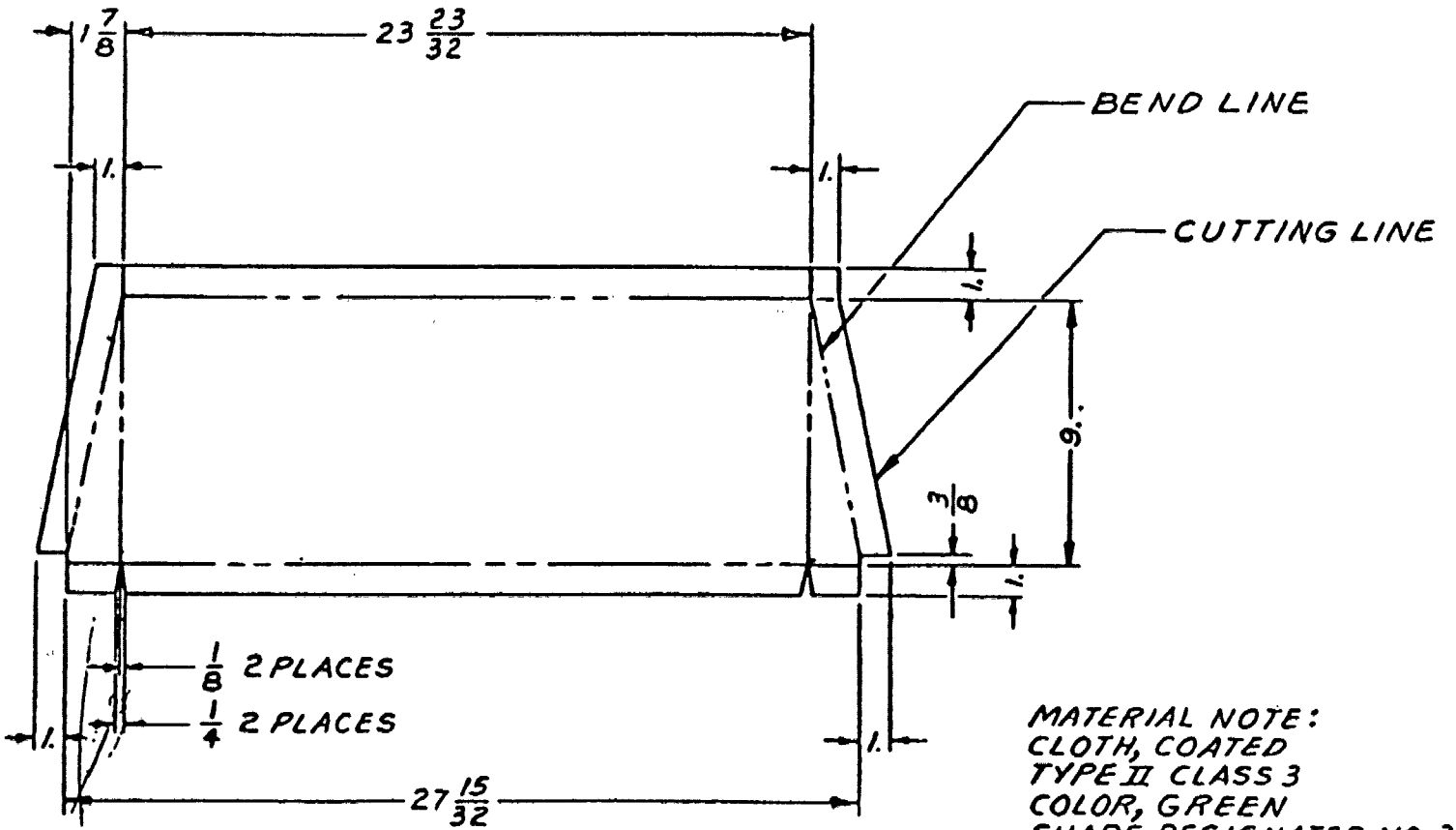
STITCHING TYPE 301
FED STD 751
6-10 STITCHES PER INCH SMALLEST NEEDLE SIZE FEASIBLE SHOULD BE USED IN ORDER TO ASSURE WEATHER PROOF SEAMS

THREAD NYLON
TYPE II CLASS I
AZEF. COLOR O.D.
SPEC V-T-295

HOOD, VENT
CLOSURE, VEHICLE
(SEWN)
FIGURE 49

PART NUMBER
10913563

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
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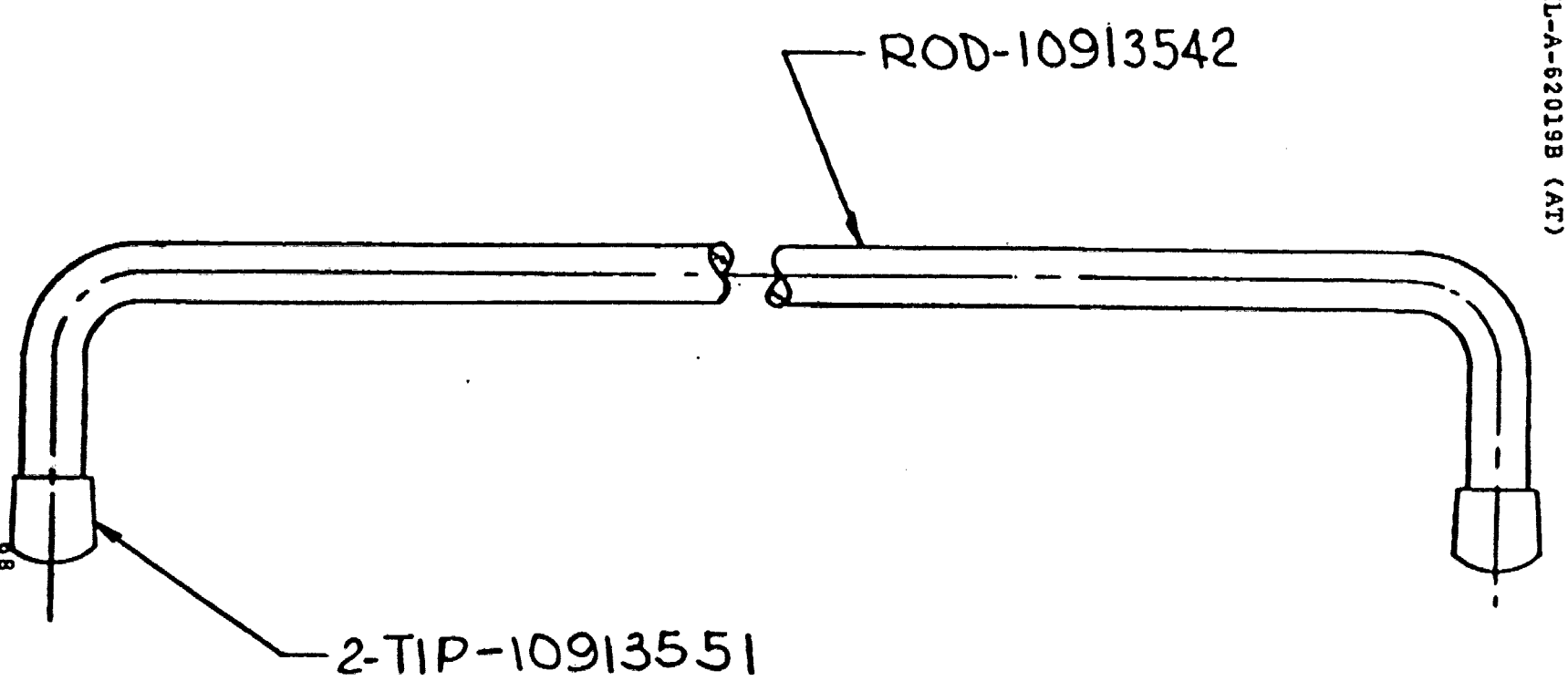


MATERIAL NOTE:
CLOTH, COATED
TYPE II CLASS 3
COLOR, GREEN
SHADE DESIGNATED NO. 34087
OF STANDARD FEDERAL STD. 595
SPEC. MIL-C-20696
WAIVE PAR. 3.5

Hood, Vent,
Closure, Vehicle
(Sewn)
FIGURE 50

PART NUMBER
10913537

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

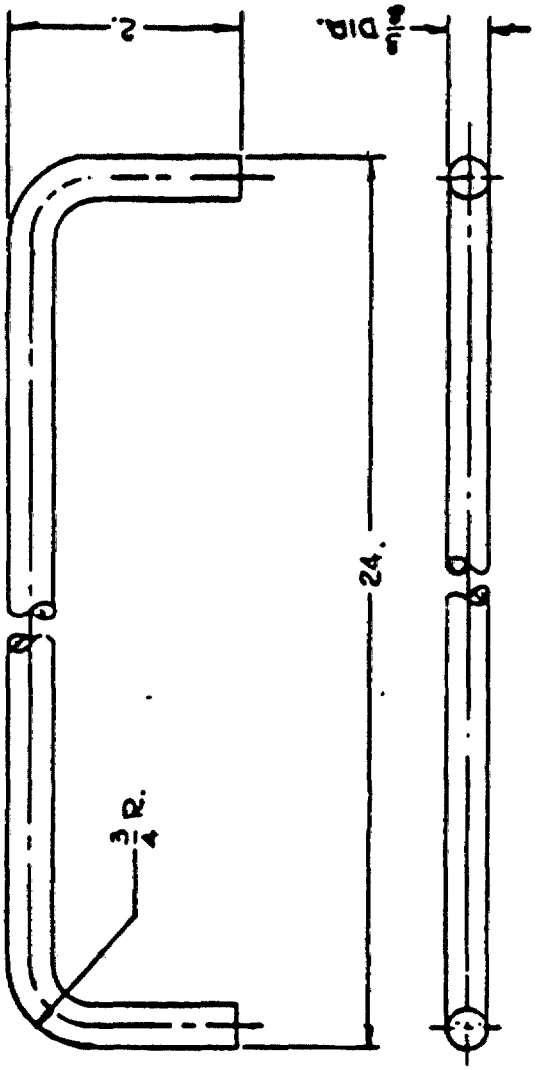


ROD ASSEMBLY,
HOOD, VENT,
CLOSURE, VEHICLE

FIGURE 51

PART NUMBER
10913561

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



MUST BE FREE OF BURRS & SHARP EDGES

OPTIONAL MATERIAL
ALUMINUM ALLOY(6061)
TEMPER O, TG AFTER FORMING
SPEC QQ-A-225/8

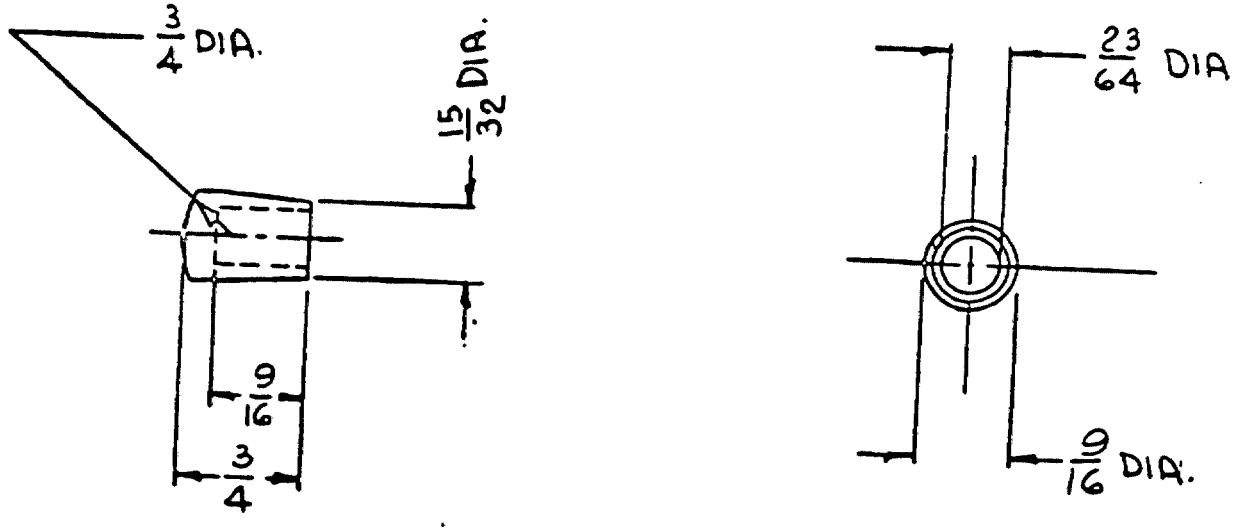
MATERIAL NOTE:
ALUMINUM ALLOY(6061 & 6062)
TEMPER O, TG AFTER FORMING
SPEC QQ-A-200/8

PART NUMBER
10913542

ROD, HOOD,
VENT, CLOSURE,
VEHICLE

FIGURE 52

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



NOTE: MARK ITEM WITH CAPTIAL LETTERS " OZ " SYMBOL (STAMPED OR MOULDED)

MATERIAL NOTE:
RUBBER
GRADE SC615B, C, F_2
SPEC. MIL-R-3065

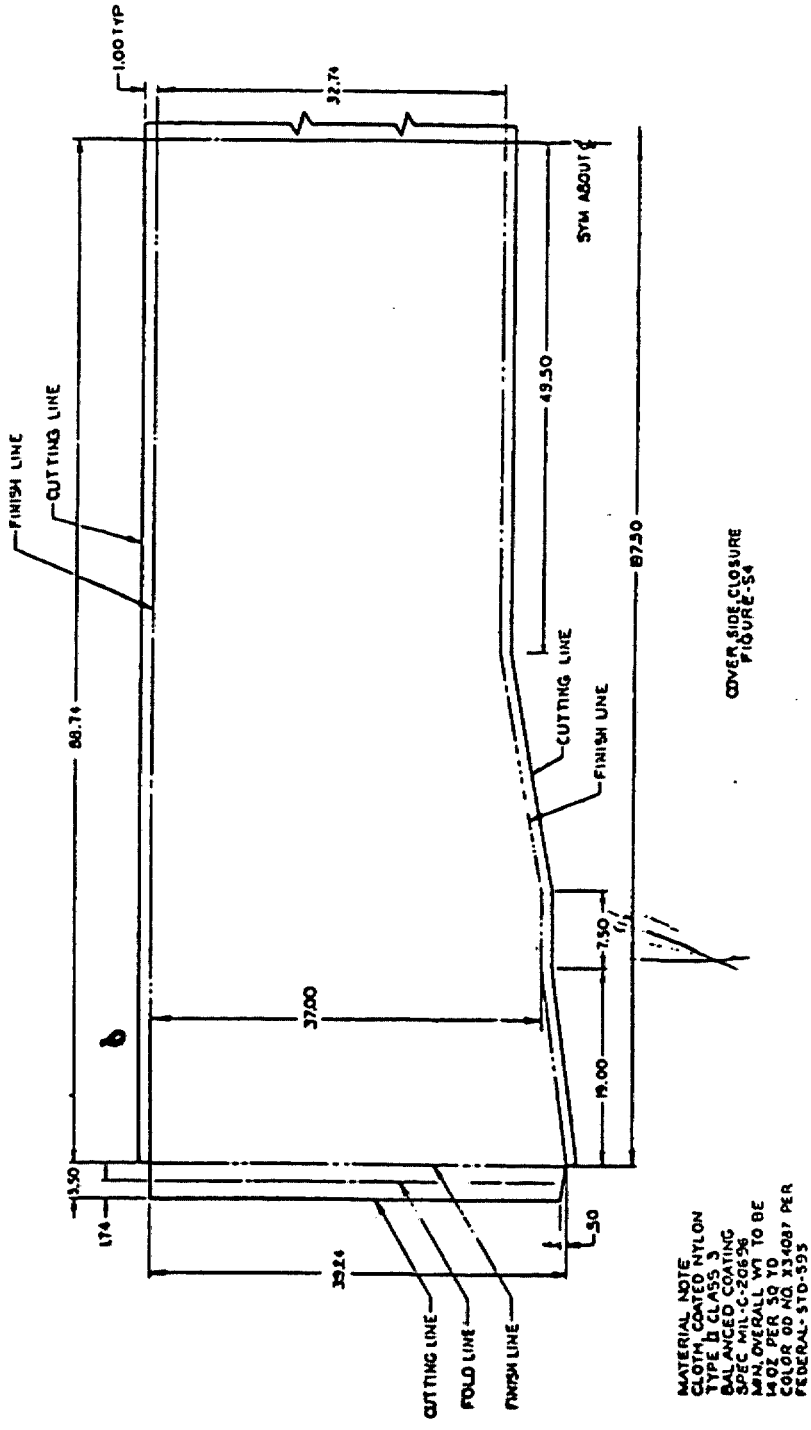
TIP, ROD END

FIGURE 53

PART NUMBER
10913551

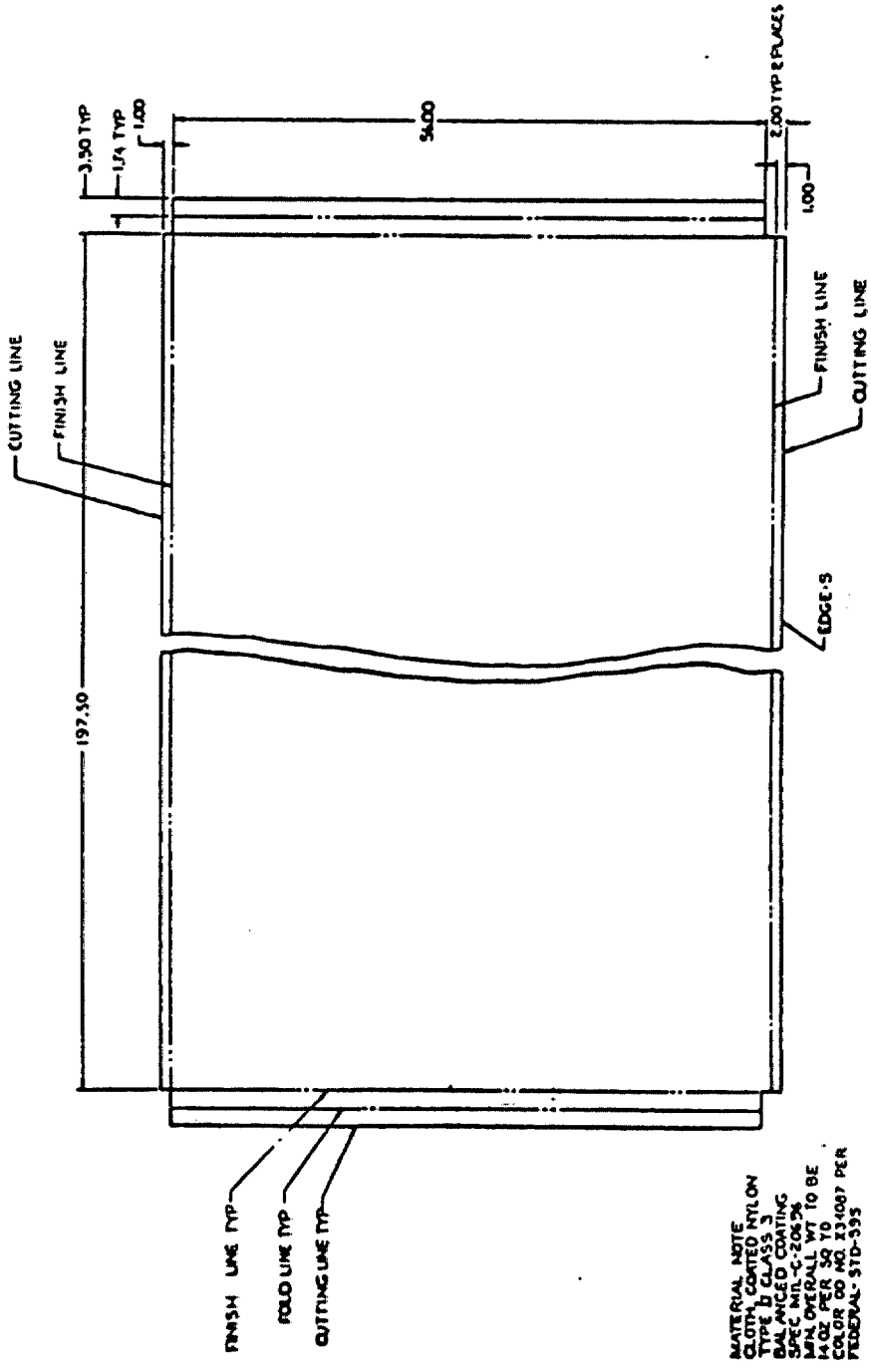
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

DRAWING NO.
F 11604881

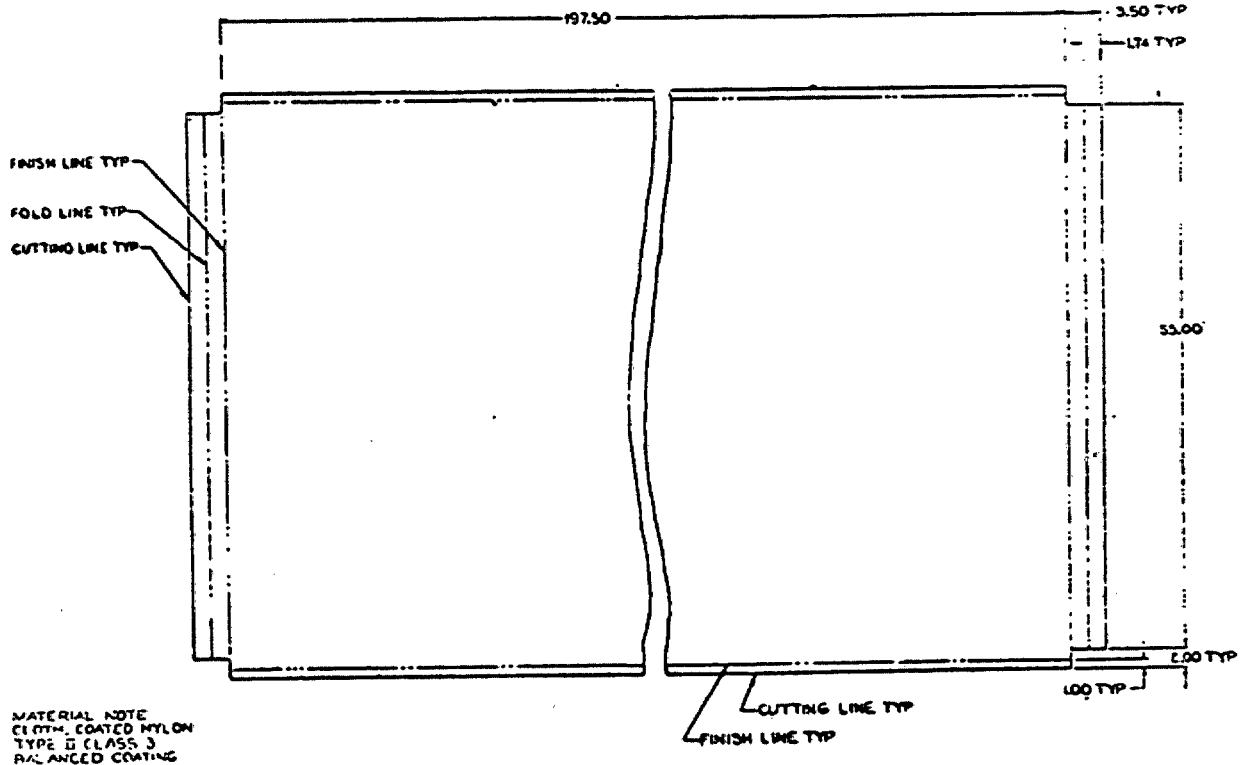


Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

DRAWING NO.
F 11604881



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



MATERIAL NOTE
CLOTH, COATED NYLON
TYPE II CLASS 3
BALANCED COATING
SPEC MIL-C-20696
MIN OVERALL WT TO BE
1.4 OZ PER SQ YD
COLOR OD NO. 34087 PER
FEDERAL-STD-595

COVER, SIDE CLOSURE
FIGURE 56

DRAWING NO.
F 11604331

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Check the source to verify that this is the current version before use.

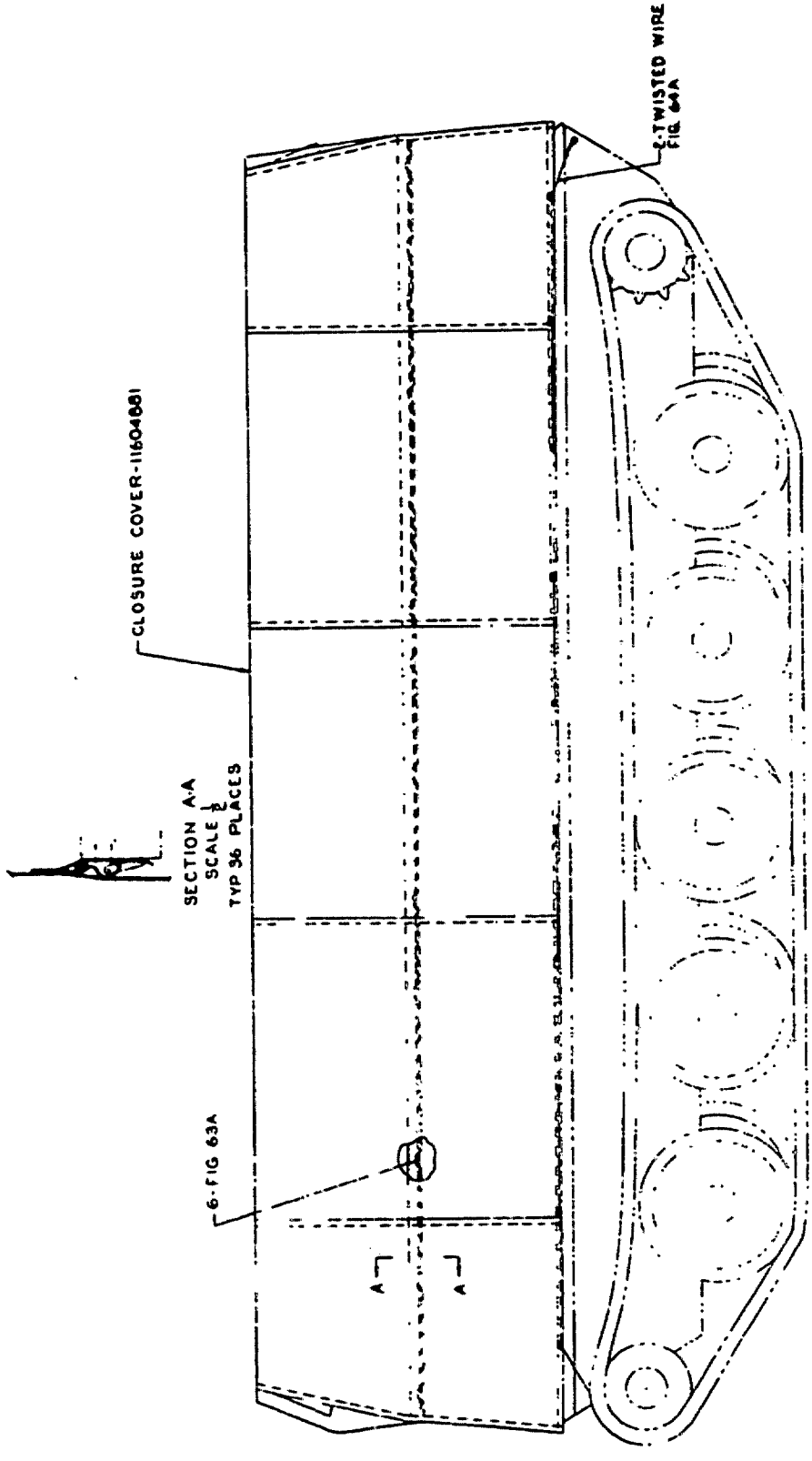
FIG NO.	QTY REQD	ARMY PART NO.	DESCRIPTION
57A	36	MS20230-WBS3	WASHER
57B	36	MS20230-GBS3	GROMMET.
57C	2	538458	FASTENER, SLIDE

HARDWARE-CLOSURE COVER
FIGURE 57

FIG. NO.	NOMENCLATURE	QTY. REQD.
59	COVER INSTALLATION-LEFT SIDE ELEV.	—
60	COVER INSTALLATION-REAR ELEVATION	—
61	COVER INSTALLATION-FRONT ELEVATION	—
62	COVER INSTALLATION-PLAN VIEW	—
63	ROD-TABULATED DRAWING	—
64	HARDWARE-SEE HARDWARE LIST	—

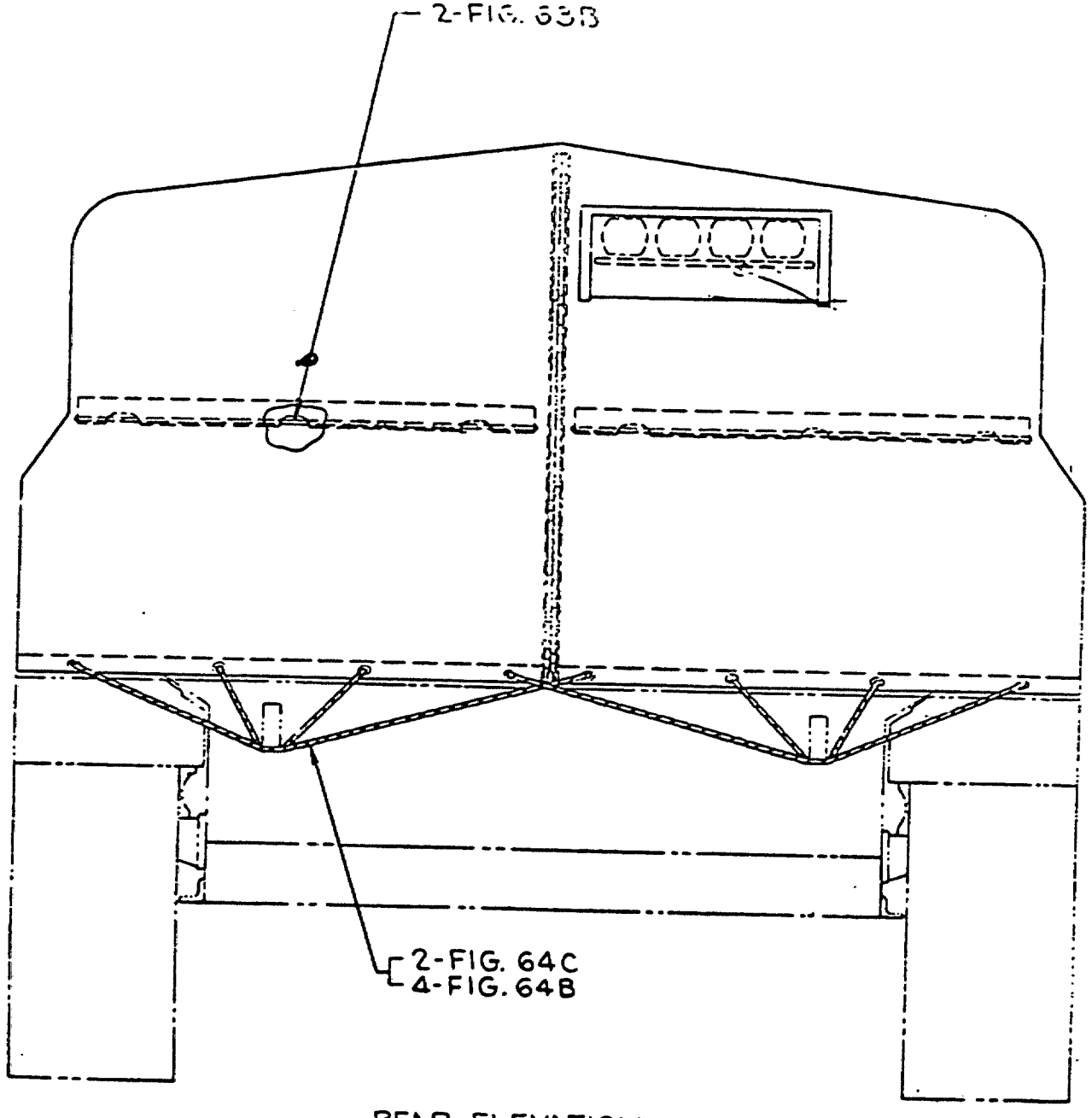
PARTS LIST-CLOSURE COVER INSTALLATION
FIGURE 58

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



LEFT SIDE ELEVATION
CLOSURE COVER INSTALLATION
FIGURE 59

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Check the source to verify that this is the current version before use.

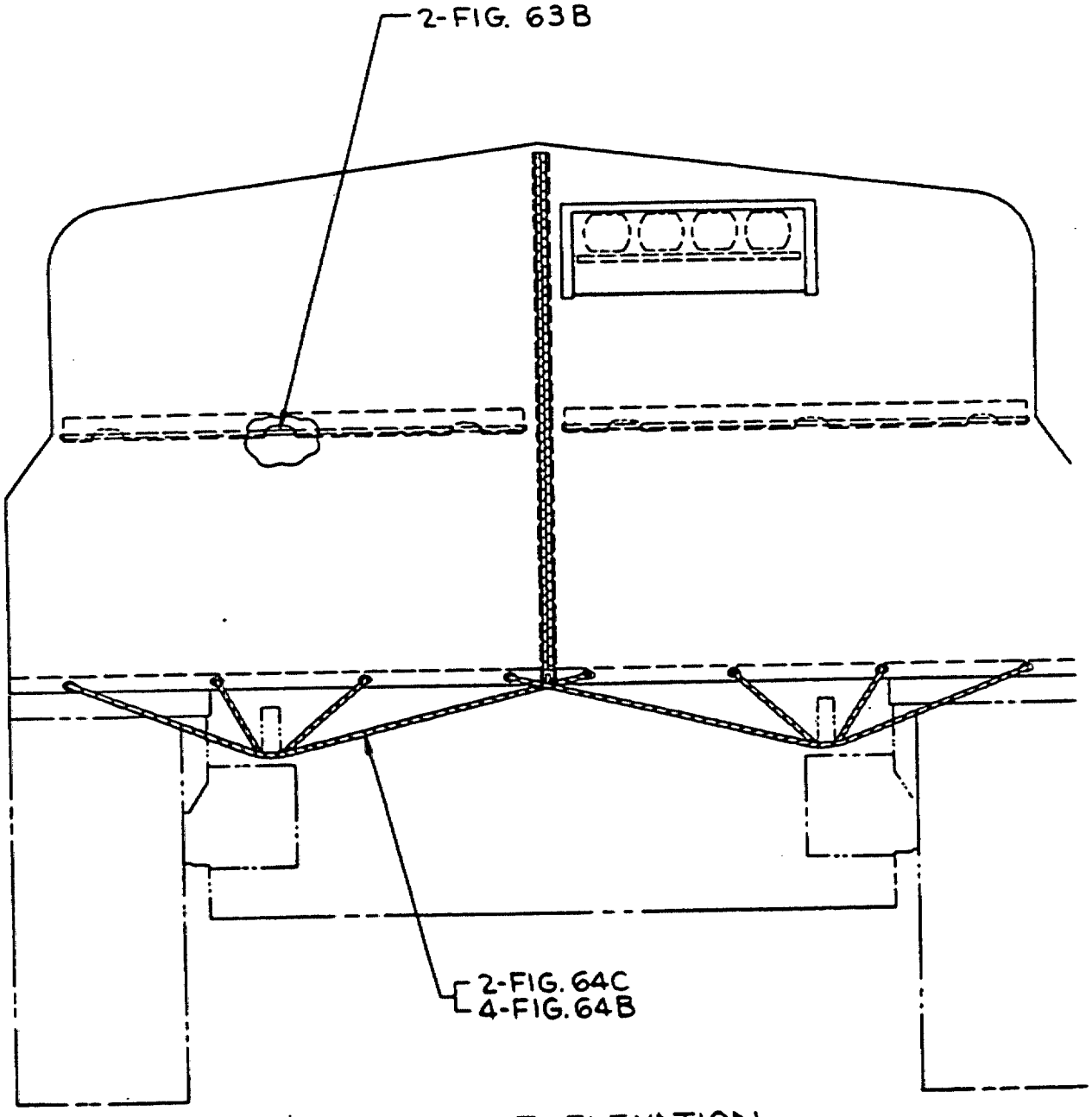


REAR ELEVATION

CLOSURE COVER INSTALLATION
FIGURE 60

DRAWING NO.
F 11604882

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Check the source to verify that this is the current version before use.



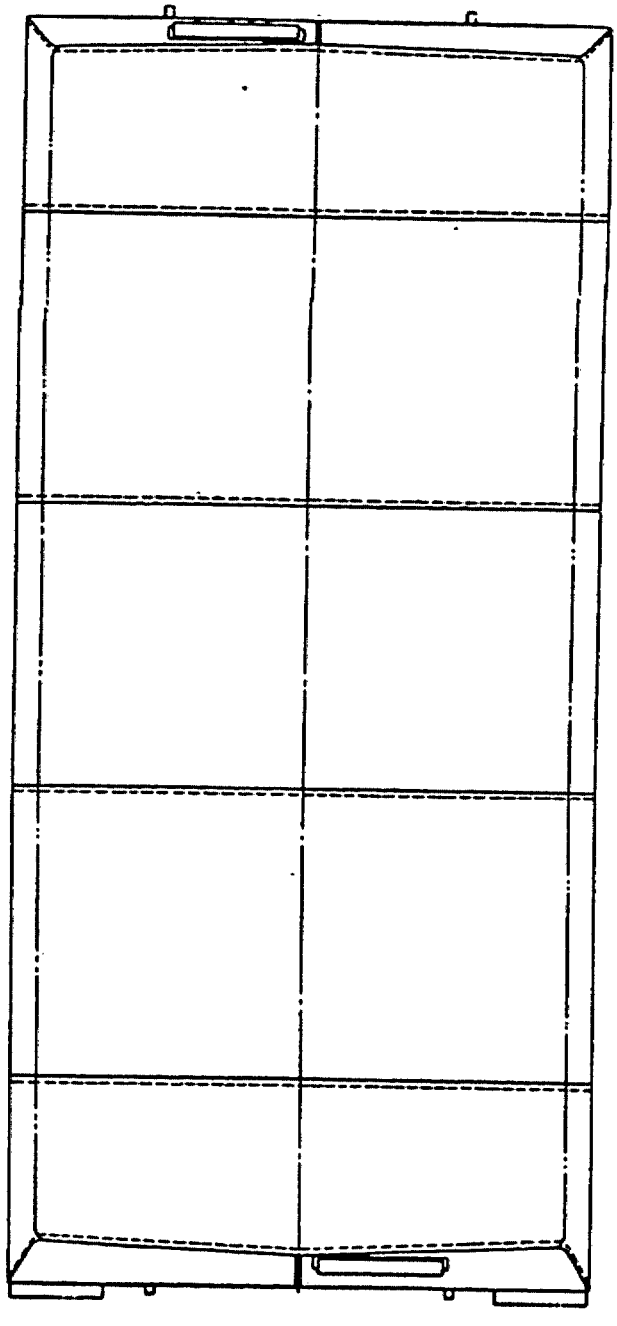
FRONT ELEVATION

CLOSURE COVER INSTALLATION
FIGURE 61

DRAWING NO.
F 11604882

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DRAWING NO.
F11604882



PLAN VIEW

CLOSURE COVER INSTALLATION
FIGURE 62

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

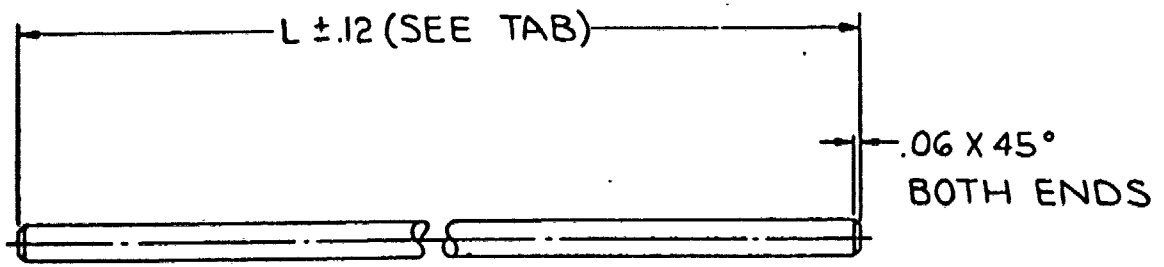


FIG. NO.	L INCHES
63A	79
63B	47.62

ROD, COVER, CLOSURE
FIGURE 63

FINISH NOTE
TREAT PER TYPE I OR III
SPEC TT-C-490
PRIME PER
SPEC TT-P-636
ENAMEL, O.D.
SPEC TT-E-529 OR TT-E-485

MUST BE FREE OF BURRS
AND SHARP EDGES

MATERIAL NOTE
STEEL, CARBON
1010 THRU 1020
SPEC QQ-S-634 OR 630
.375 DIA

DRAWING NO.
11604882

MIL-A-62019B (AT)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
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FIG. NO.	QTY. REQD.	PART NO.	DESCRIPTION
64A	AS REQD	MS20995-F91	WIRE, GALVANIZED, SPEC QQ-W-461
64B	AS REQD	7359274	CLAMP, ROPE
64C	AS REQD	SEE DWG 8710565	ROPE, COTTON, TWISTED, FEDERAL STOCK NO. 21-R-60

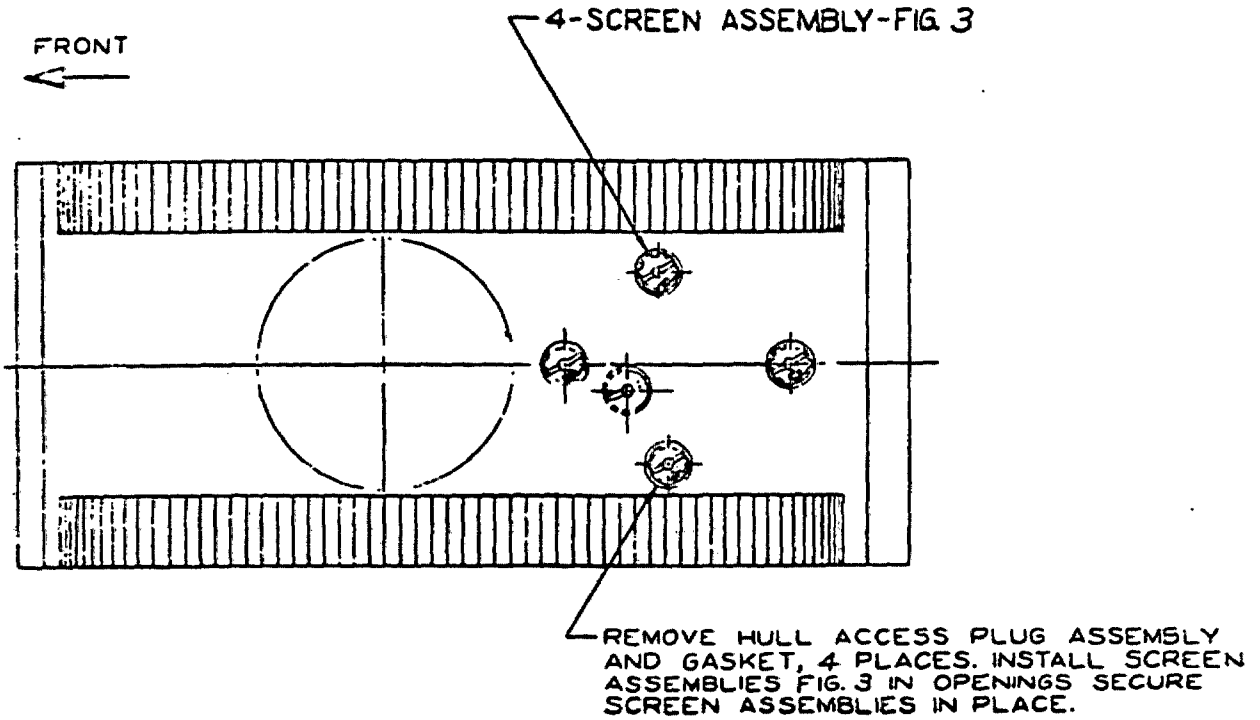
HARDWARE - CLOSURE COVER INSTALLATION
FIGURE 64

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

FIG. NO.	NOMENCLATURE	QTY REQD	PART NO.
2	BOTTOM VIEW OF VEH	-	-
3	SCREEN ASSEMBLY	4	
4	RETAINER (WELDMENT)	4	
5	BAR	4	
6	RING	4	
7	SPACER	4	
8	SCREEN	4	
9	PLATE	4	10953399
10	HARDWARE - SEE HARDWARE LIST		

PARTS LIST- VENTILATION KIT
FIGURE 1

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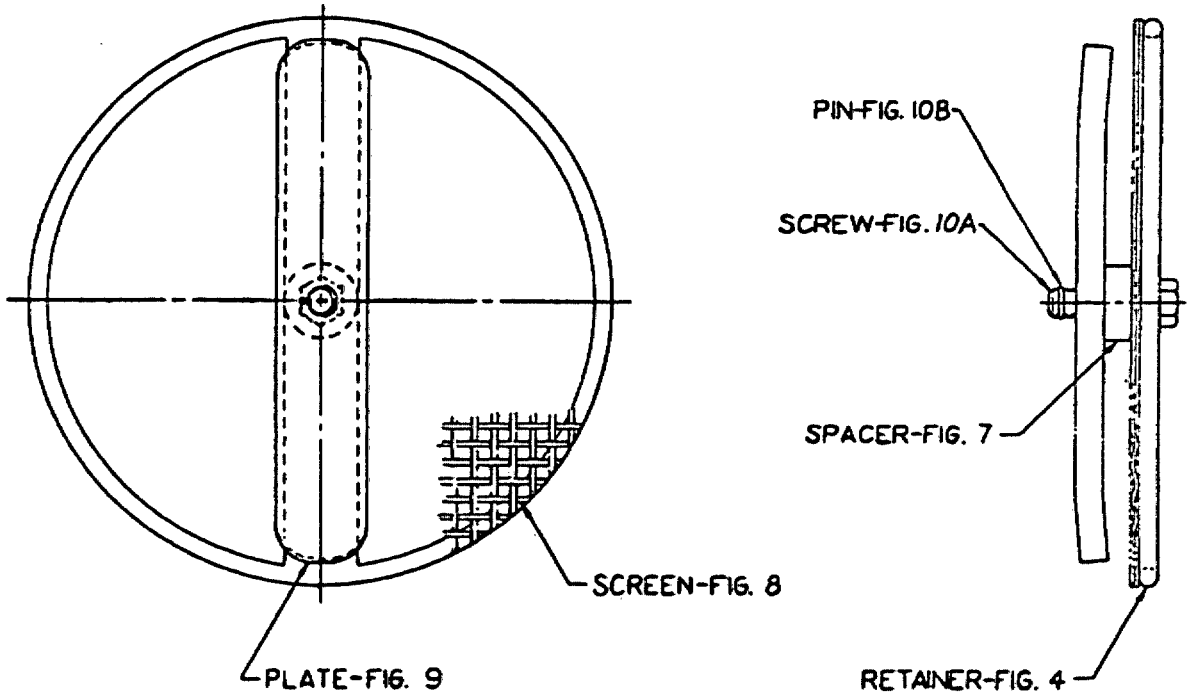


BOTTOM VIEW OF VEHICLE
FIG. 2

UNPAINTED METAL SURFACES OF PLUG ASSEMBLIES SHALL BE COATED WITH PRESERVATIVE CONFORMING TO GRADE 2 OF SPEC MIL-C-16173. PLUG ASSEMBLIES AND GASKETS SHALL BE PLACED IN A BOX CONFORMING TO SPEC PPP-B-636 IDENTIFIED, AND STOWED WITHIN VEHICLE IN A MANNER TO PREVENT MOVEMENT IN TRANSIT. USING WHITE OR YELLOW PAINT CONFORMING TO SPEC MIL-P-13983, THE FOLLOWING INFORMATION SHALL BE STENCILED ON THE EXTERIOR OF THE VEHICLE "REMOVE SCREEN ASSEMBLIES, INSTALL ACCESS PLUG ASSEMBLIES AND GASKETS BEFORE VEHICLE OPERATION."

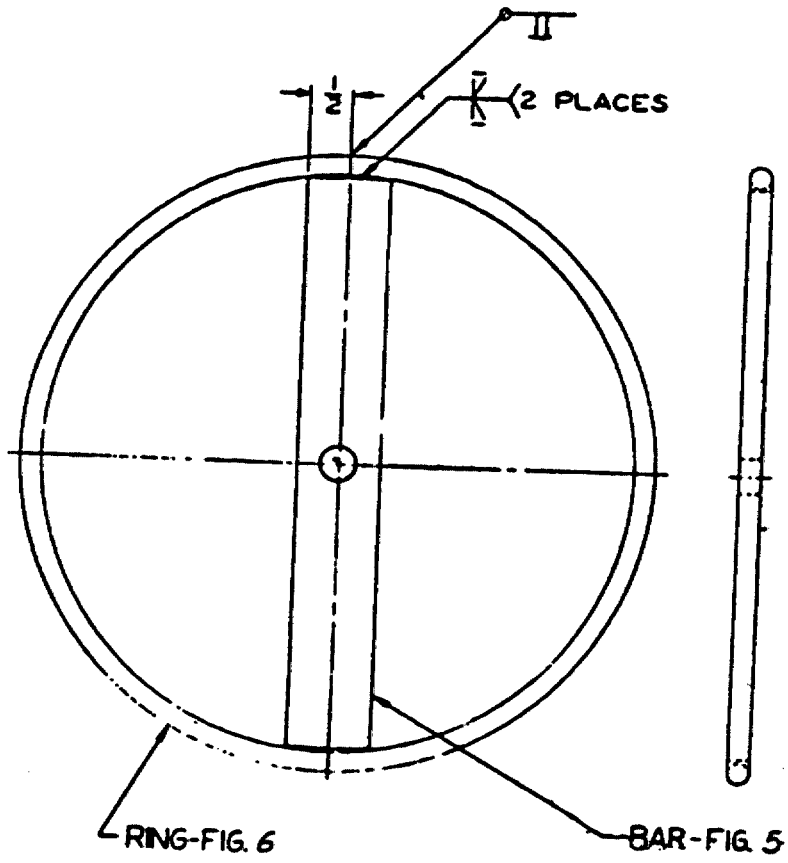
10954535

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Check the source to verify that this is the current version before use.



SCREEN ASSEMBLY-FIG. 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



RETAINER (WELDMENT)-FIG. 4

FINISH NOTE
TREAT PER TYPE I OR III
SPEC TT-C-490
PRIME PER
SPEC TT-P-636
ENAMEL OD
SPEC TT-E-529 OR TT-E-485

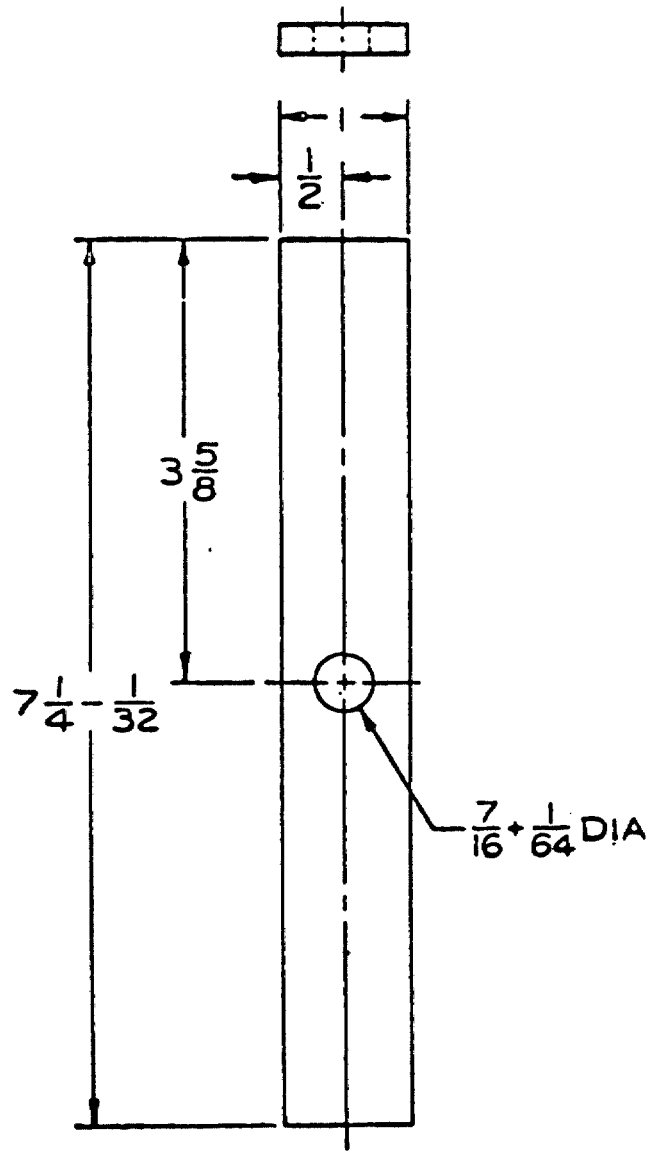
STEEL, WELD PER
MIL-STD-1261

MUST BE FREE OF BURRS
AND SHARP EDGES

DIMENSIONS AND TOLERANCES
PER MIL-STD-8

10954539

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
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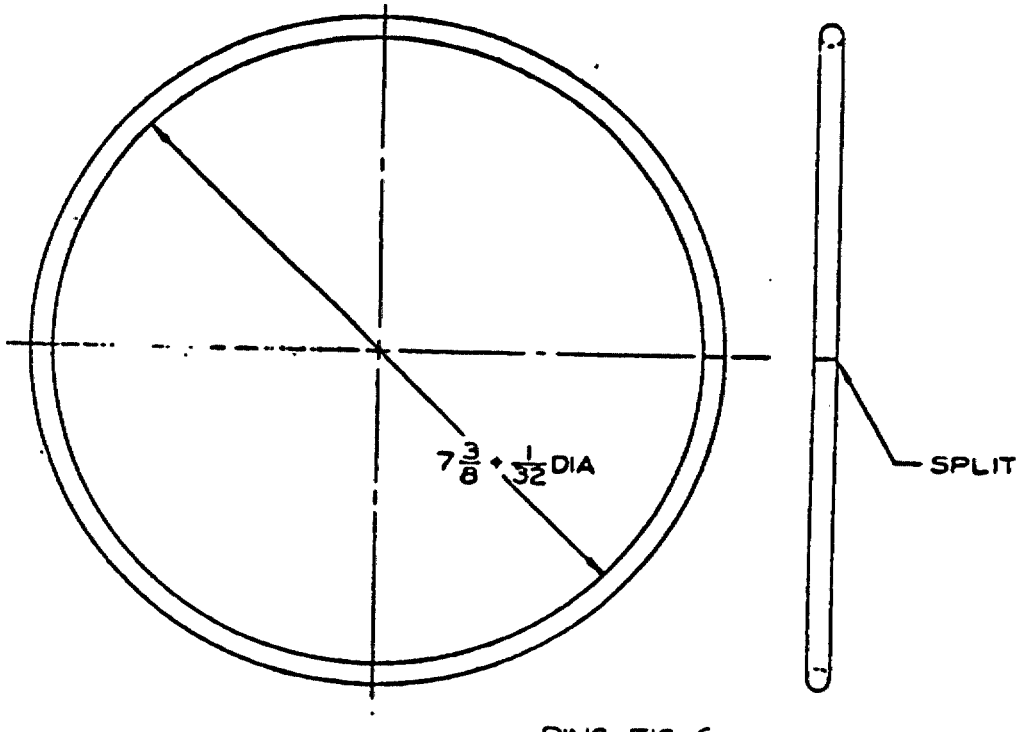


BAR - FIG. 5

MATERIAL NOTE
STEEL, CARBON
COMPOSITION C1010-C1025
CONDITION OPTIONAL
SPEC QQ-S-634
.250 THICK

10954538

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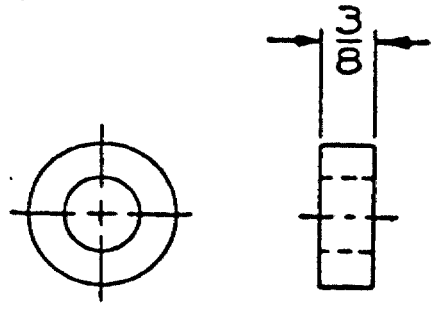
RING - FIG. 6

MATERIAL NOTE
STEEL, CARBON
COMPOSITION C1010-C1025
CONDITION OPTIONAL
SPEC QQ-S-634
.250 DIA

MUST BE FREE OF BURRS
AND SHARP EDGES

DIMENSIONS AND TOLERANCES
PER MIL-STD-8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.



SPACER-FIG. 7

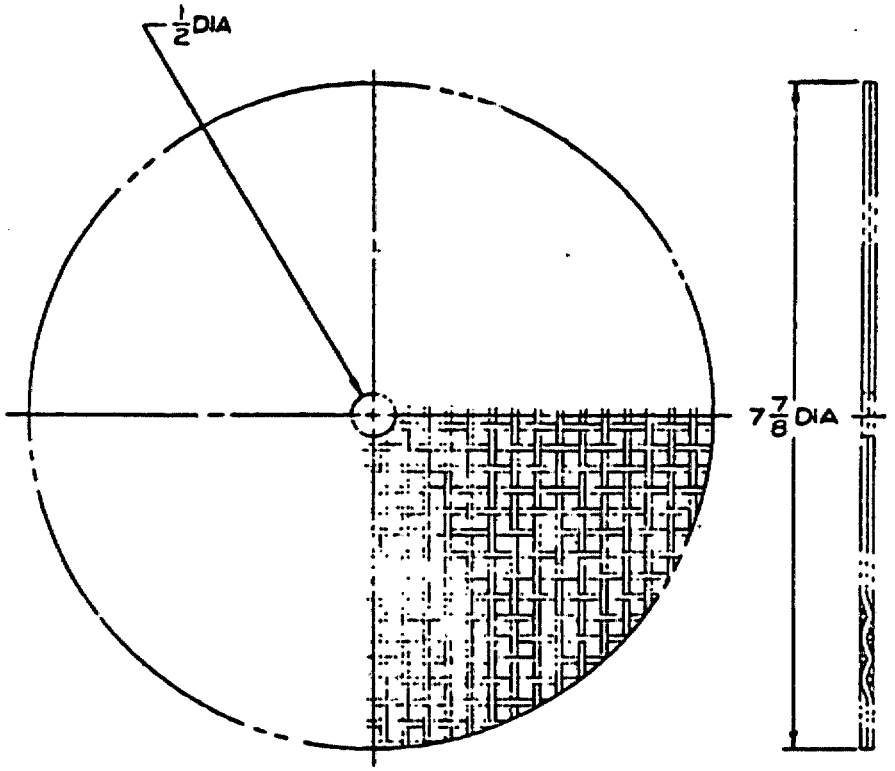
FINISH NOTE
TREAT PER TYPE I OR III
SPEC TT-C-490
PRIME PER
SPEC TT-P-636
ENAMEL OD
SPEC TT-E-529 OR TT-E-485

OPTIONAL MATERIAL
STEEL, CARBON
COMPOSITION OPTIONAL
CONDITION OPTIONAL
SPEC QQ-S-634
1.000 DIA

MATERIAL NOTE
TUBE, CARBON STEEL
COMPOSITION MT1010-1025
TYPE SEAMLESS
CONDITION OPTIONAL
SPEC QQ-T-830
1.000 OD x.250 WALL THICKNESS

10954533

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
Check the source to verify that this is the current version before use.

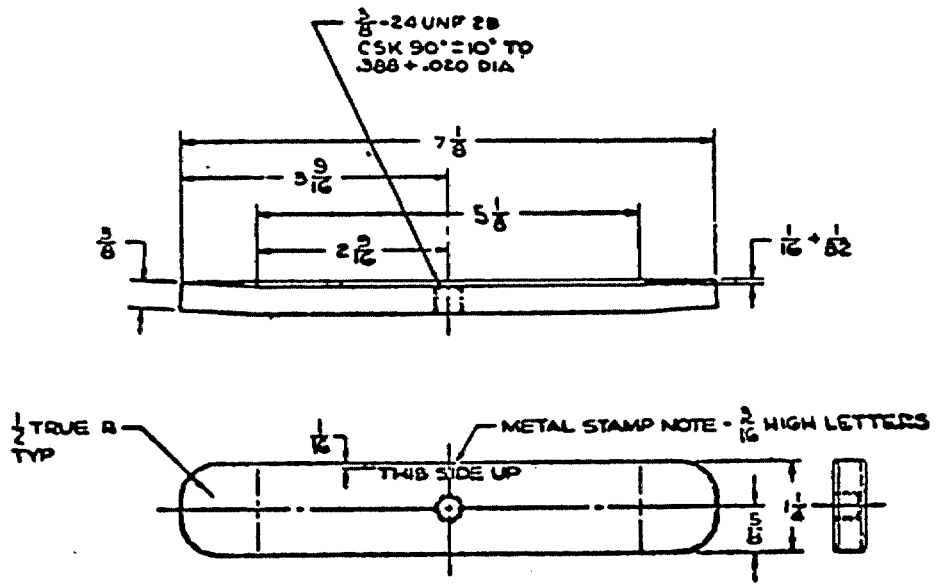


SCREEN-FIG. 8

MATERIAL NOTE
WIRE CLOTH
STEEL, LOW CARBON
HOT GALVANIZED
AFTER WEAVING
4×4 MESH
TYPE I CLASS I
SPEC RR-W-360
.047 DIA WIRE

10994535

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FINISH NOTE
PLATING, CADMIUM
CLASS 2 TYPE II
SPEC QQ-P-416
EMBRITTLEMENT RELIEVE
IMMEDIATELY AFTER PLATING

MUST BE FREE OF
BURRS AND SHARP EDGES

MATERIAL NOTE
STEEL, CARBON
C1060 THRU C1095
SPEC QQ-S-681

THREADS PER MIL-STD-9

DIMENSIONS AND TOLERANCES
PER MIL-STD-8

PLATE
FIGURE 9

10953599

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FIG. NO.	QTY. REQ'D	PART NO.	DESCRIPTION	MATERIAL
10A	4	MS51108-62	SCREW, MACHINE - 3/8-24UNF-2A X 1-1/2	ALLOY STEEL
10B	4	MS24665-283	PIN, COTTER - 3/32 DIA. X 3/4 LONG	CARBON STEEL

HARDWARE - VENTILATION KIT ASSEMBLY
FIGURE 10

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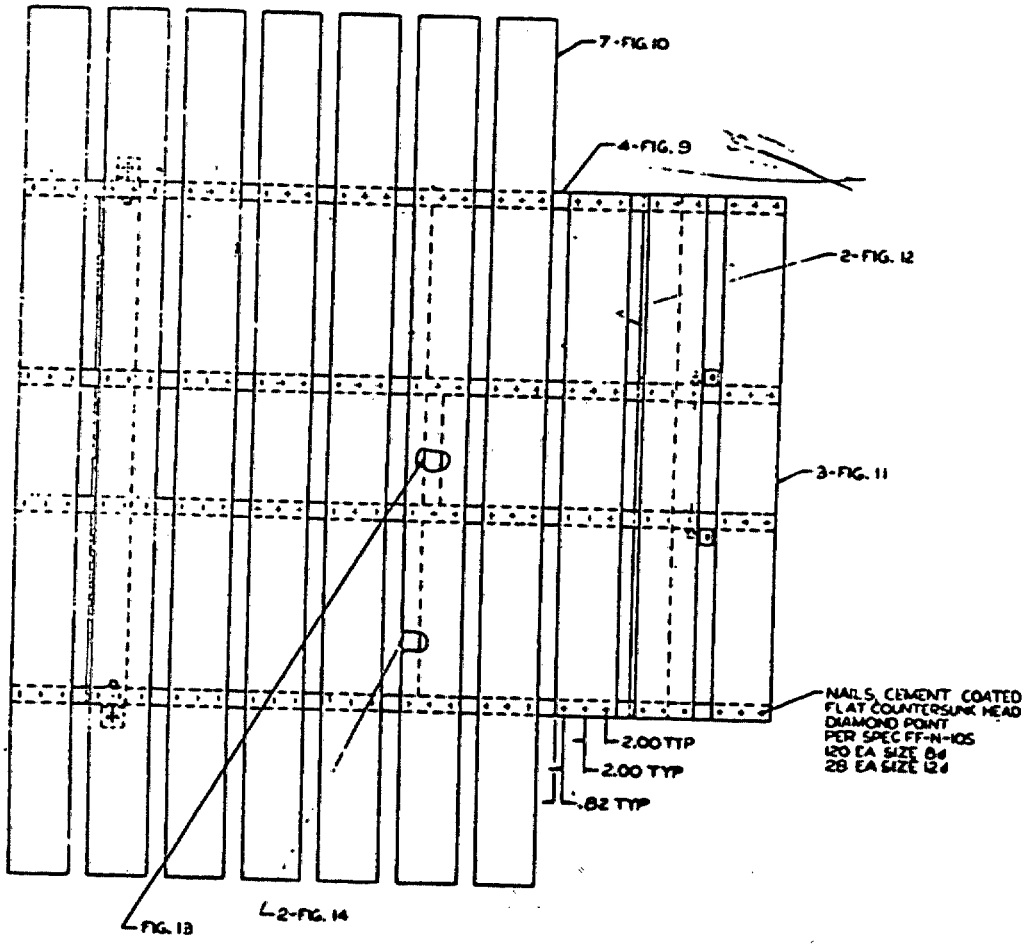
FIG. NO.	NOMENCLATURE	QTY REQD
2	MATERIAL LIST	—
3	SADDLE ASSEMBLY- PLAN VIEW	—
4	SADDLE ASSEMBLY-SIDE VIEW	—
5	SADDLE ASSEMBLY-END VIEW	—
6	BRACKET, LEFT, REAR	1
7	BRACKET, RIGHT, REAR	1
8	BRACKET, SIDE	2
9	SUPPORT, PLATFORM	4
10	PLANK, PLATFORM	7
11	PLANK, PLATFORM	3
12	BRACE	2
13	BRACE	1
14	BRACE	2
15	HARDWARE, OVE SADDLE ASSEMBLY SEE HARDWARE LIST	—

PARTS LIST-OVE SADDLE ASSEMBLY
FIGURE 1

DESCRIPTION	QTY REQD
1 X 6 WOOD (COMMERCIAL SIZE)	62 FEET LONG
2 X 4 WOOD (COMMERCIAL SIZE)	13 FEET LONG
2 X 6 WOOD (COMMERCIAL SIZE)	25 FEET LONG
1/4 X 2 STEEL STRIP	19 INCHES LONG
1 1/2 X 1 1/2 X 1/4 ANGLE, STEEL	38 INCHES LONG

MATERIAL LIST-OVE SADDLE ASSEMBLY
FIGURE 2

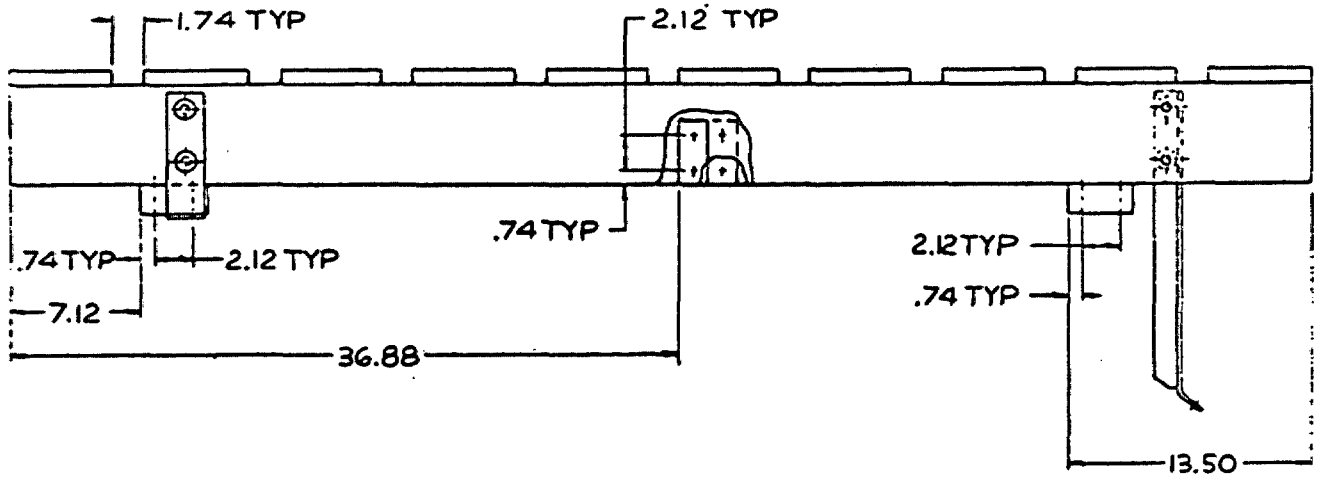
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PLAN VIEW
FIGURE 8

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F 11604898

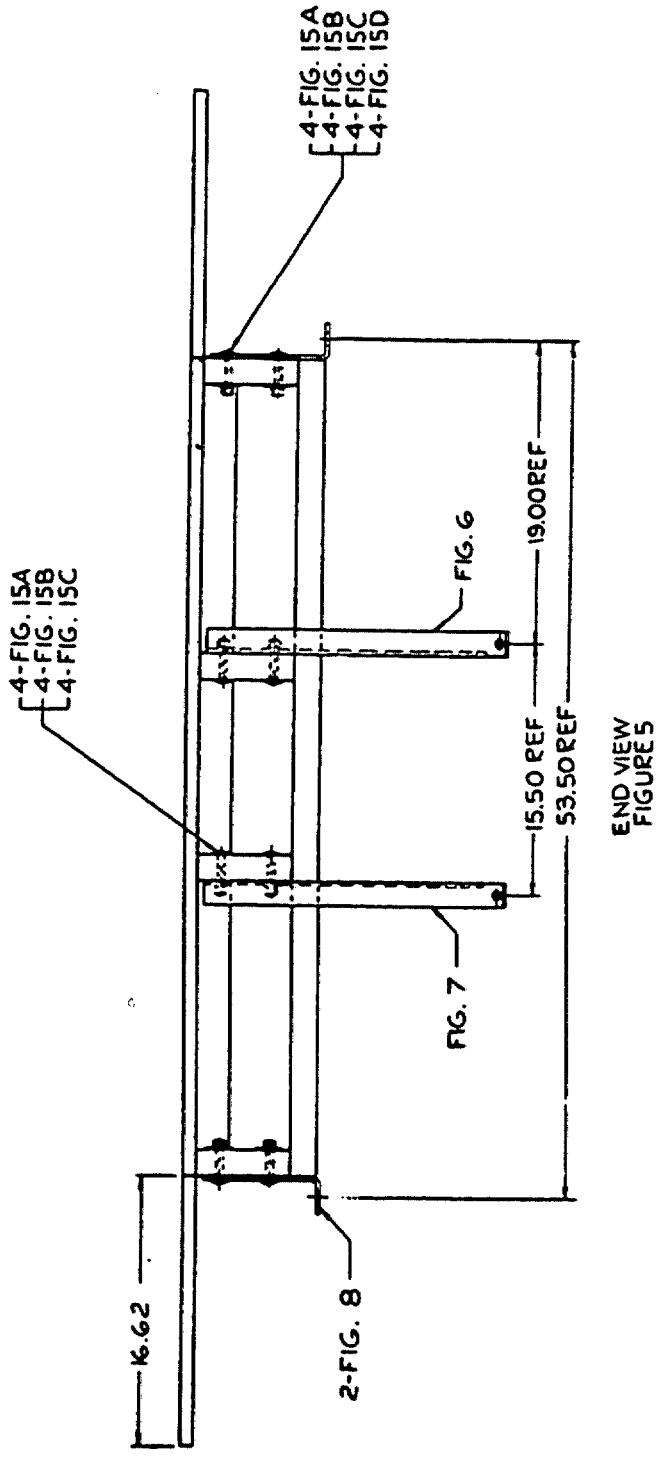
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SIDE VIEW
FIGURE 4

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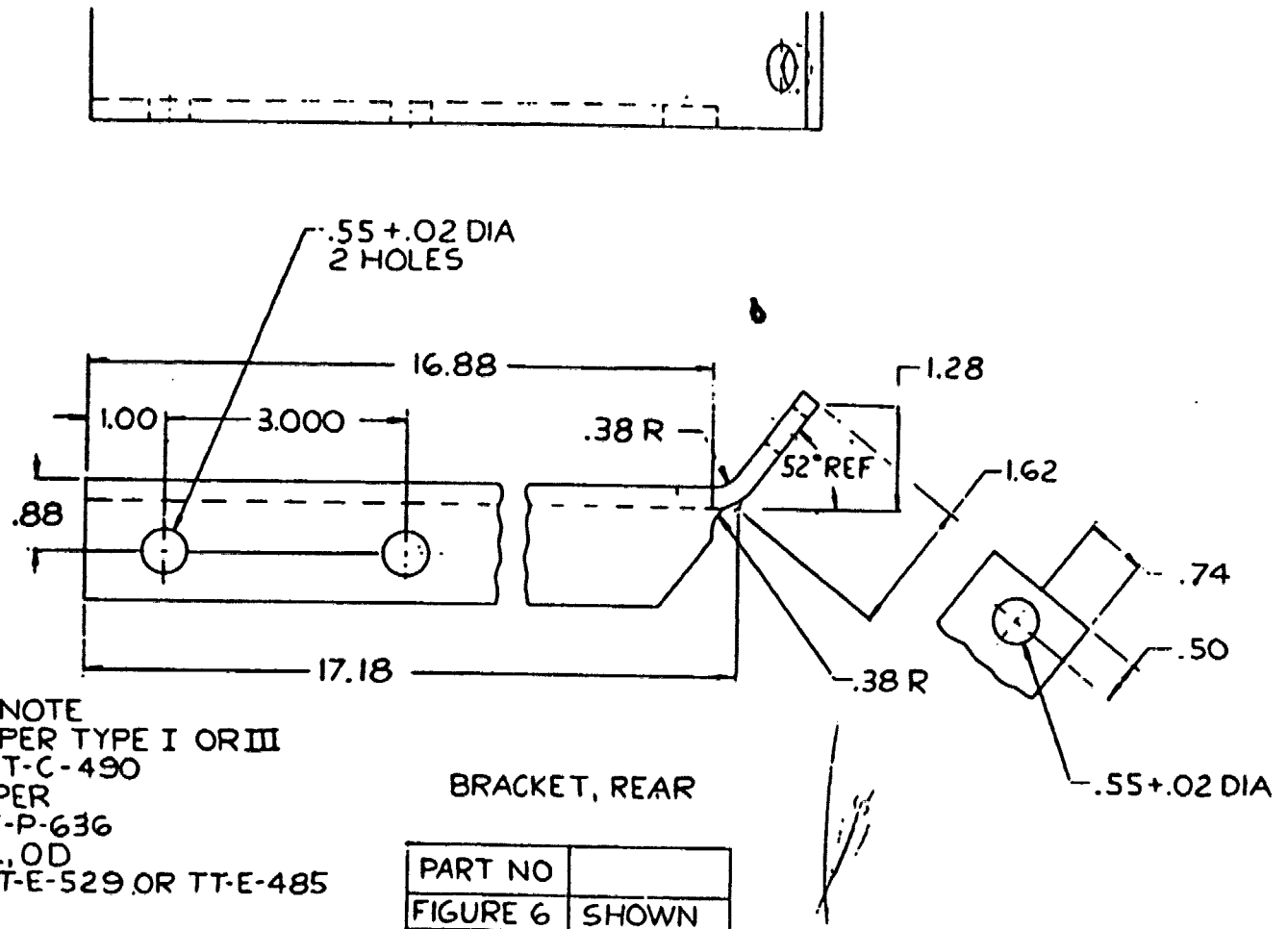
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END VIEW
FIGURE 5

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FINISH NOTE
TREAT PER TYPE I OR III
SPEC TT-C-490
PRIME PER
SPEC TT-P-636
ENAMEL, OD
SPEC TT-E-529 OR TT-E-485

MUST BE FREE OF BURRS
AND SHARP EDGES

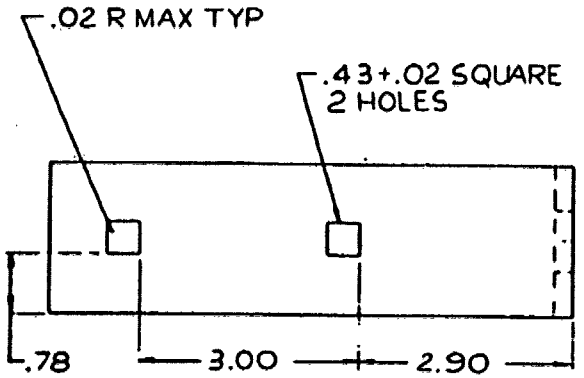
MATERIAL NOTE
STEEL, CARBON
GRADE A
SPEC QQ-S-741
$1\frac{1}{2} \times 1\frac{1}{2} \times \frac{1}{4}$ ANGLE

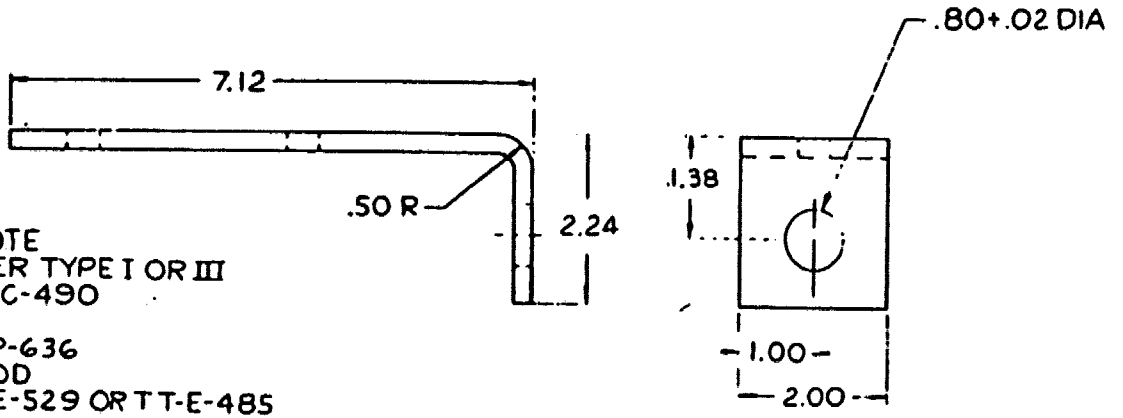
BRACKET, REAR

PART NO	
FIGURE 6	SHOWN
FIGURE 7	OPPOSITE

DRAWING NO.
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FINISH NOTE
TREAT PER TYPE I OR III
SPEC TT-C-490
PRIME PER
SPEC TT-P-636
ENAMEL, OD
SPEC TT-E-529 OR TT-E-485

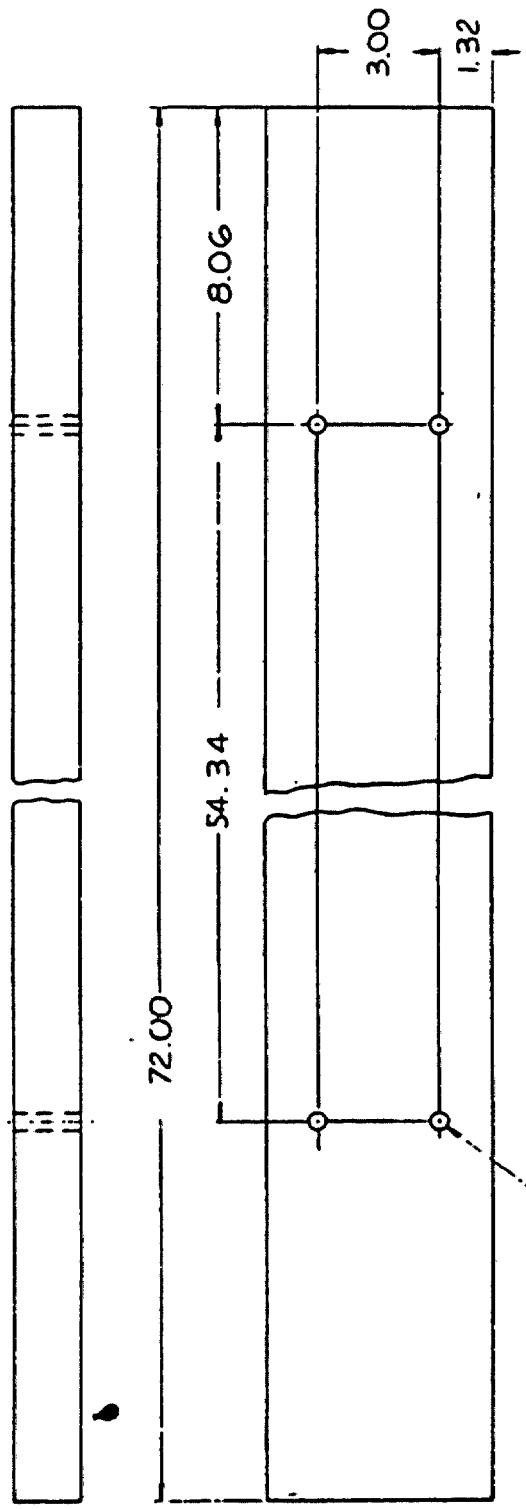
MUST BE FREE OF BURRS
AND SHARP EDGES

MATERIAL NOTE
STEEL, CARBON
C1010 THRU C1020
CONDITION OPTIONAL
SPEC QQ-S-630 OR 634
.250 THICK

BRACKET, SIDE
FIGURE 8

DRAWING NO.
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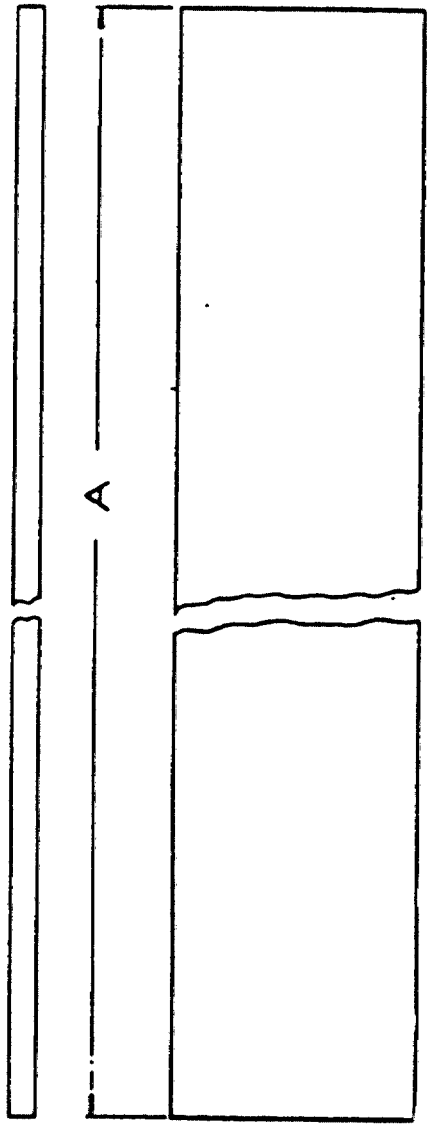


SUPPORT, PLATFORM
FIGURE 9

DRAWING NO.
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DRAWING NO
F 11604898



PLANK, PLATFORM

FIGURE NO	DIM A
10	84.00
11	50.74

MATERIAL NOTE
WOOD CLASS B
MIL-W-3912
COMMERCIAL SIZE 1 X 6

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BRACE

FIGURE NO	DIM A
12	50.74
13	10.74
14	16.74

MATERIAL NOTE
WOOD CLASS B
MIL-W-3912
COMMERCIAL SIZE 2 X 4

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FIG. NO.	ARMY PART NO.	DESCRIPTION	MATERIAL	QTY REQD
15A	MS35338-65	WASHER, LOCK, SPLIT 3/8	CARBON STEEL	8
15B	MS51967-9	NUT, PLAIN, HEX 3/8-16UNC-2B	CARBON STEEL	8
15C	MS35752-45	BOLT, SQ. NECK, 3/8-16UNC-2A X 2 1/2	CARBON STEEL	8
15D	MS63040-6	WASHER, FLAT 3/8	CARBON STEEL	4

HARDWARE-OVE SADDLE ASSEMBLY
FIGURE 15

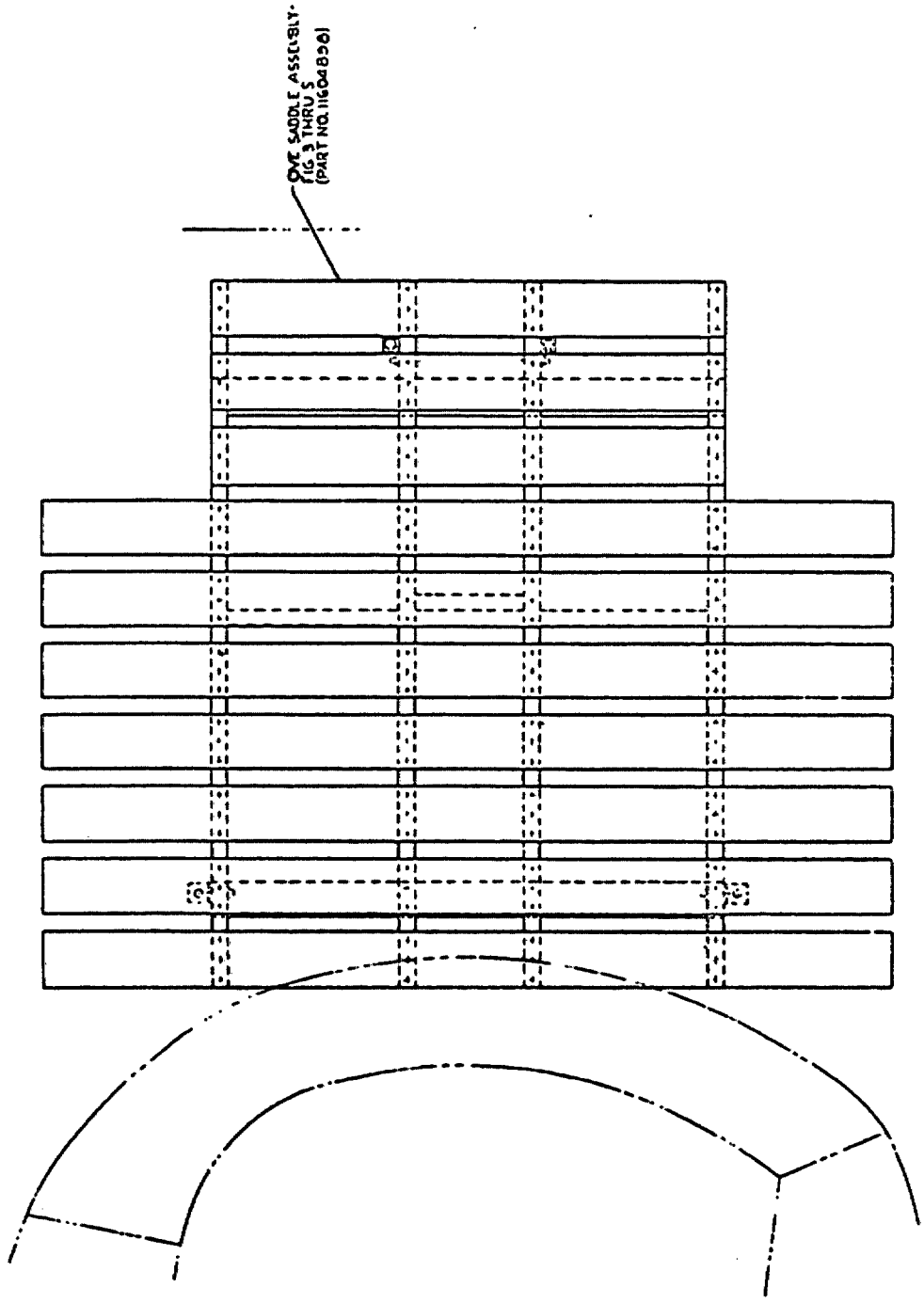
FIG. NO.	NOMENCLATURE	QTY REQD
17	OVE SADDLE INSTALLATION-PLAN VIEW	—
18	OVE SADDLE INSTALLATION-SIDE VIEW	—
19	OVE SADDLE INSTALLATION-END VIEW	—
20	HARDWARE-SEE HARDWARE LIST	—

FIGURE 16. PARTS LIST-OVE SADDLE INSTALLATION

DRAWING NO.
F 11604000

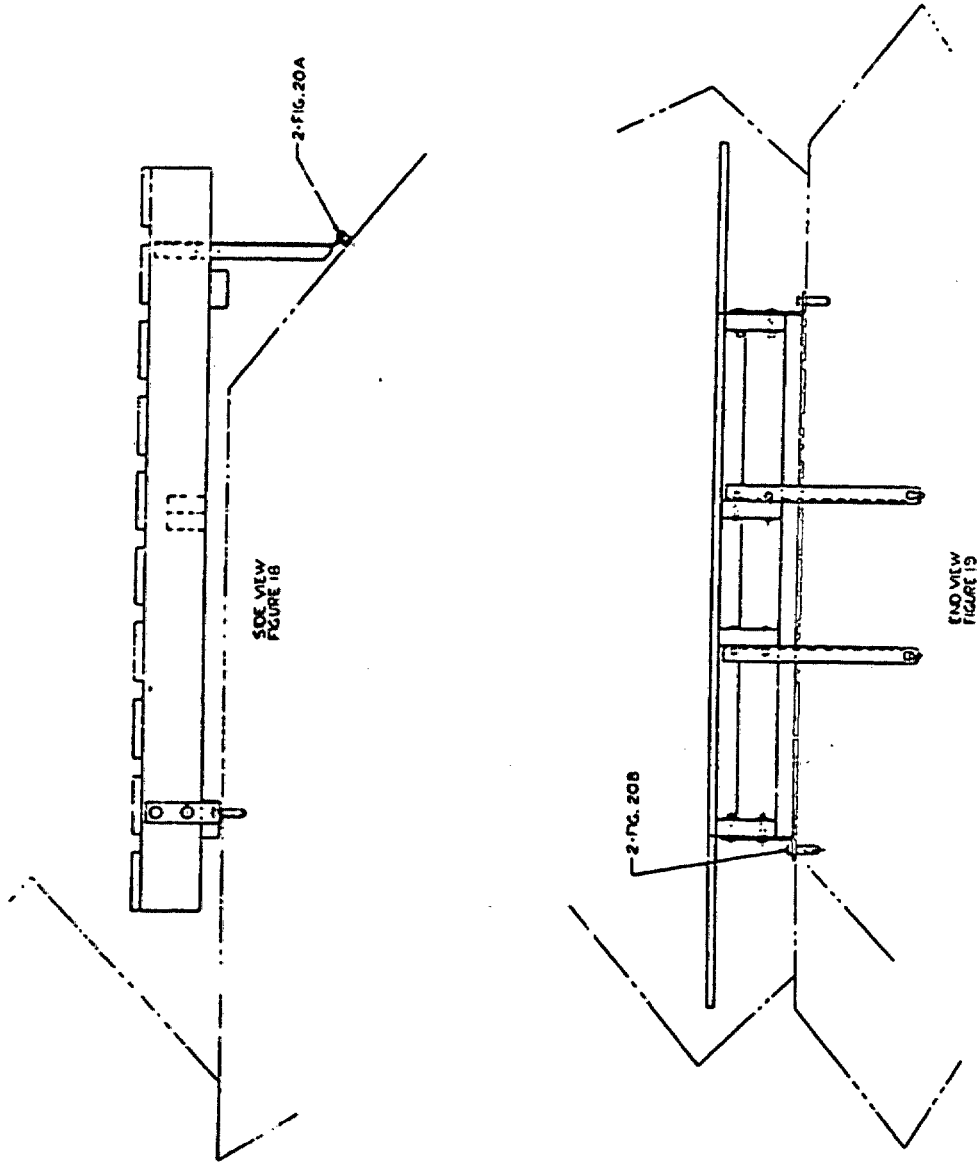
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DRAWING NO
F 11604899



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DRAWING NO.
F 11604899



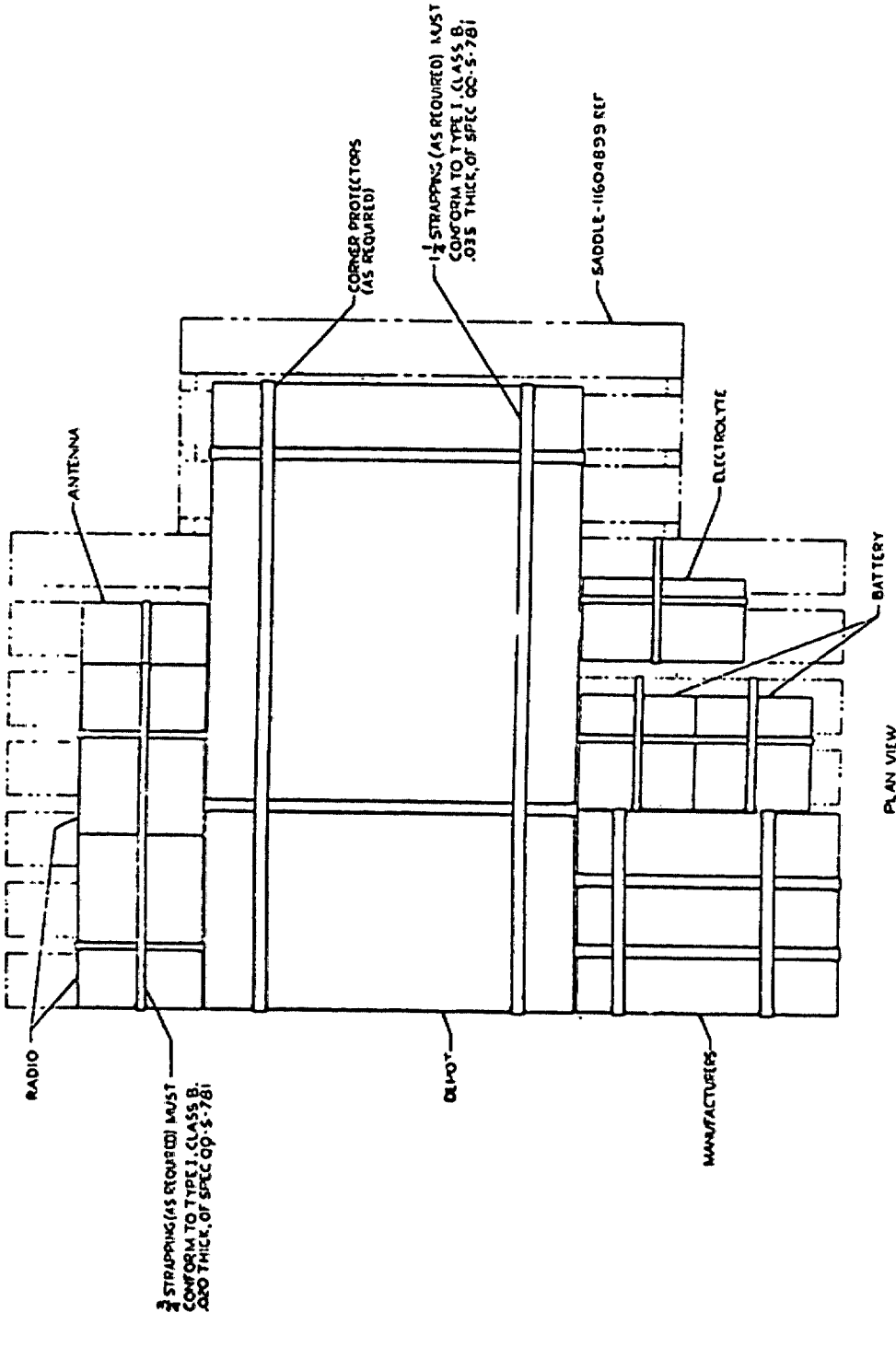
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:22Z
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FIG. NO.	ARMY PART NO.	DESCRIPTION	MATERIAL	QTY REQD
20A	MS90728-108	SCREW, CAP 1/2-13UNC-2A X 7/8	ALLOY STEEL	2
20B	MS90727-189	SCREW, CAP 3/4-16UNF-2A X 2 1/2	ALLOY STEEL	2

HARDWARE-OVE SADDLE INSTALLATION
FIGURE 20

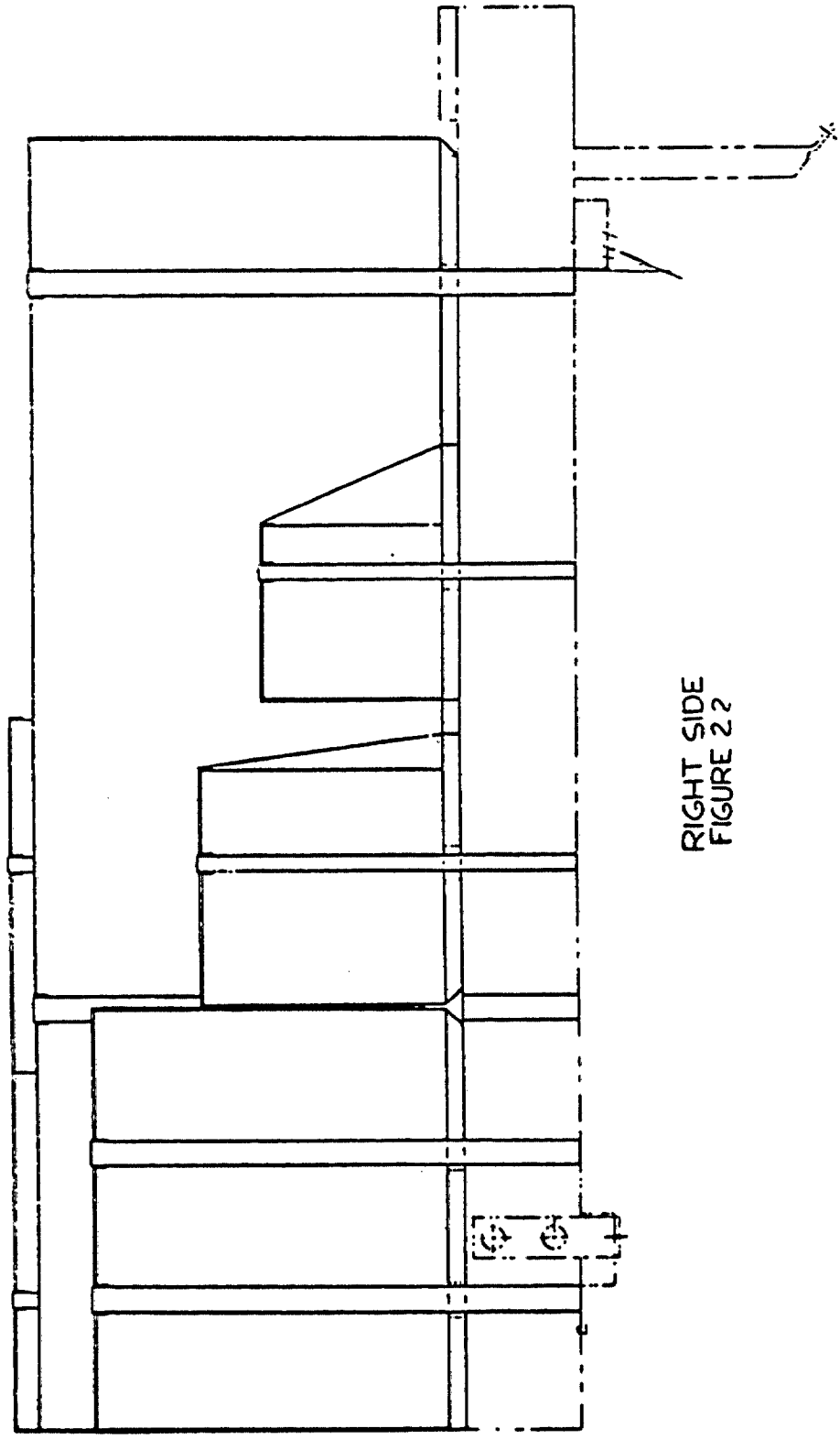
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F 11604900



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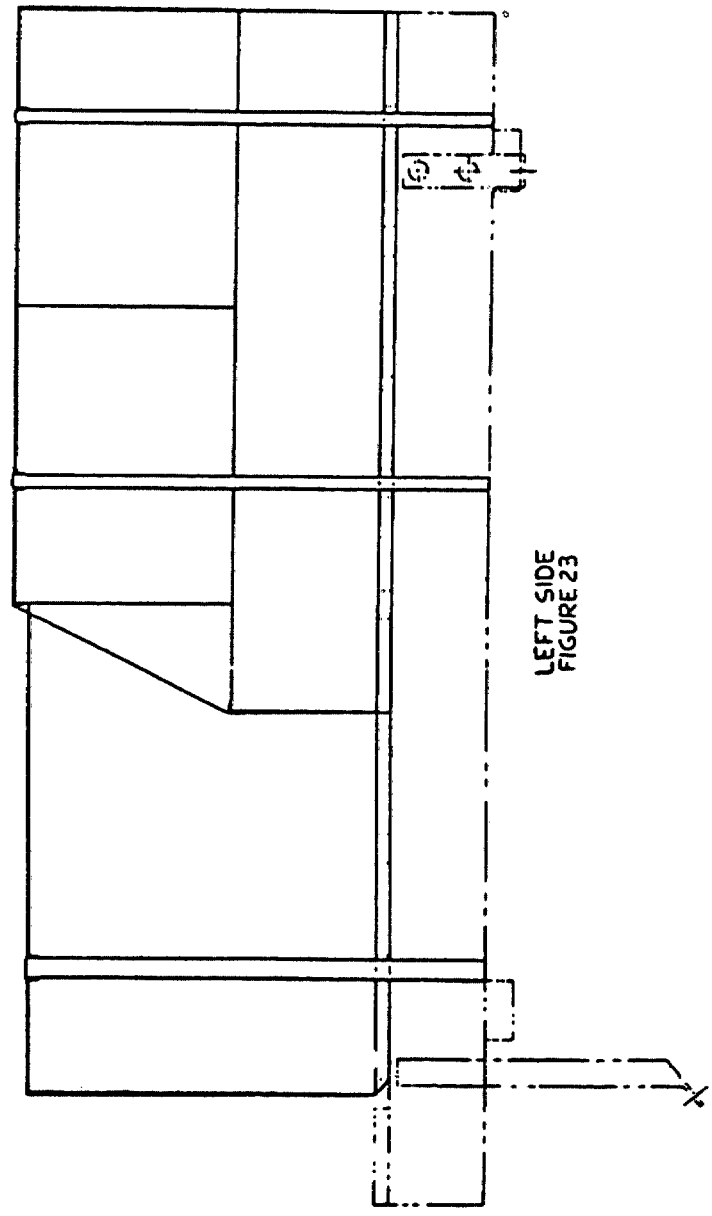
DRAWING NO.
F 11604900



RIGHT SIDE
FIGURE 22

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LEFT SIDE
FIGURE 23

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